

Computer Graphics

COMP3421 and COMP9415

Never Stand Still

Unreal Engine (How to develop a 3D maze)

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School of Computer Science and Engineering
Term 3, 2023

Content overview

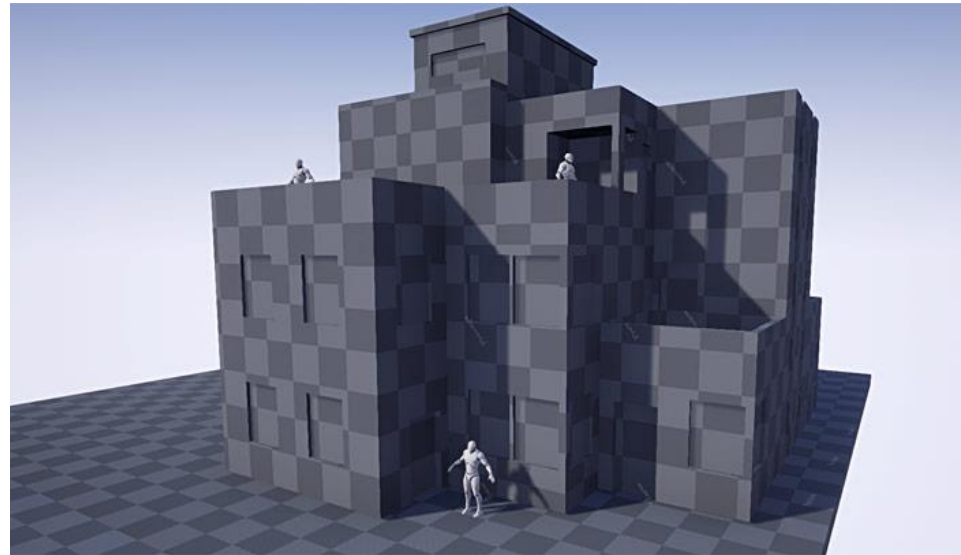
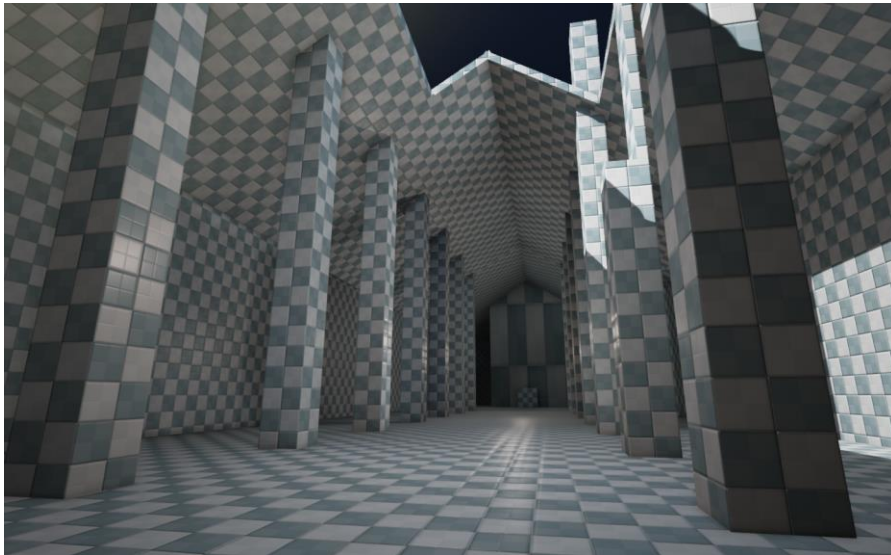
- Transforming the floor using keyboard and mouse input
- Add content to the level
- Add material to the objects
- Apply physics
- Geometry Brushes
- Static mesh and collision detection
- Create a simple win condition for levels
- Using trigger box
- Restarting and loading a level
- Publishing

Maze breakdown



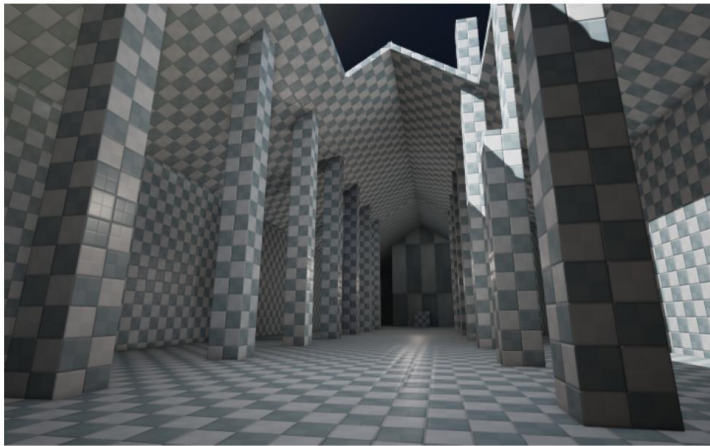
Geometry Brushes

- Geometry Brushes are the most basic tools for level construction in Unreal.
- Conceptually, it is best to think of a Geometry Brush as filling in and carving out volumes of space in your level.



Geometry Brushes

- Block out and path level
- Play test flow and gameplay
- Modify layout and repeat testing
- Initial meshing pass
- Initial lighting pass
- Play test for collision and performance issues
- Polish pass



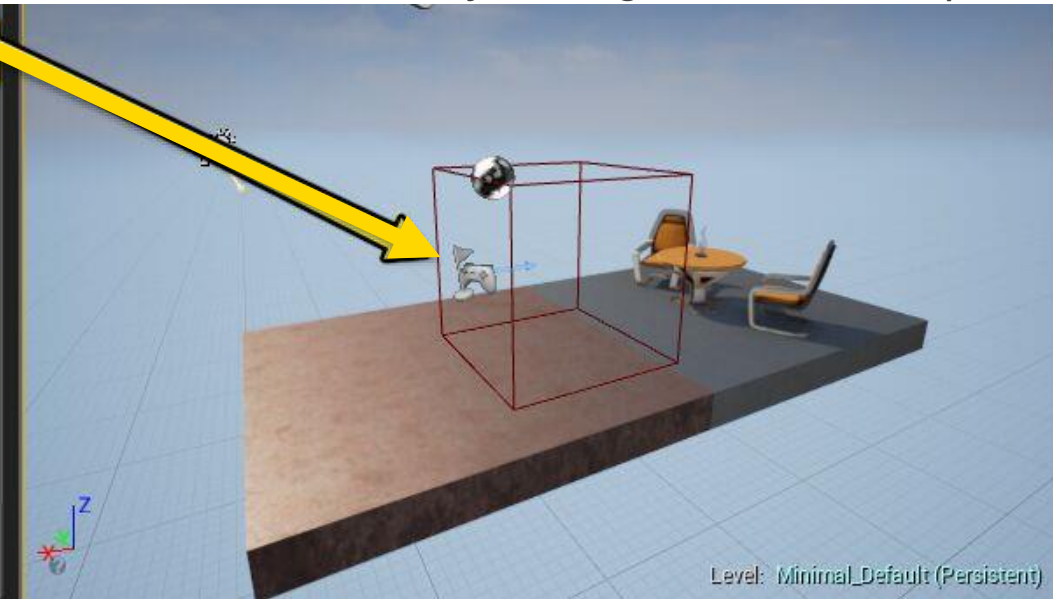
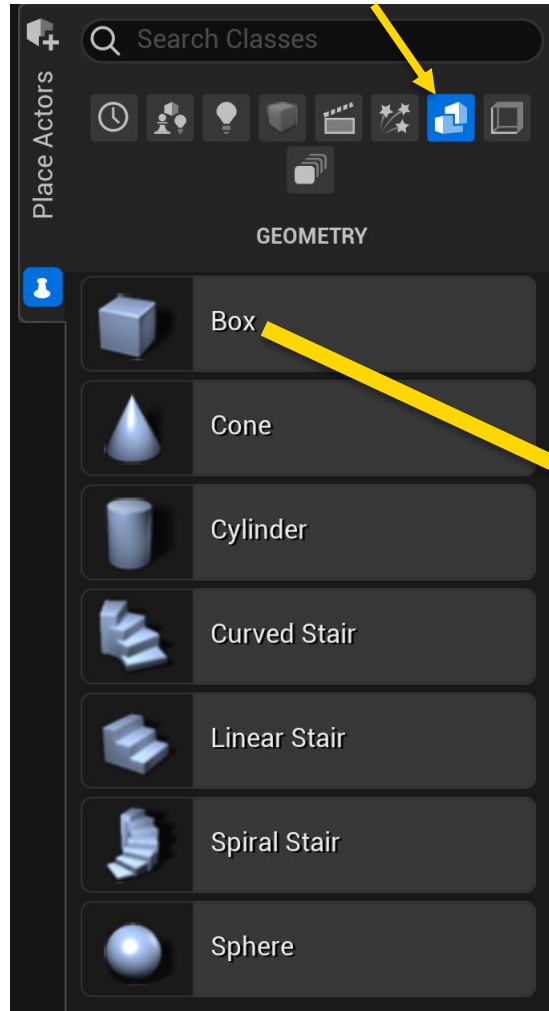
Brush Blocking / Rough-In



Final level

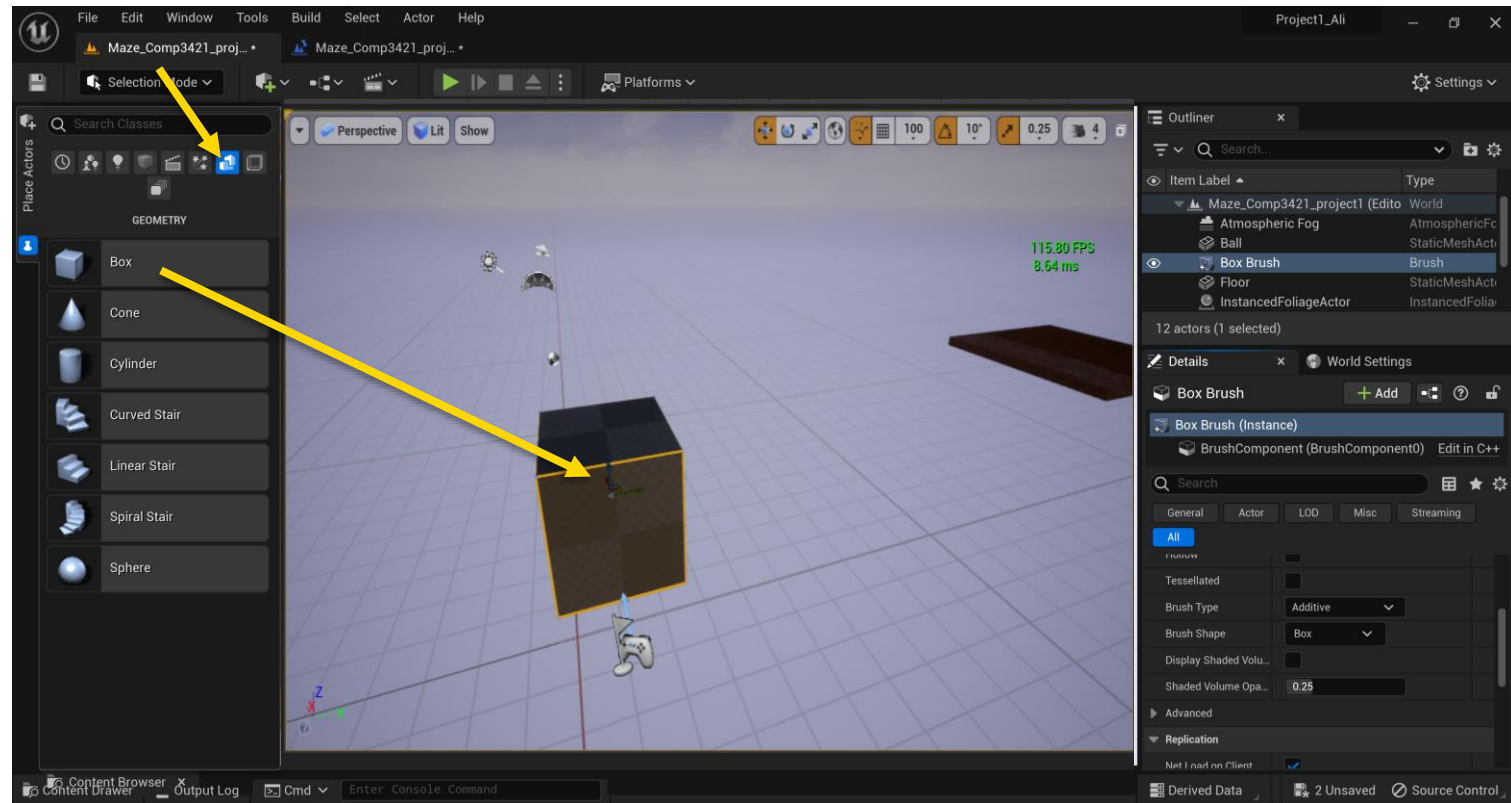
Geometry Brushes

1. Add a brush using the Geometry tab in the Place Actor panel.
2. Choose the Brush Type from the **Details** panel (additive or subtractive)
3. Drag and drop one of the primitive types from the list into the Viewport.
4. Modify the Brush Settings, use the transform tools, or activate Geometry Editing Mode to reshape and resize.



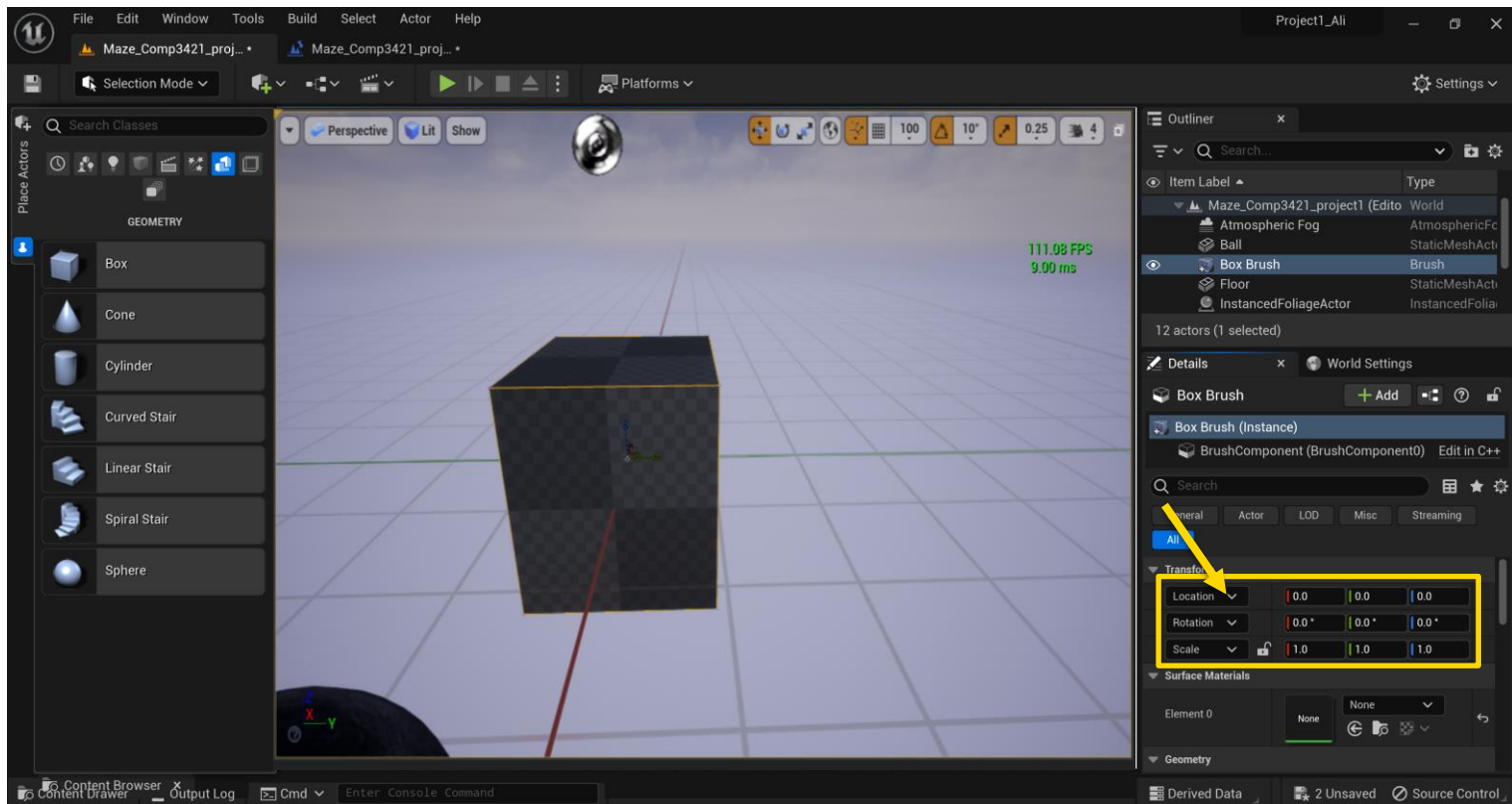
Create the base floor using Geometry Brushes

- Place Actor panel > Geometry tab > Box brush



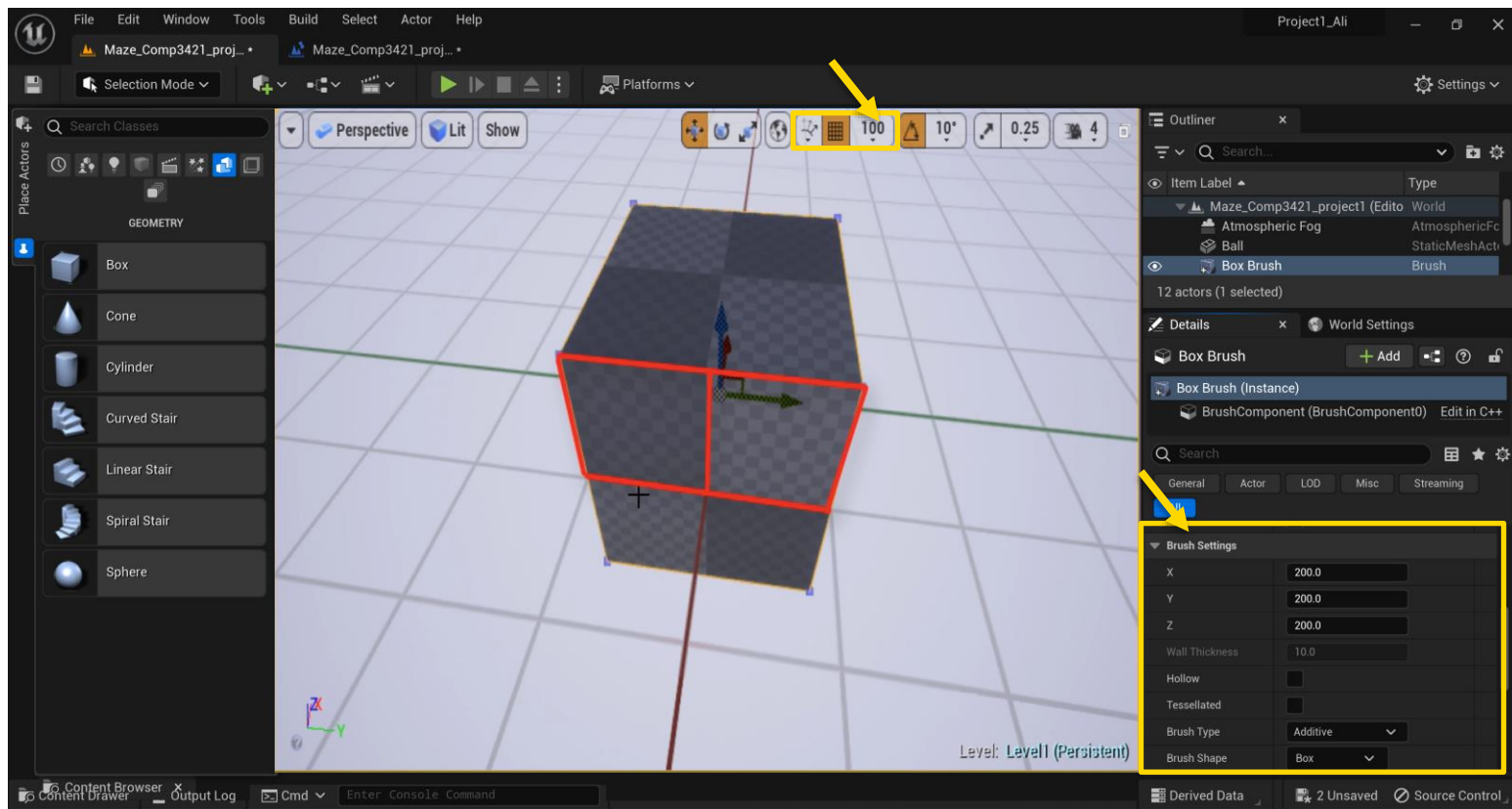
Create the base floor using Geometry Brushes

- Put the Box brush to the center of the game world
- Go to the Transform and type zero for X, Y and Z axes.



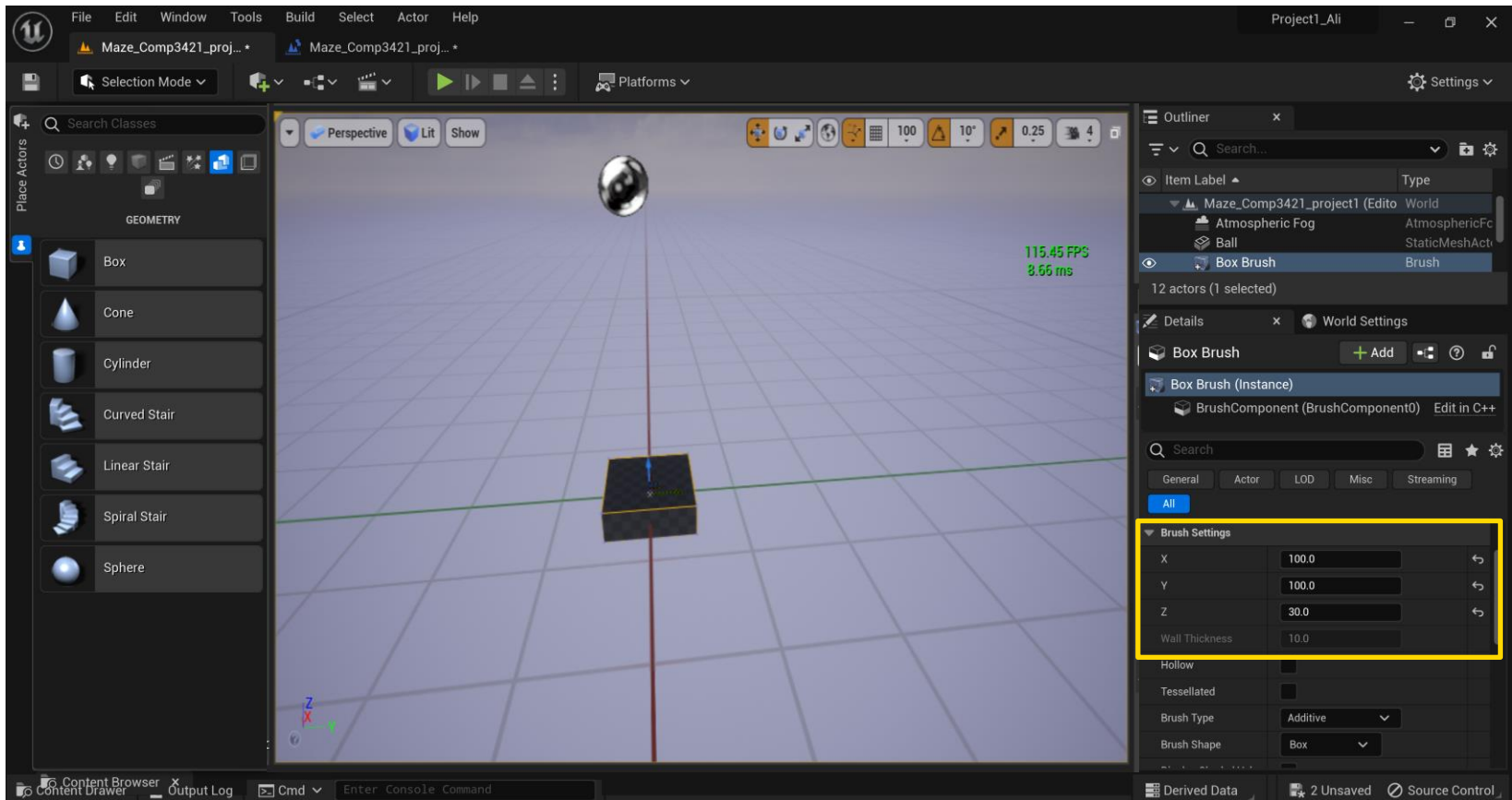
Create the base floor using Geometry Brushes

Based on the snapping size that is set to 100, each side of the square is 200 units in size.



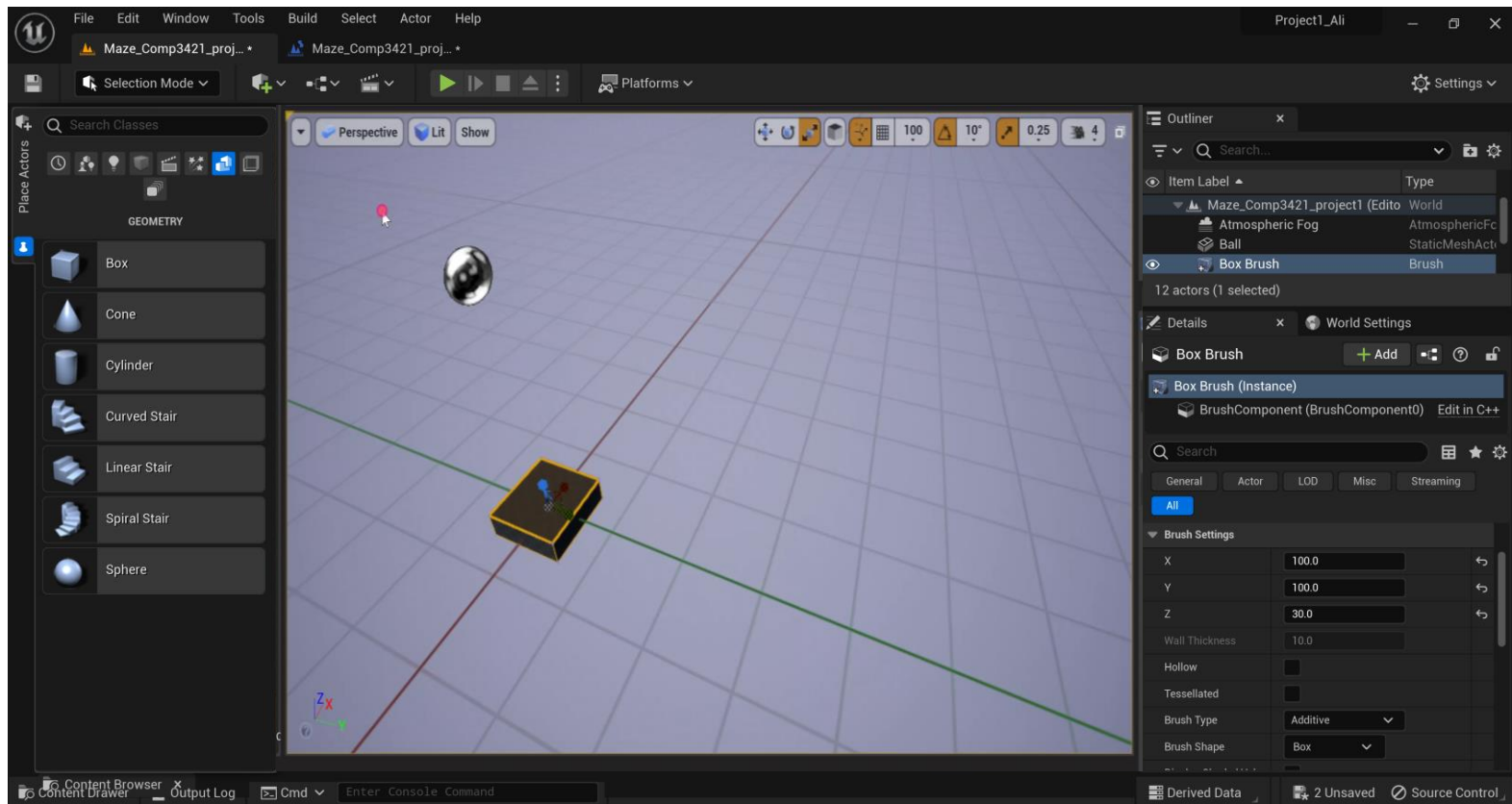
Create the base floor using Geometry Brushes

- Change box brush size > Details panel > Brush settings > Type 100 on the X > Type 100 on the Y > Type 30 on the Z.



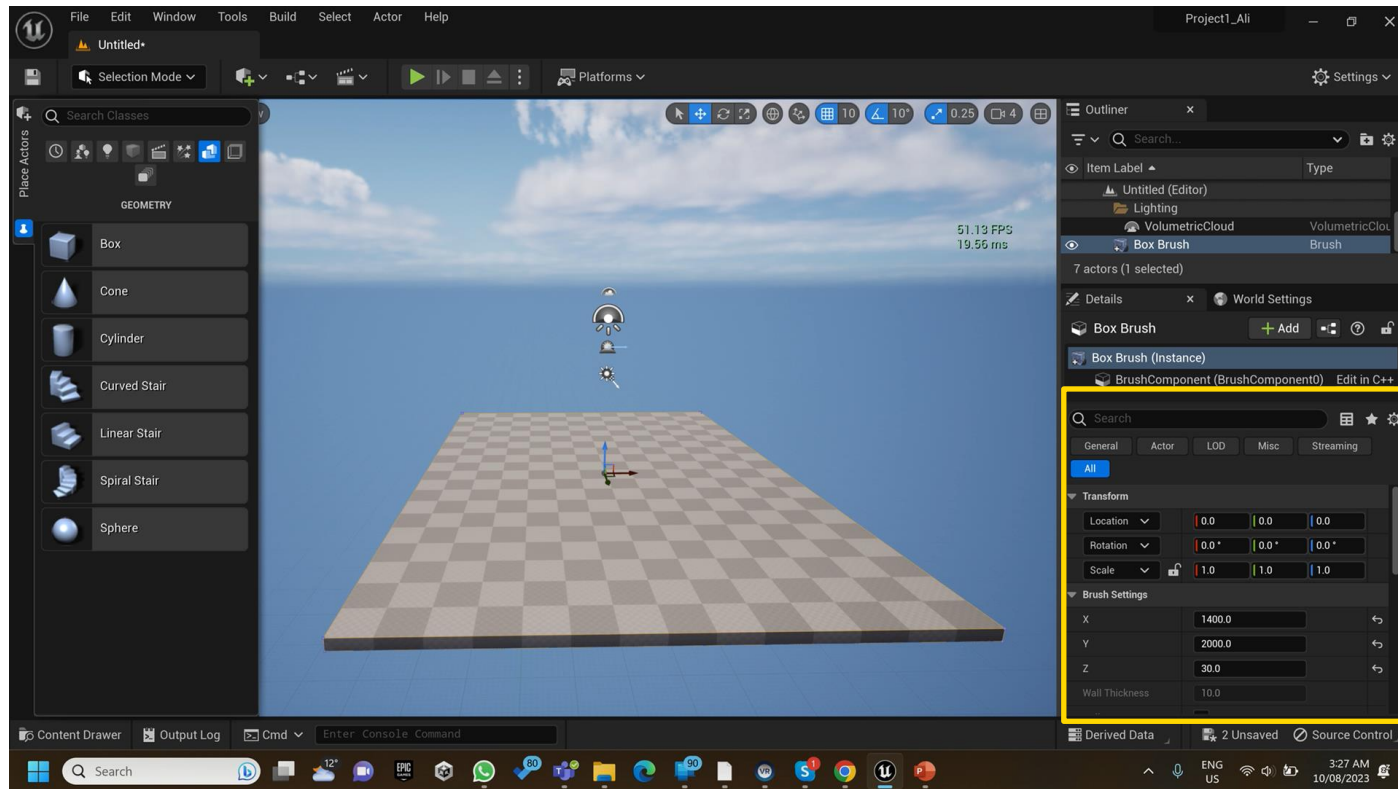
Create the base floor using Geometry Brushes

Avoid scaling the brush with the Scale Transform tool in the viewport.



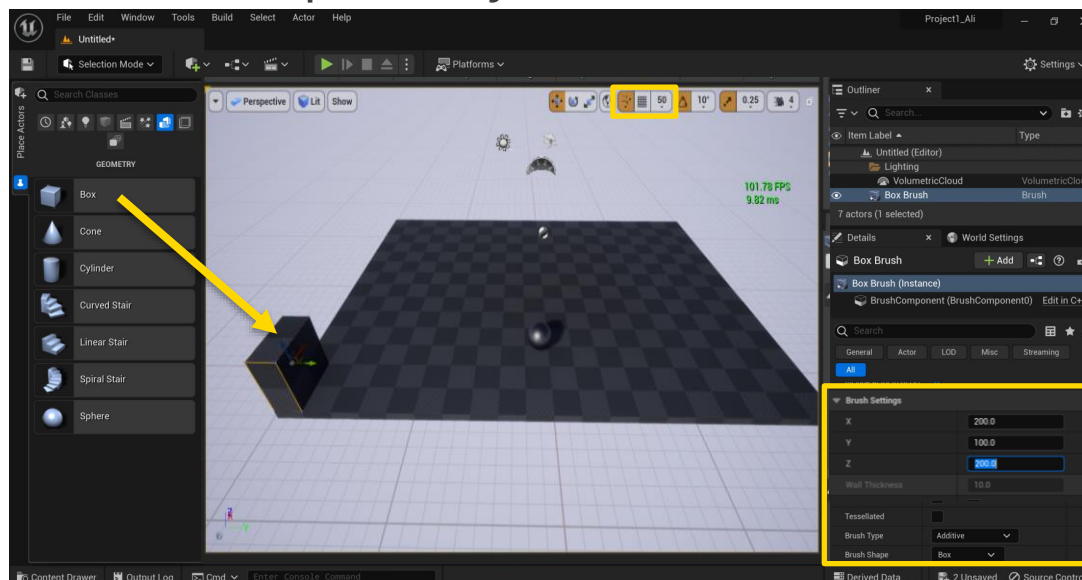
Create the base floor using Geometry Brushes

Details panel > Brush settings > Type 1400 on the X
> Type 2000 on the Y > Type 30 on the Z.

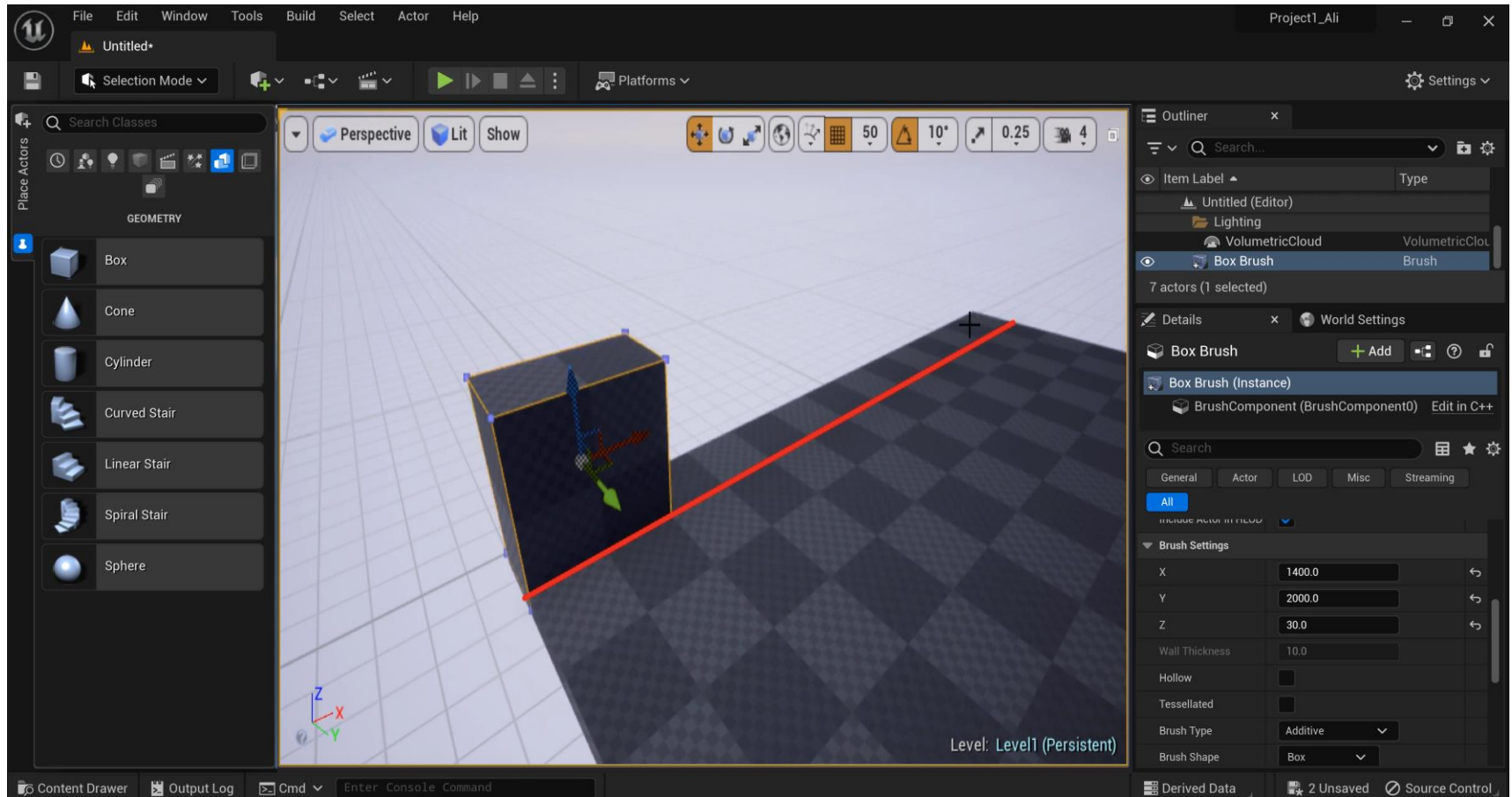


Create the maze walls using Geometry Brushes

- Drag out another box brush
- Details panel > Brush settings > Type 200 on the X > Type 100 on the Y > Type 200 on the Z.
- Put the wall on the corner of the floor.
- Change the snapping size to 50 or disable snapping to the grid so that you can move the wall precisely.

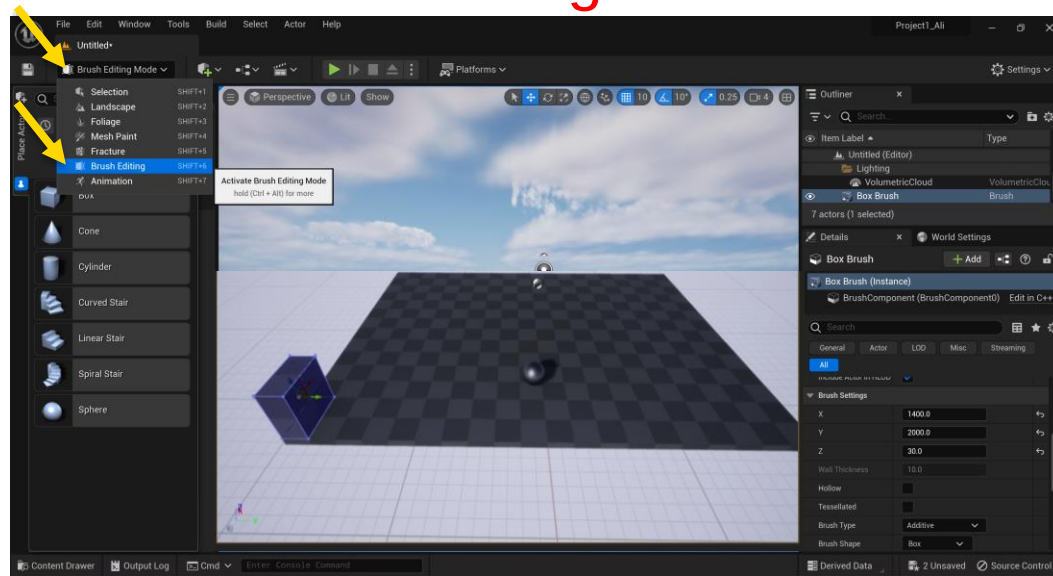


Create the maze walls using Geometry Brushes



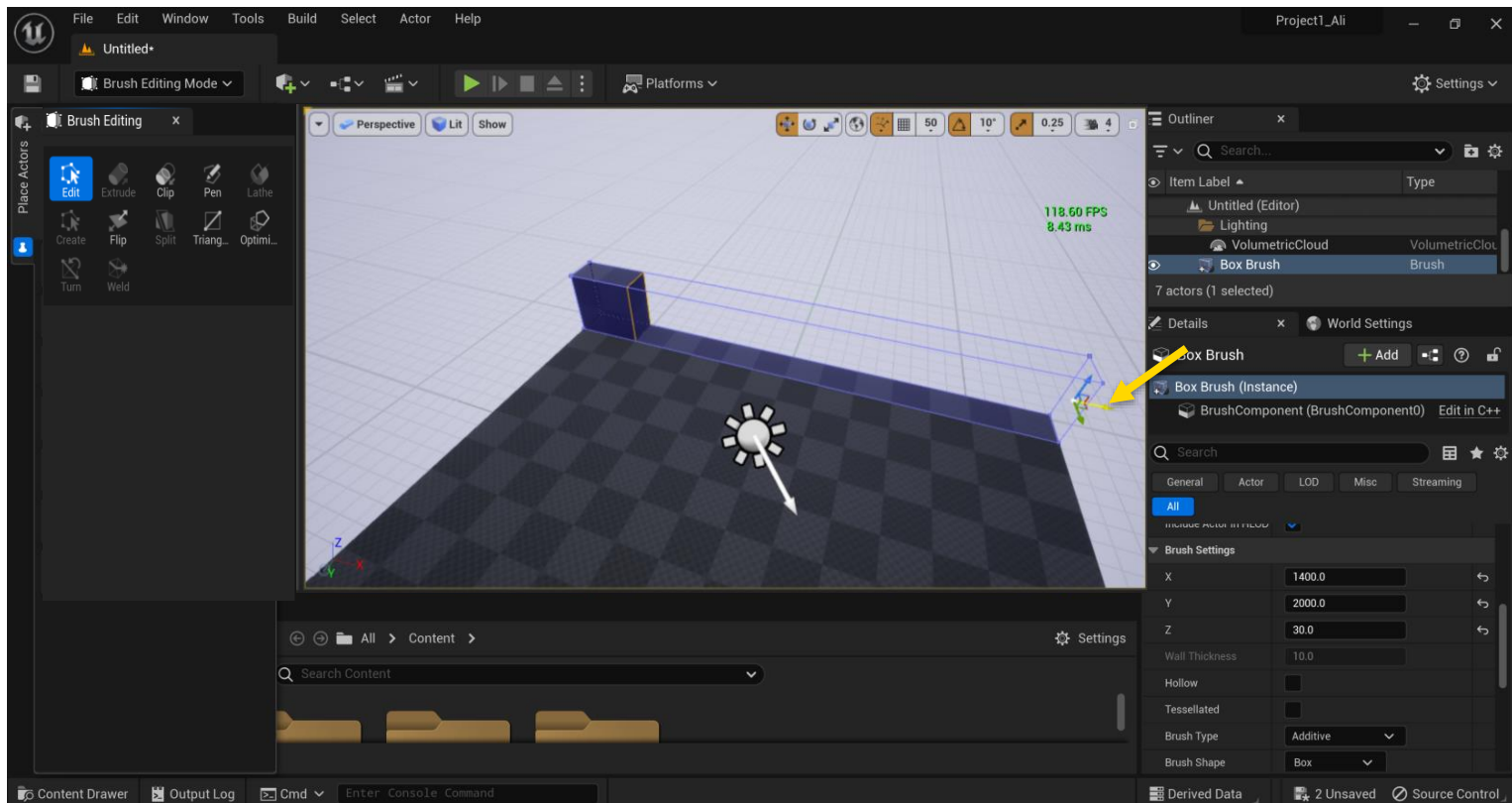
Create the maze walls using Geometry Brushes

- Two solutions to change the brush size:
 - a. Details panel > Brush settings
 - b. Changing the **Editing Mode** to **Brush Editing** and use drag and drop to resize the wall.
- In order to use brush editing mode, click the 'Modes' button and click '**Brush Editing**'.



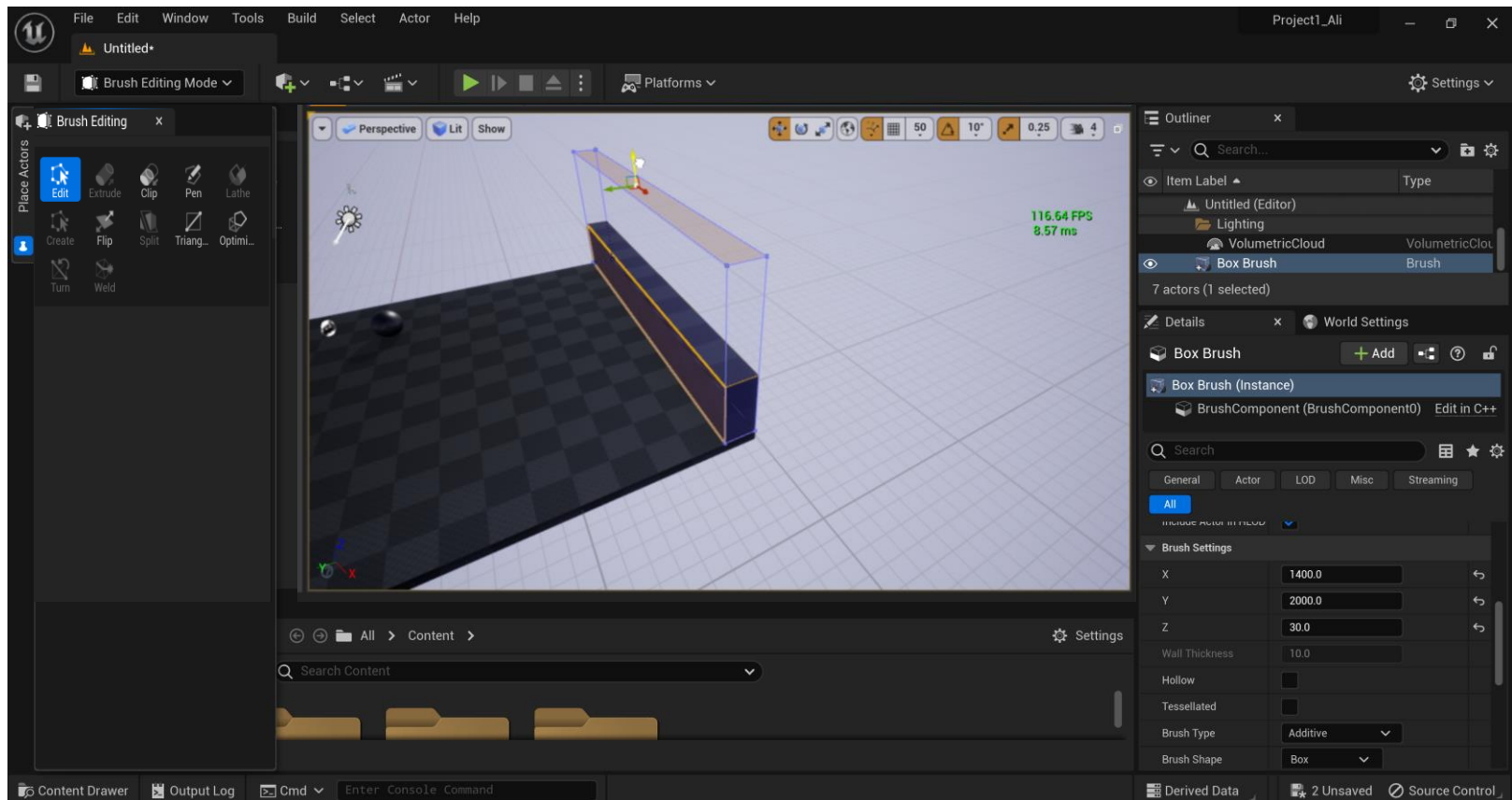
Create the maze walls using Geometry Brushes

Select the desired face of the box brush and use the translate tool to change the length.

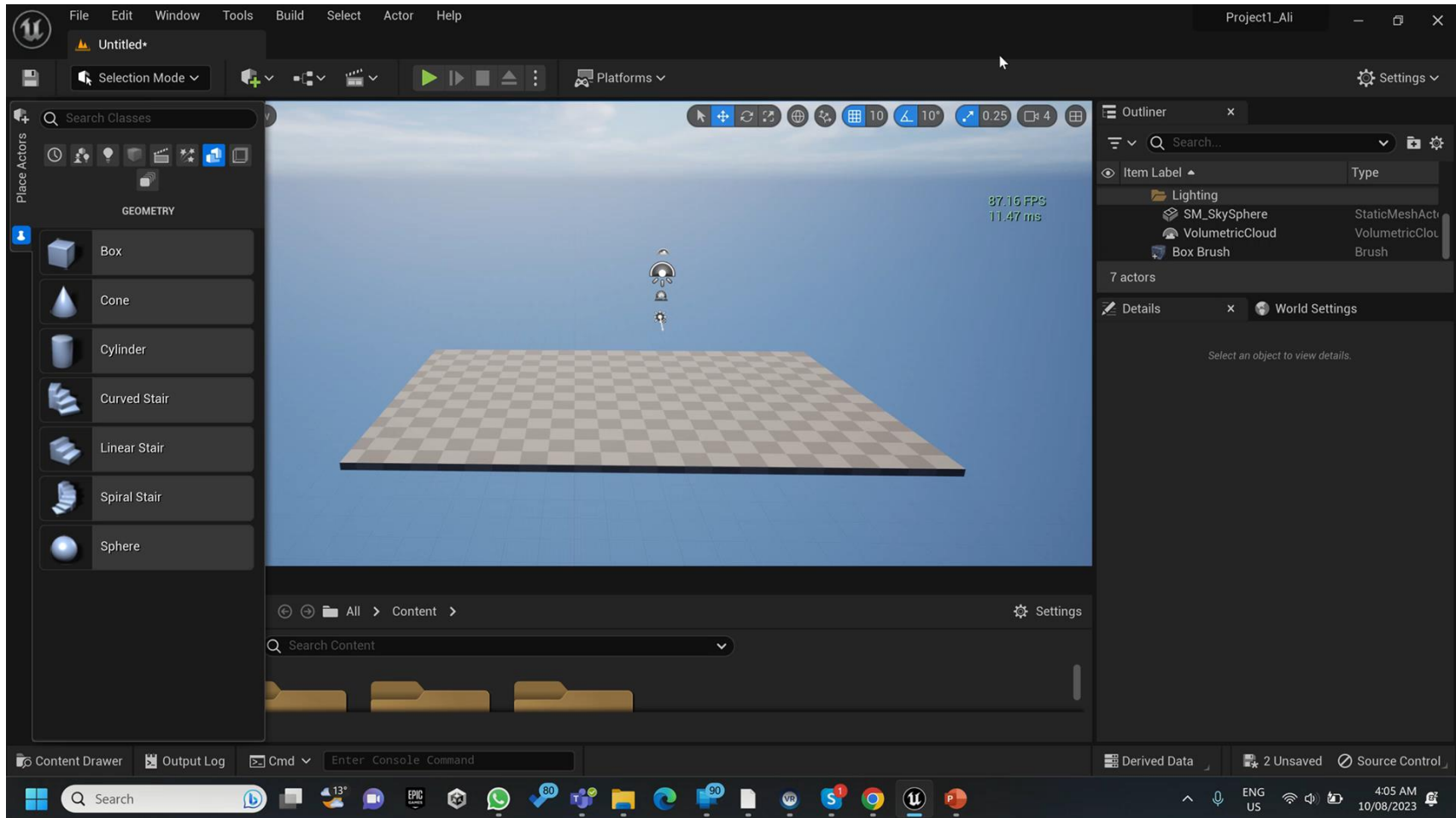


Create the maze walls using Geometry Brushes

Set the wall height by dragging the Y axis.



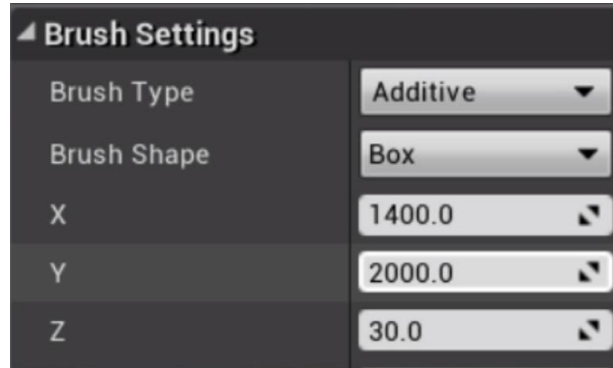
Create the maze walls using Geometry Brushes



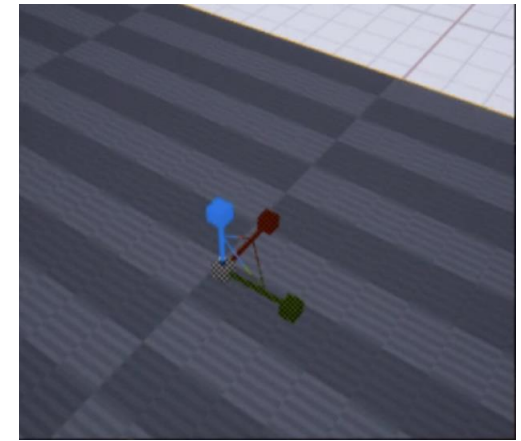
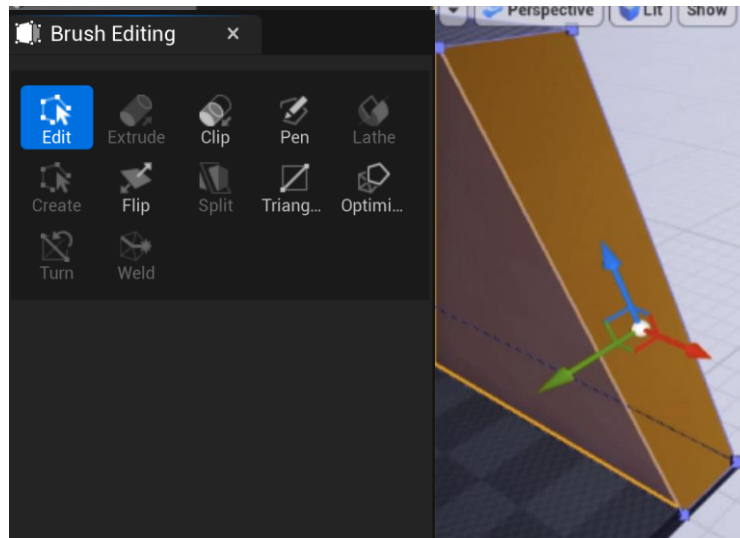
Create the maze walls using Geometry Brushes



Brush settings
(details panel)



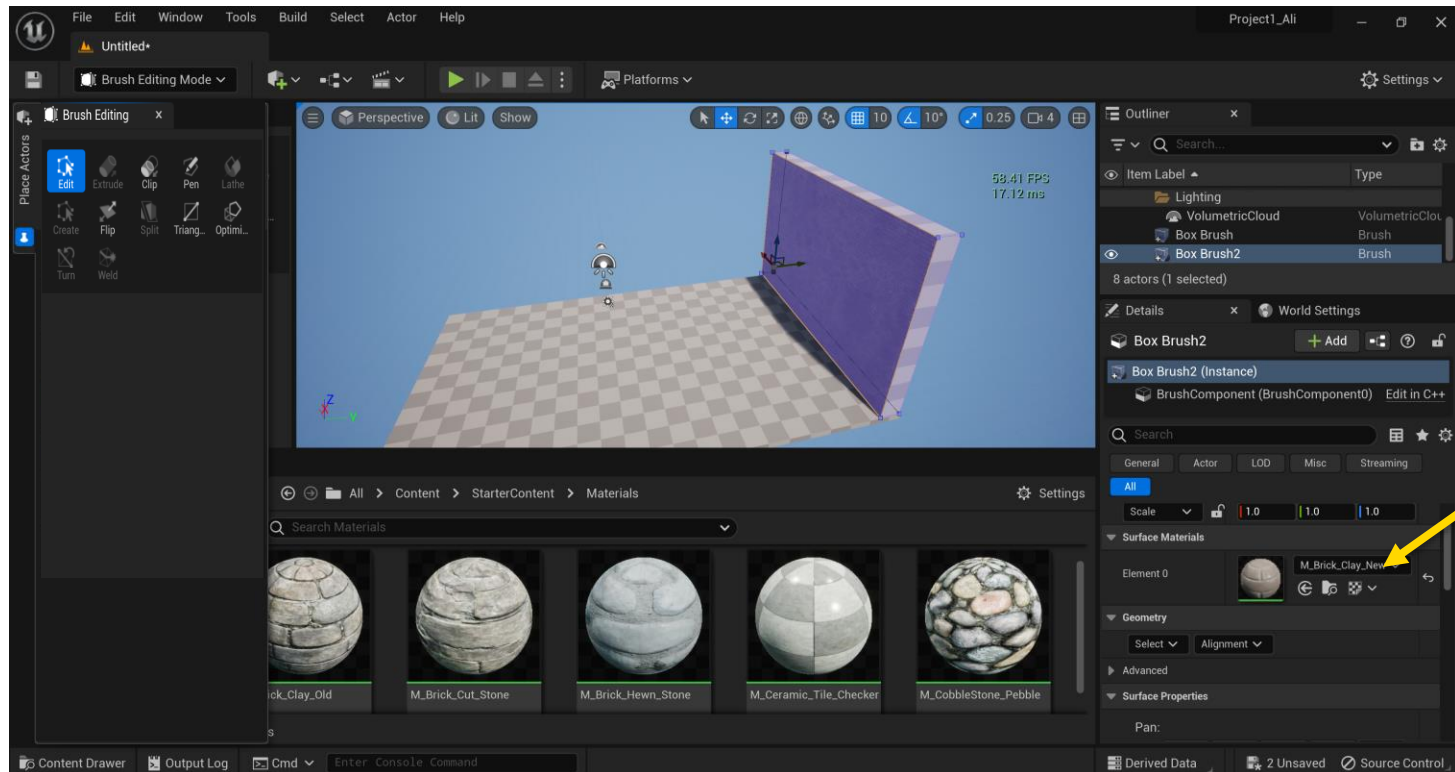
Geometry editing
(modes panel)



Scale tool
(Causes stretching of materials)

Applying material to the Geometry Brushes

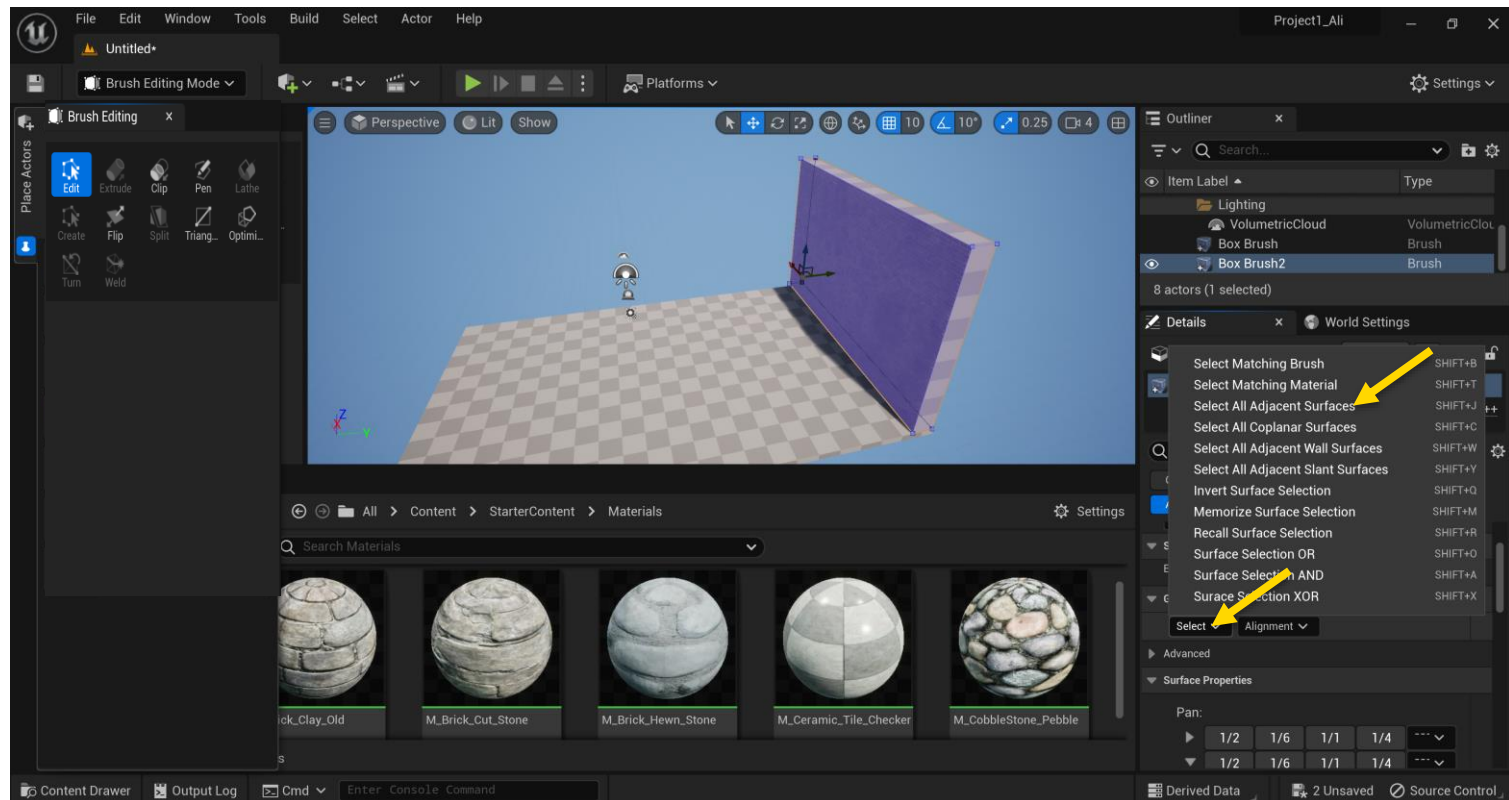
In contrast with the static meshes, when we change the material of the brushes, the material will be only applied on the selected side not the whole shape.



Applying material to the Geometry Brushes

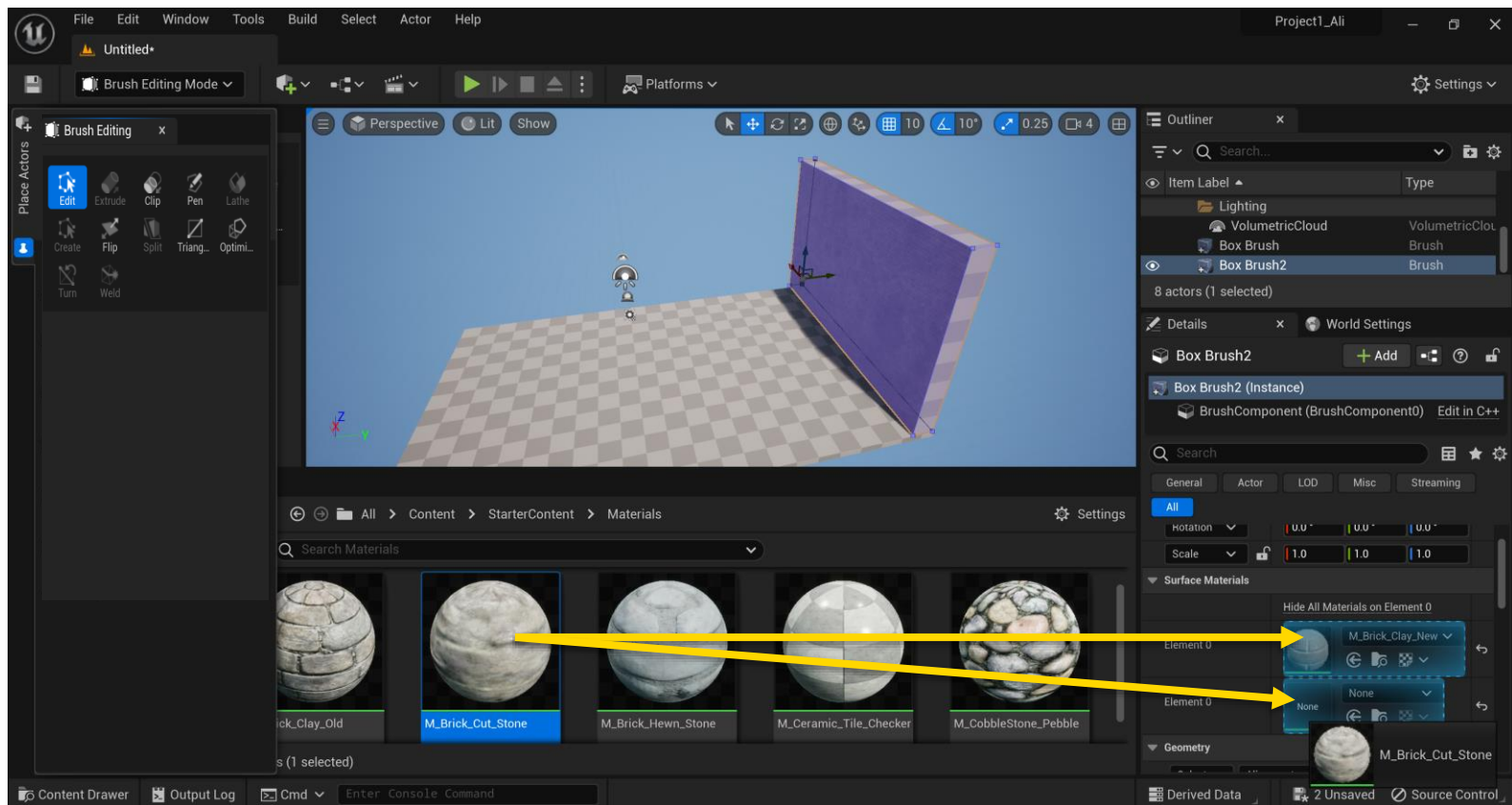
To select all sides of the brush:

Select the geometry brush > Details panel > Geometry category > Click the select button > Select 'All adjacent surfaces'.



Applying material to the Geometry Brushes

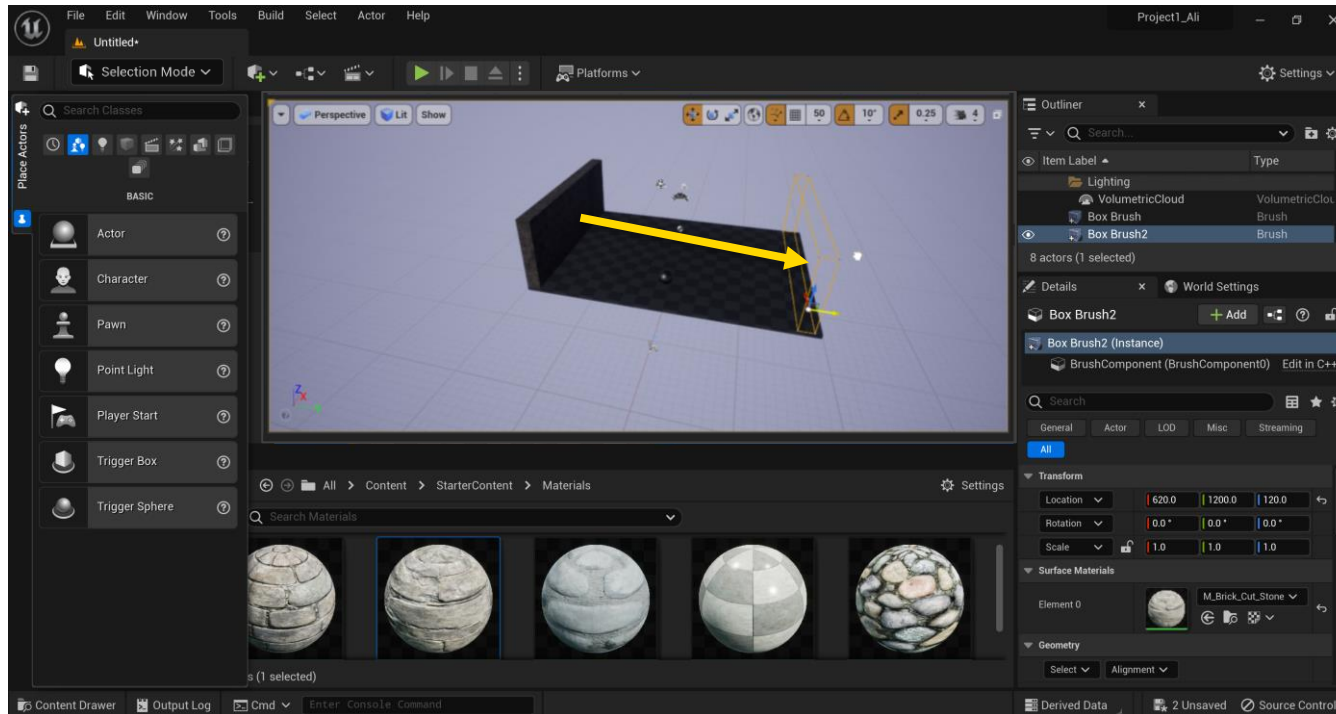
Drag and drop a single material onto the surface materials to combine all into one material.



Creating the other walls of the maze

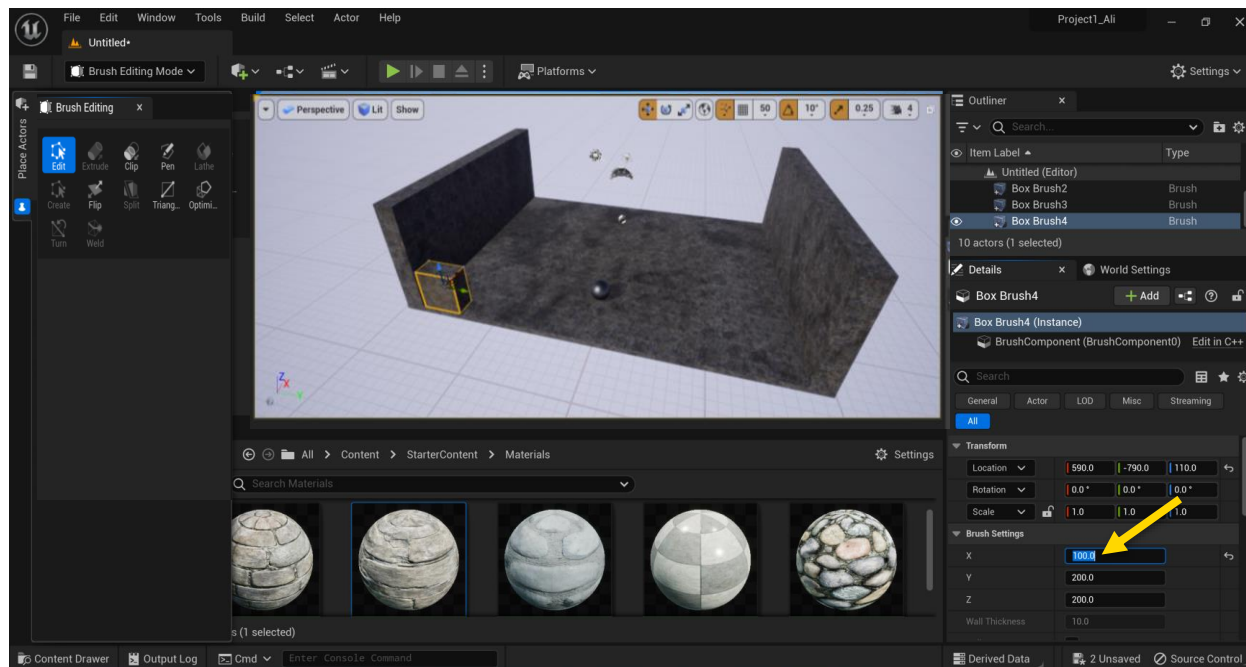
In order to duplicate the wall:

- Change the mode from Brush Editing to Select
- Copy the existing wall by: holding the **alt** key and using the translate tool and dragging the X axis.



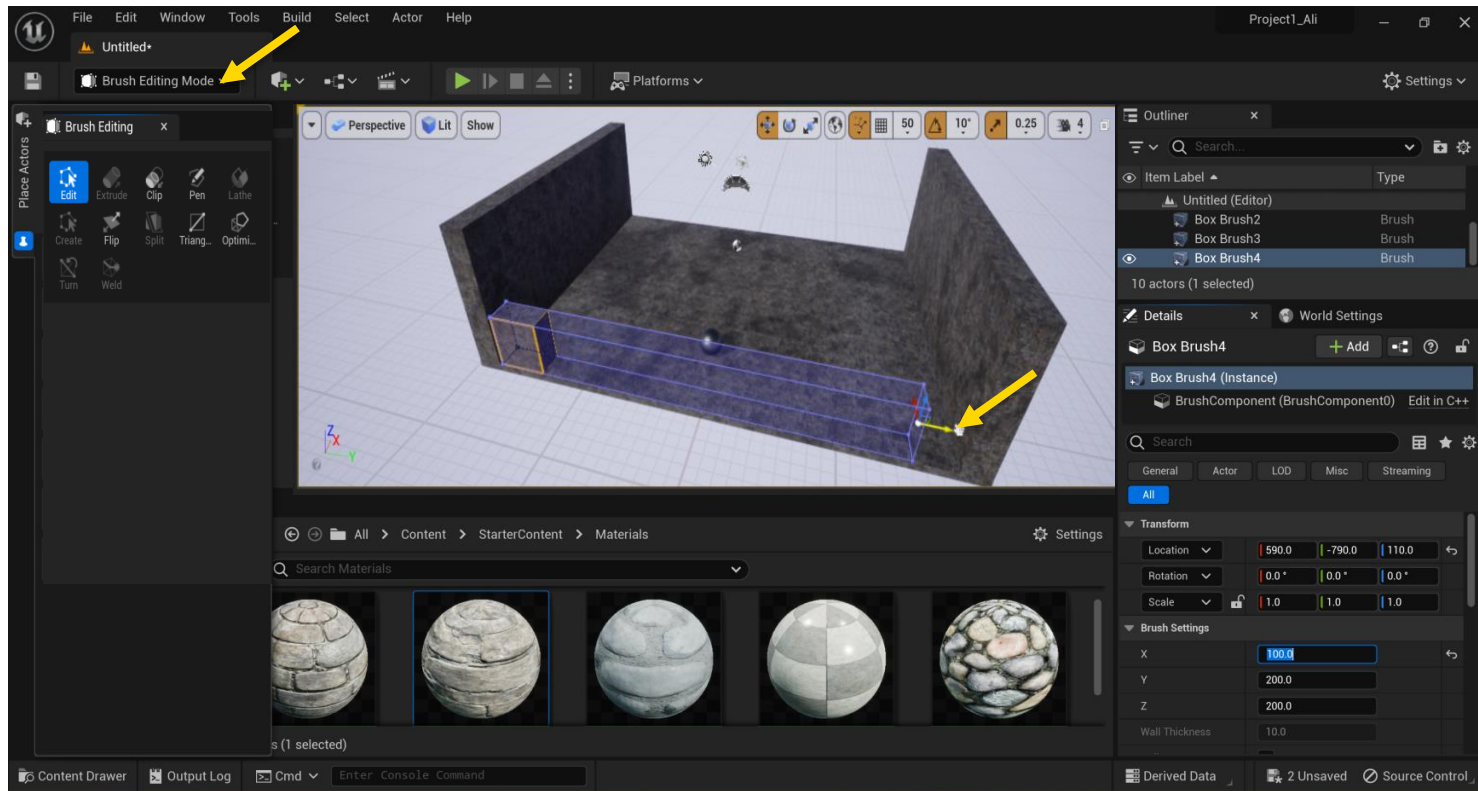
Creating the other walls of the maze

- In order to add the side walls, drag out another box brush.
- Details panel > Brush settings > Type 100 on the X > Type 200 on the Y > Type 200 on the Z.
- Put the wall on the corner of the floor.



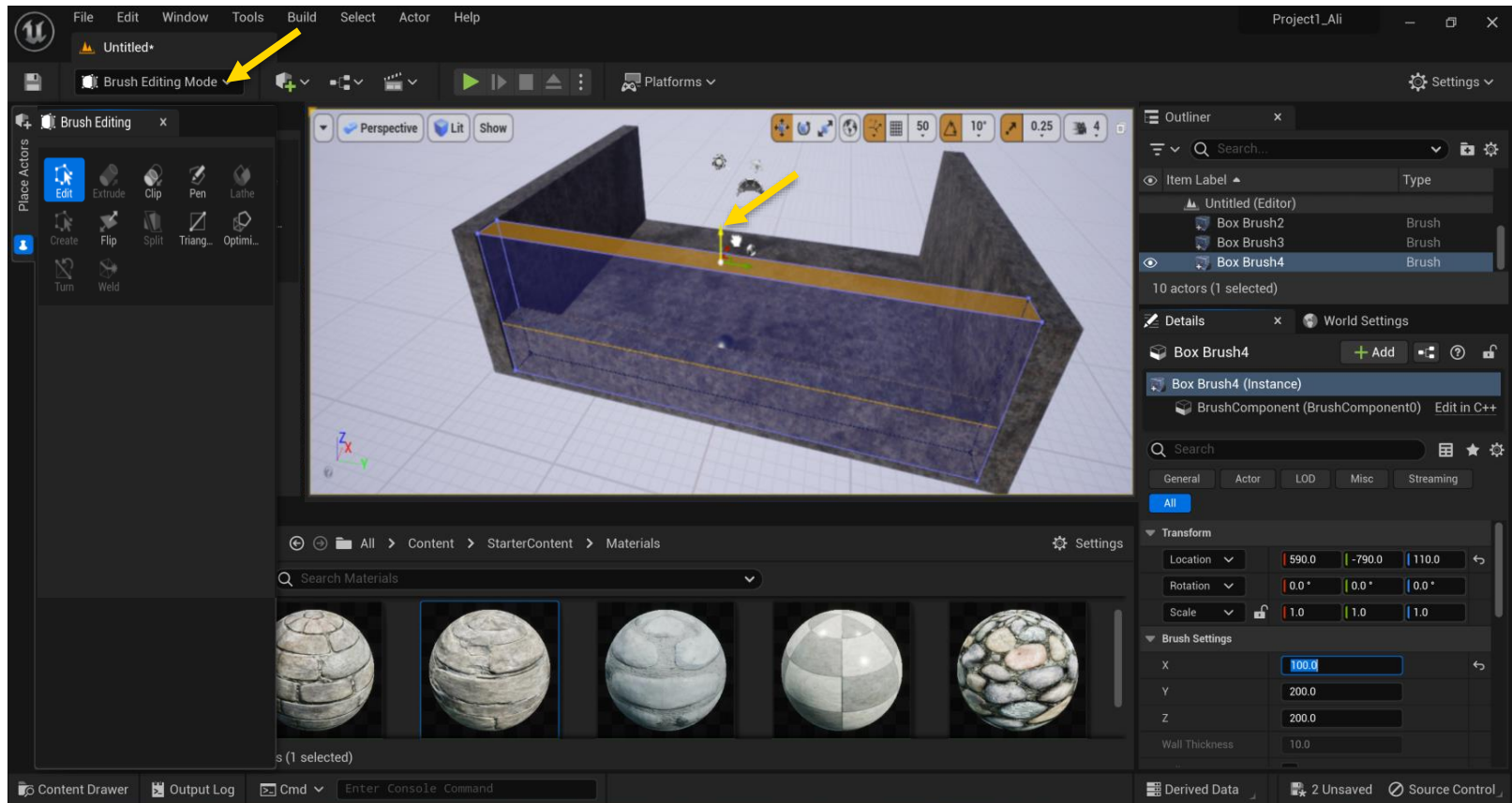
Creating the other walls of the maze

- Go to '**Brush Editing**' mode.
- Select the side face of the box brush and use the translate tool to change the length.



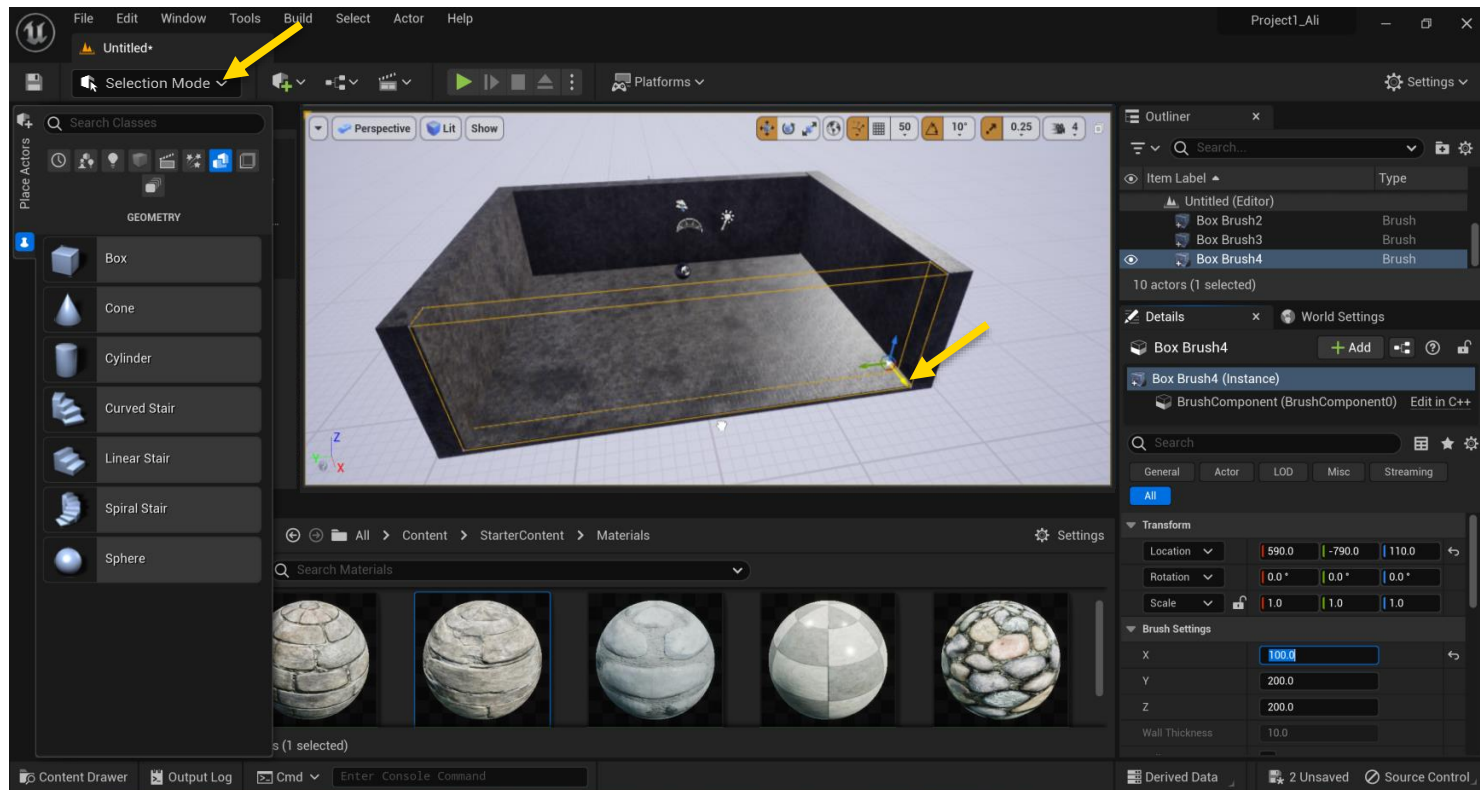
Creating the other walls of the maze

Set the wall height by dragging the Y axis.

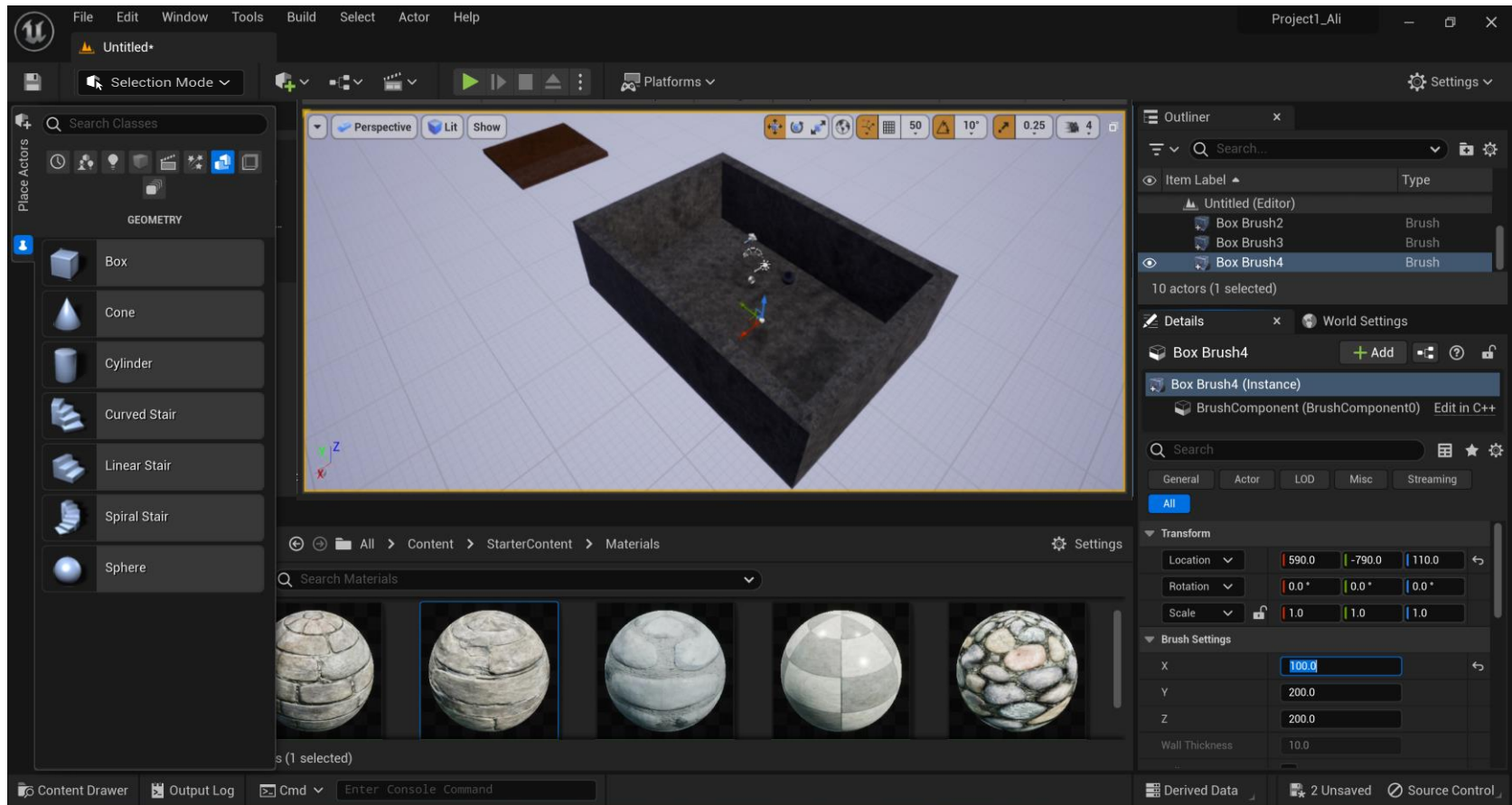


Creating the other walls of the maze

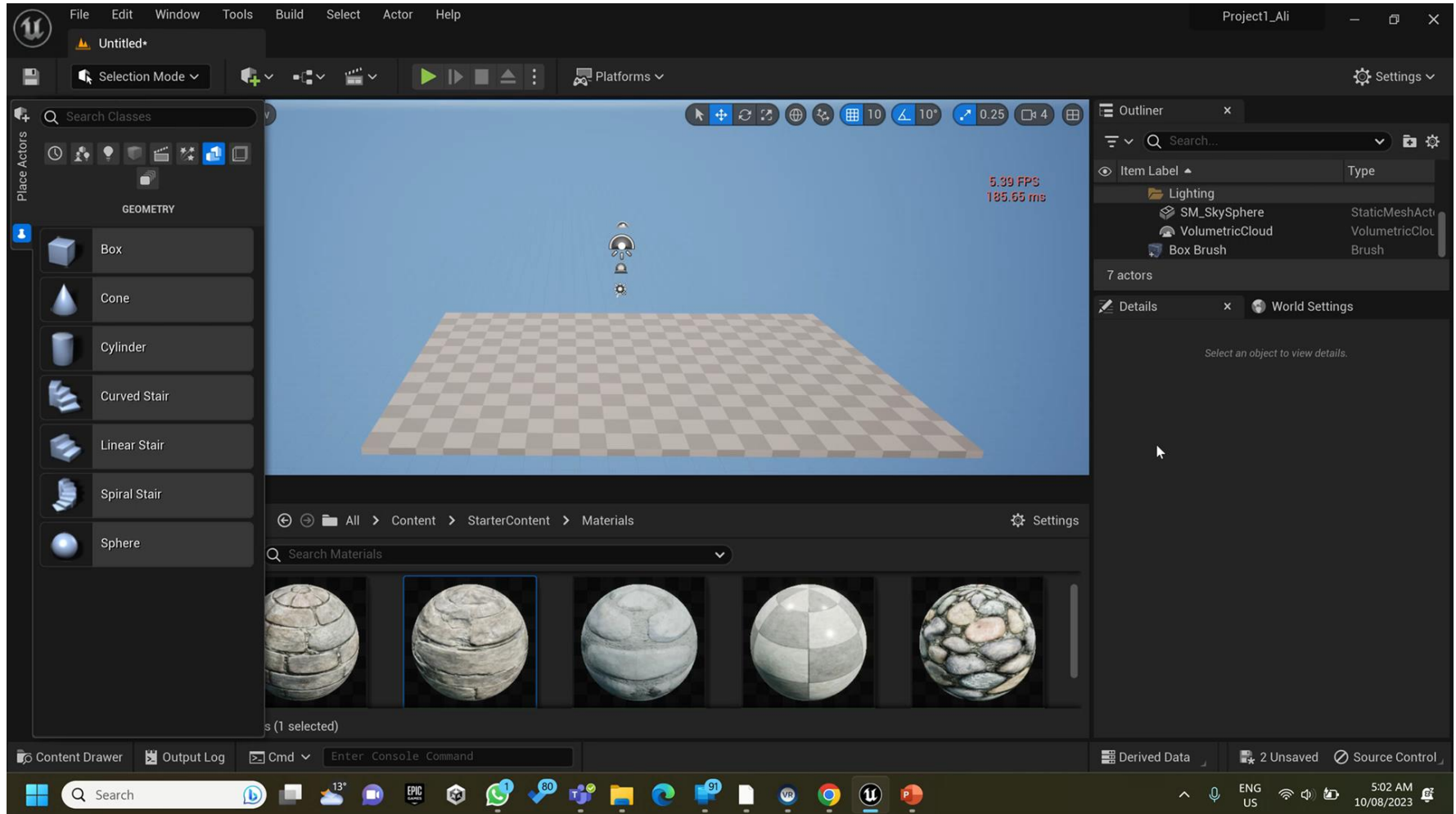
- Change the mode to the **Selection** mode.
- Copy the existing wall by holding the **alt** key and using the translate tool and dragging the X axis.



Creating the other walls of the maze



Creating the other walls of the maze



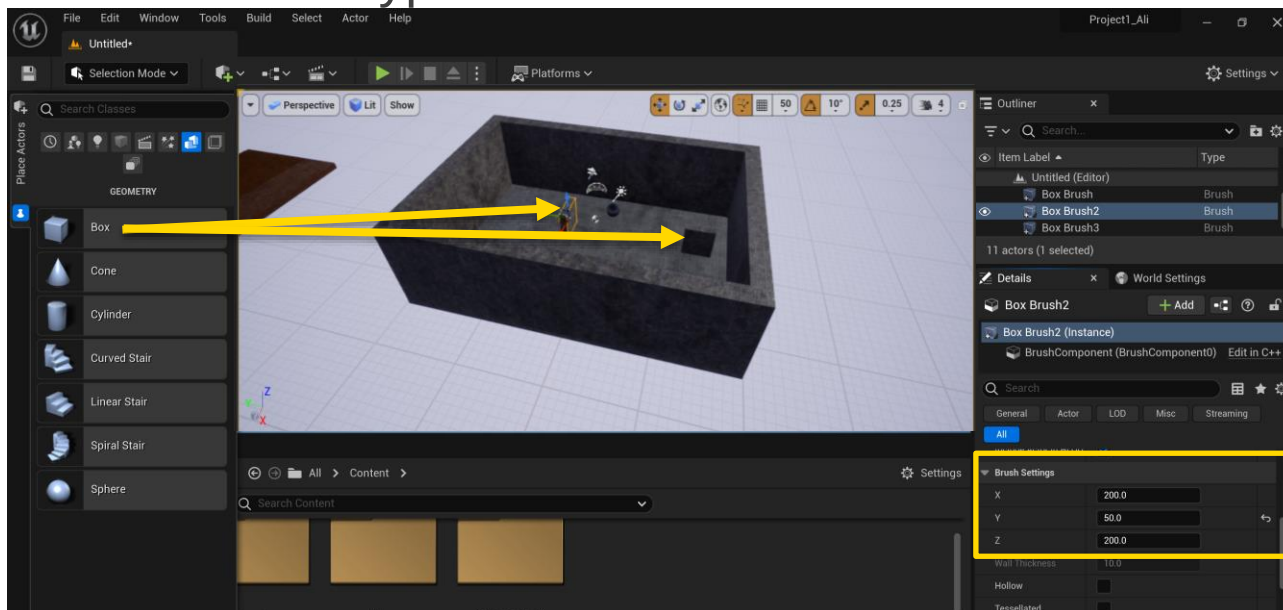
Lecture activity 1

- Create the maze floor and the walls Using geometry brushes.
- Change the material of the floor and the walls.



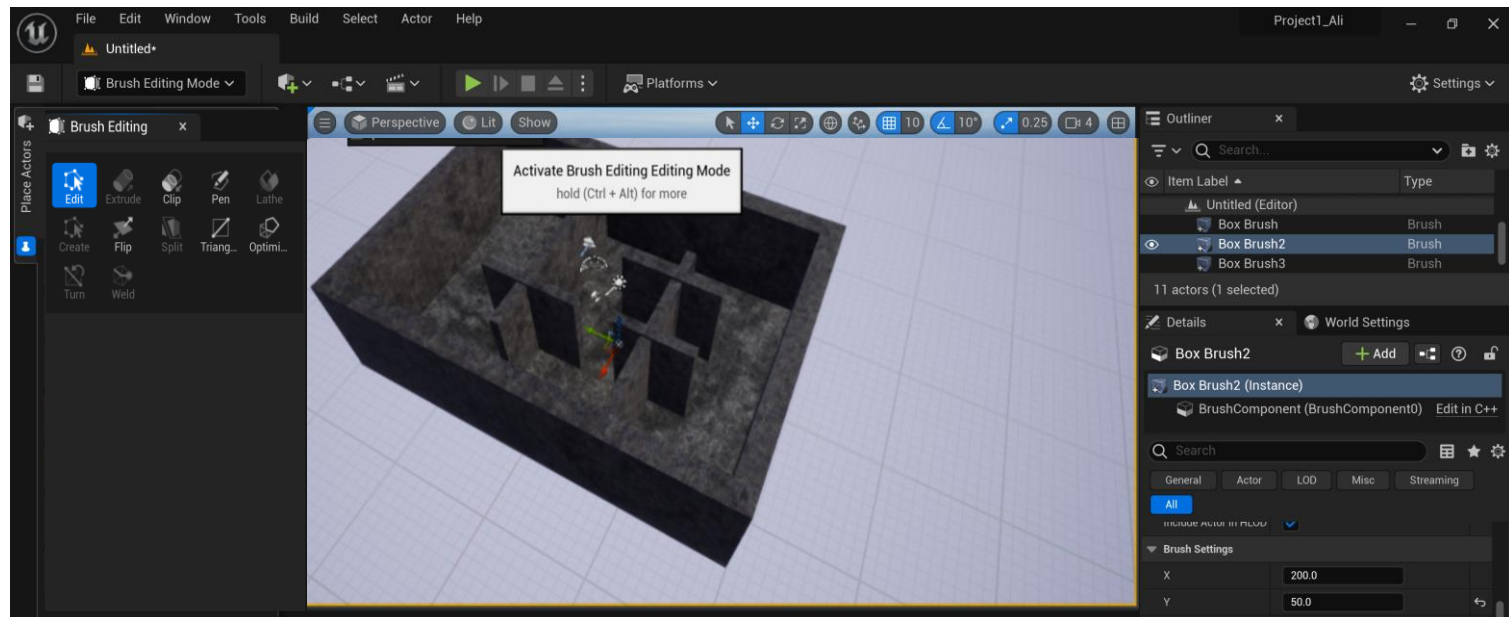
Creating the interior walls of the maze

- Drag out two box brushes
- The first box > Details panel > Brush settings > Type 200 on the X > Type 50 on the Y > Type 200 on the Z.
- The second box > Details panel > Brush settings > Type 50 on the X > Type 200 on the Y > Type 200 on the Z.

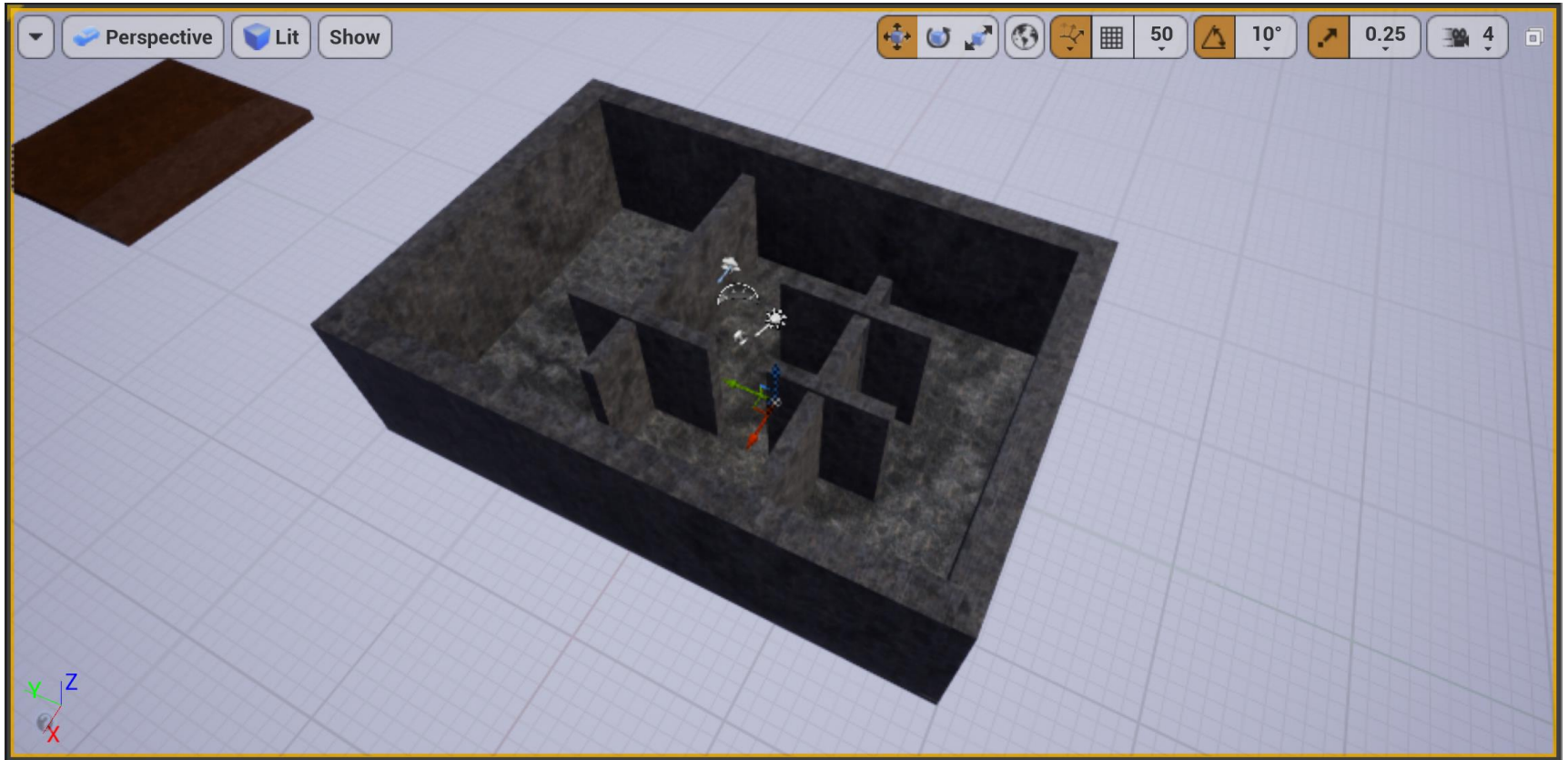


Creating the interior walls of the maze

- Modes > Brush editing
- Adjust the size of the walls.
- Copy the created walls and resize them for different parts of the maze.

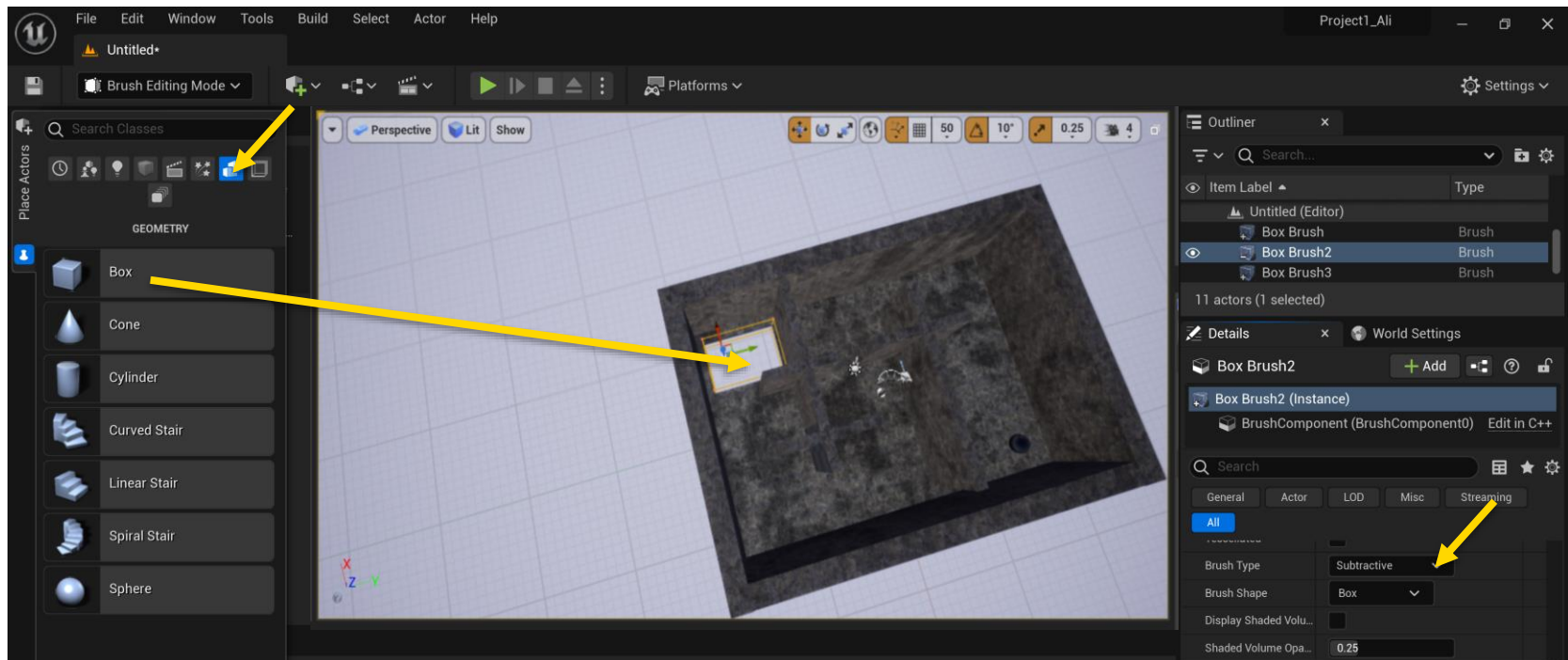


Creating the interior walls of the maze

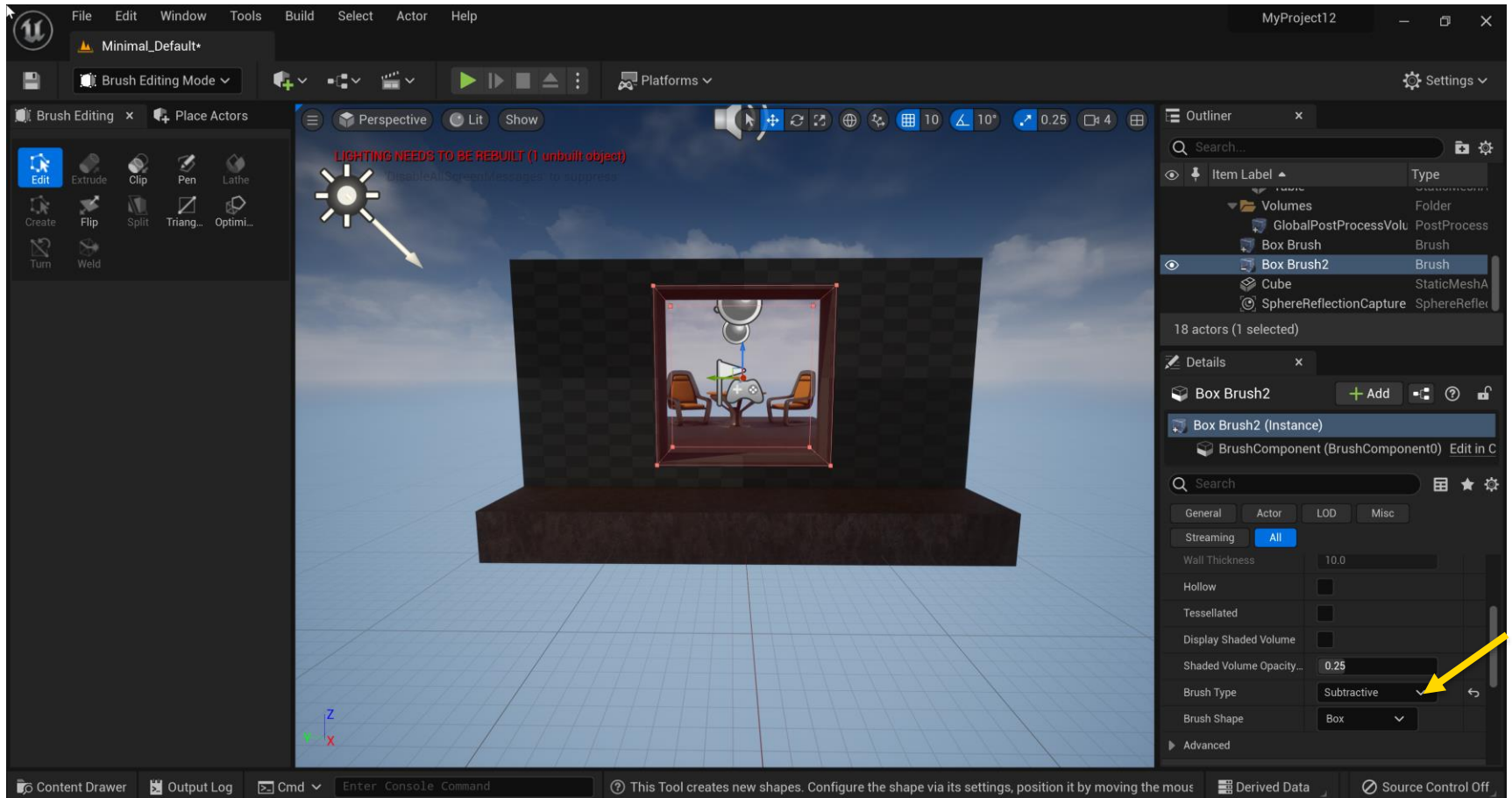


End point for the maze

In order to create a hole in the floor > Add a box brush at the end of the maze > Details panel > Change the Brush Type to **Subtractive** > Resize it to create the hole.

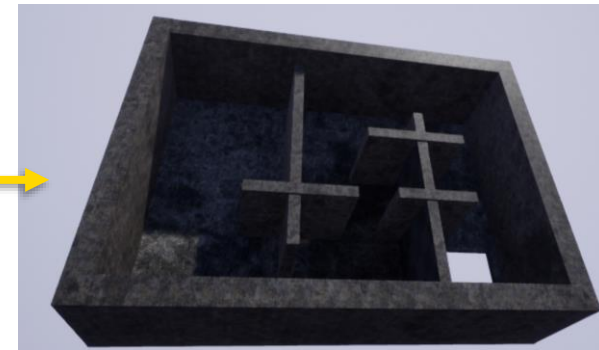
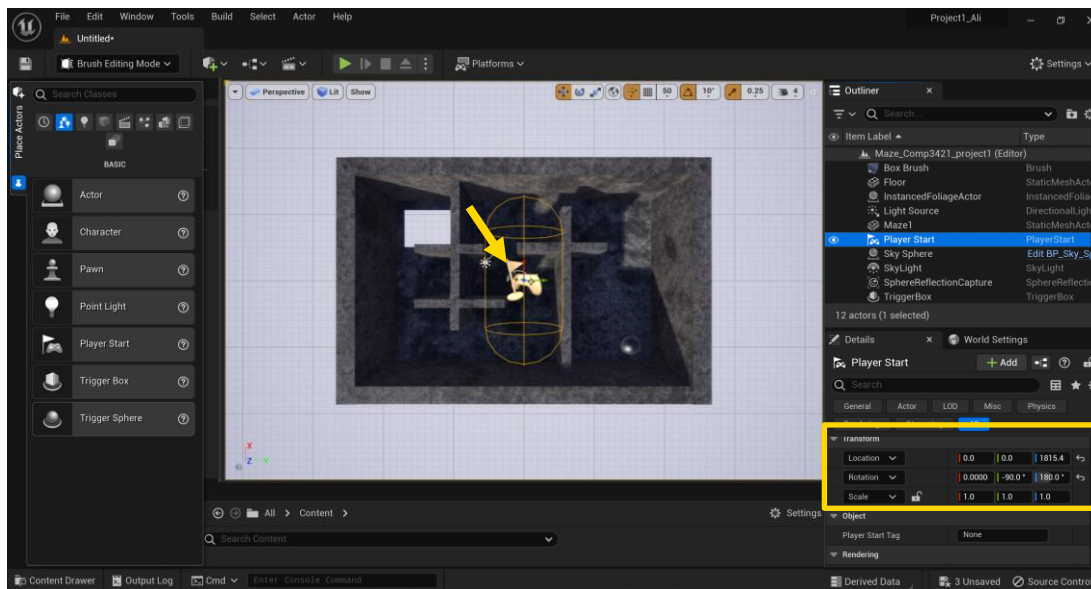


Subtract brush usage



Ball location and the starting point

In order to get a complete view of the maze > put the starting point on the middle and top of the maze
> Set the rotation Y value to -90 and the rotation Z value to 180.

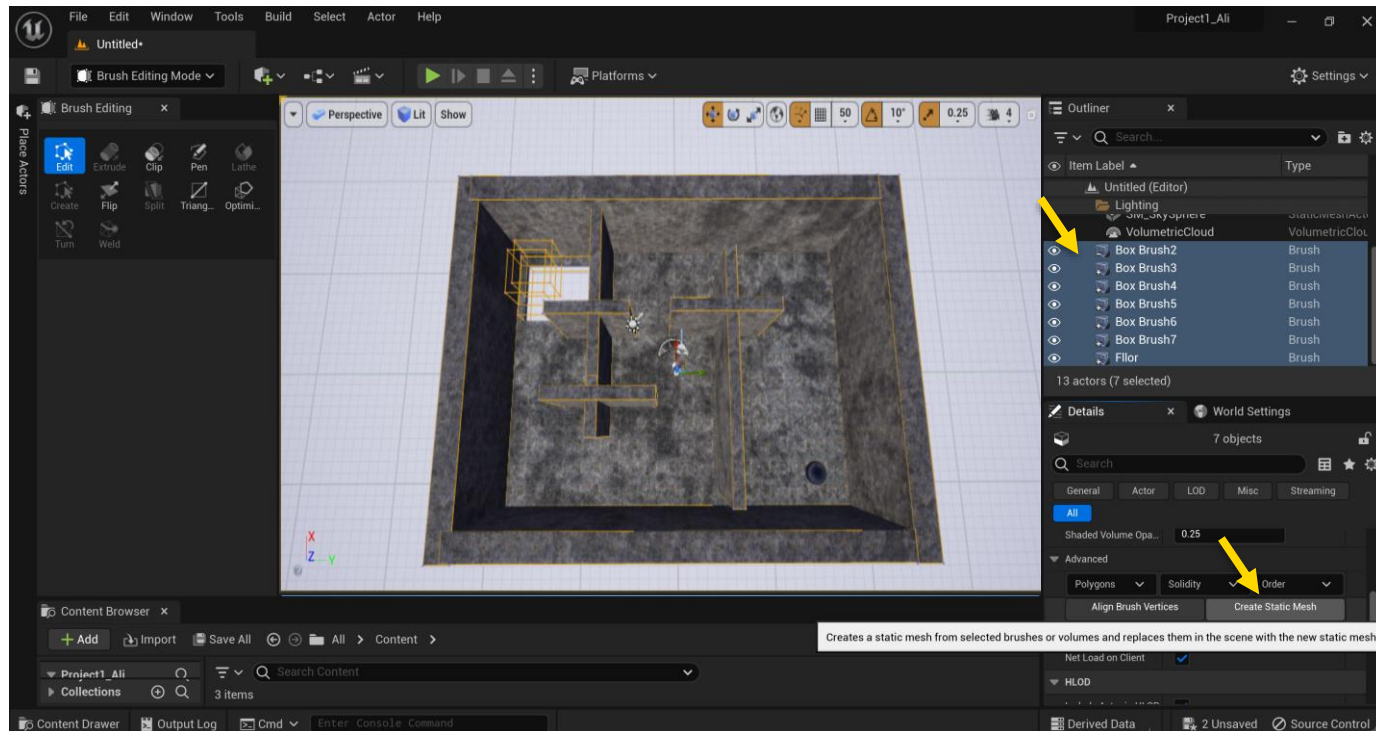


Converting geometry brush-based objects to static mesh-based objects

- The geometry brushes cannot be used in blueprints.
- We should convert the objects that are created using geometry brushes to static meshes.

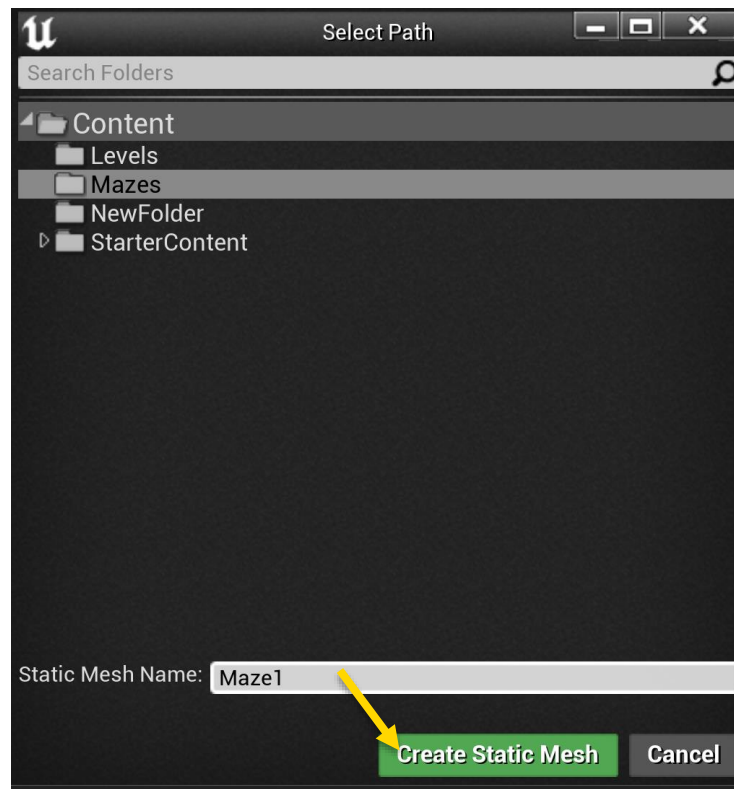
Converting geometry brush-based objects to static mesh-based objects

Select all the box brushes from the **Outliner** > Details panel
> Expand the brush settings > Click the **Create Static Mesh** button.



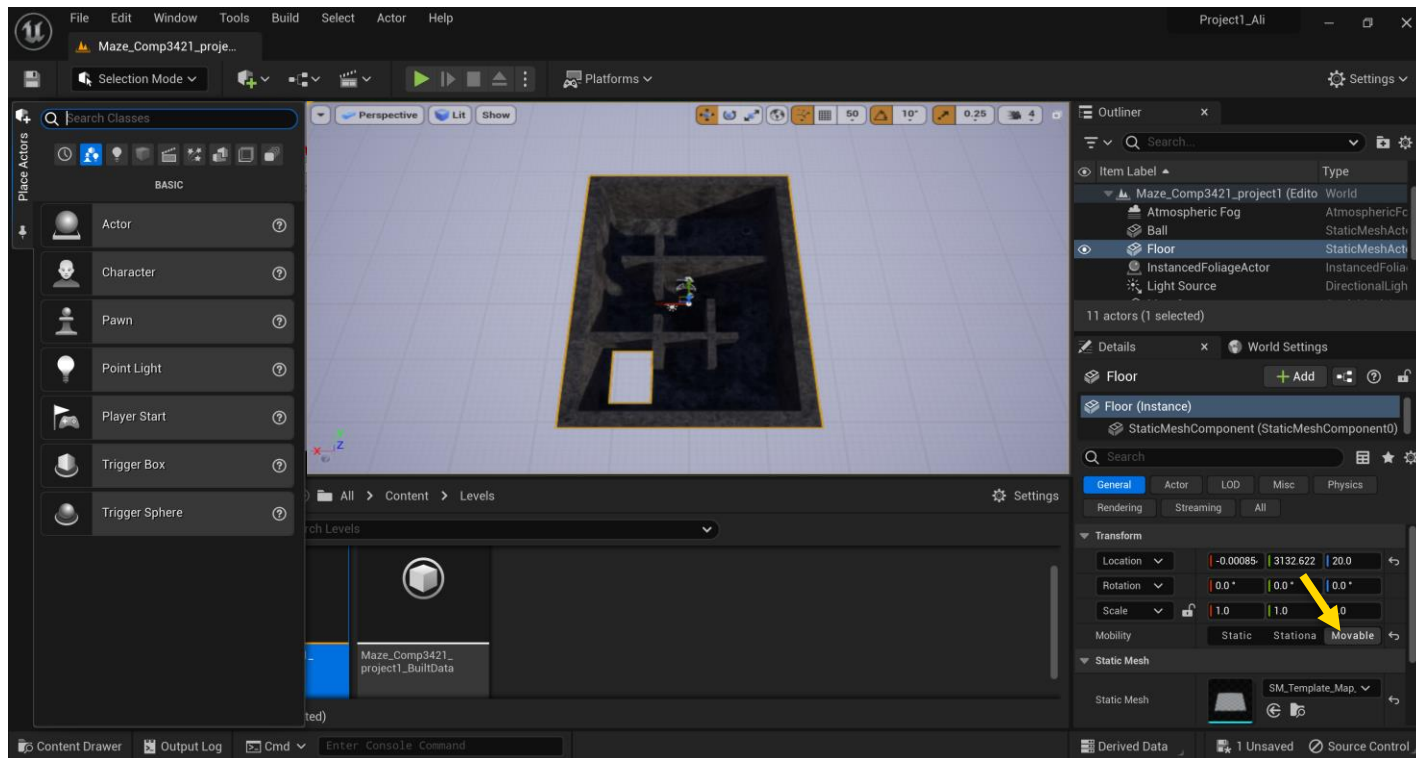
Converting geometry brush-based objects to static mesh-based objects

Save the static mesh as a new object called: 'Maze1'.



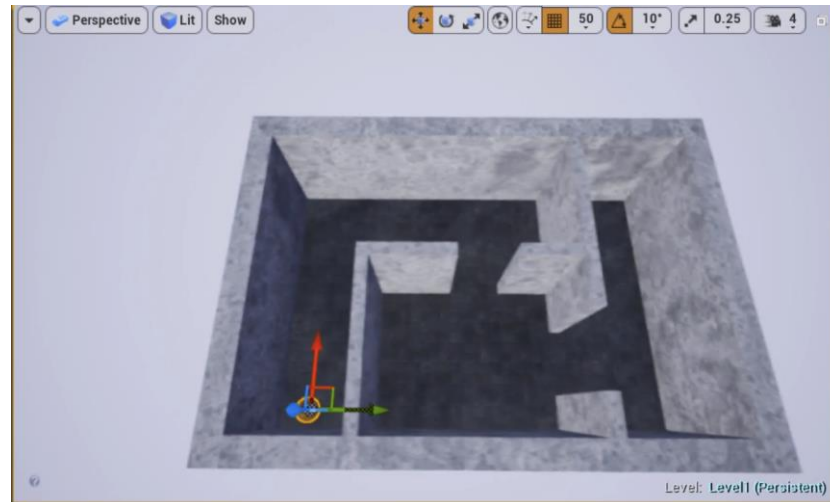
Make the maze movable with mouse input

Details panel > Transform section > Change the mobility from Static to Movable.



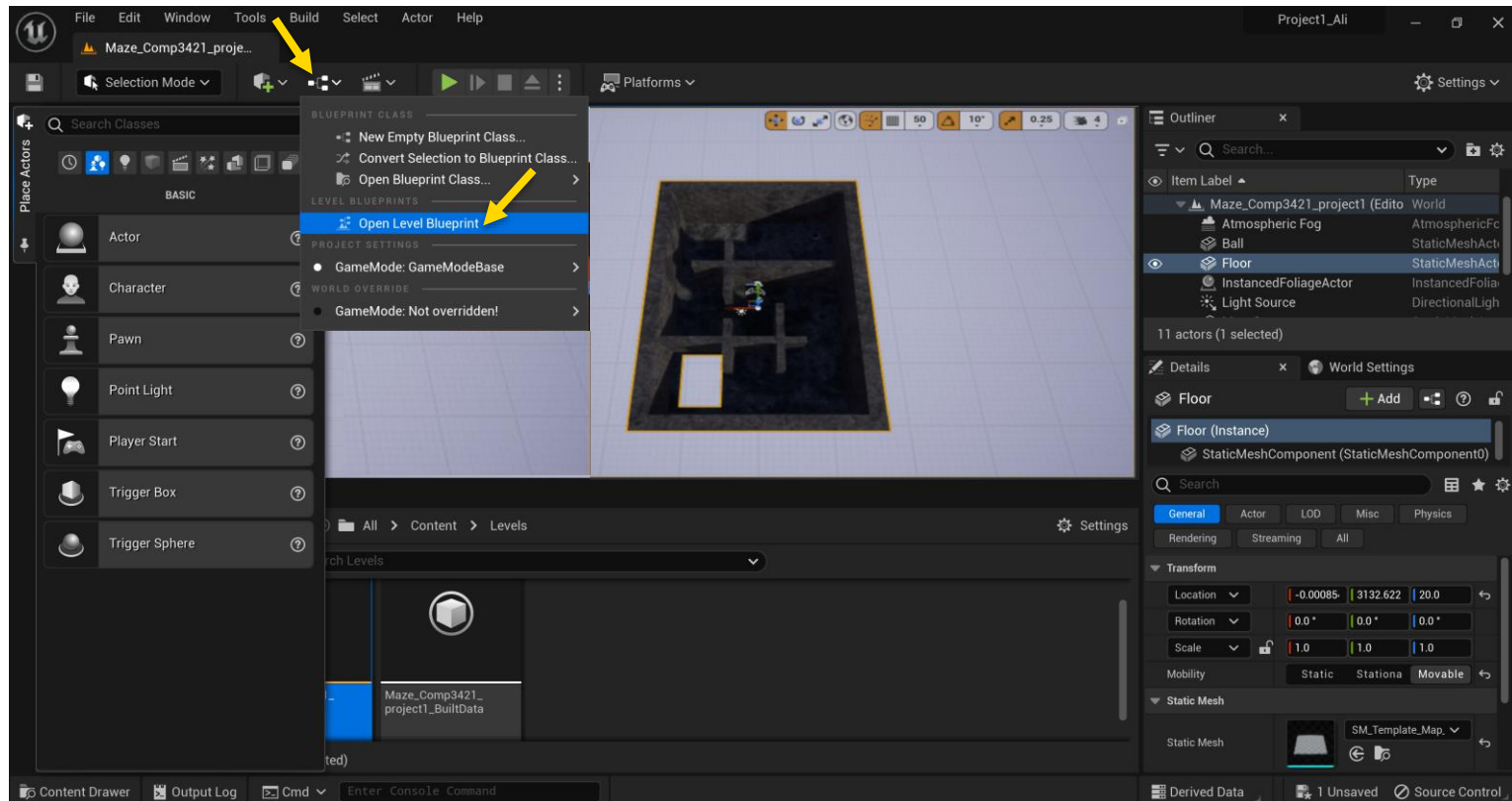
Lecture activity 2

- Create the maze interior walls using geometry brushes.
- Set a starting point location for your maze to get a complete view.
- Create your maze escape hole.
- Convert the objects that you created using geometry brushes to a single static mesh.



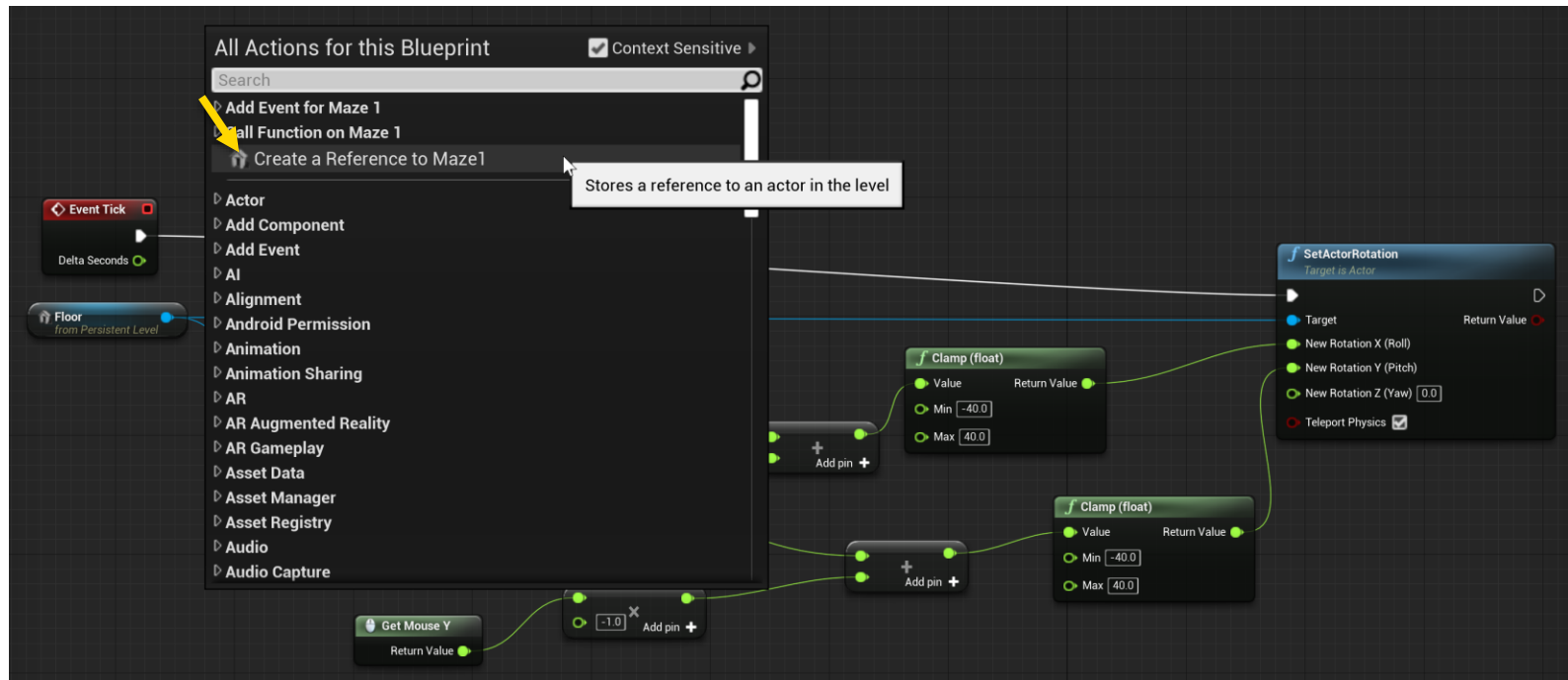
Make the maze movable with mouse input

- Go to the level blueprints



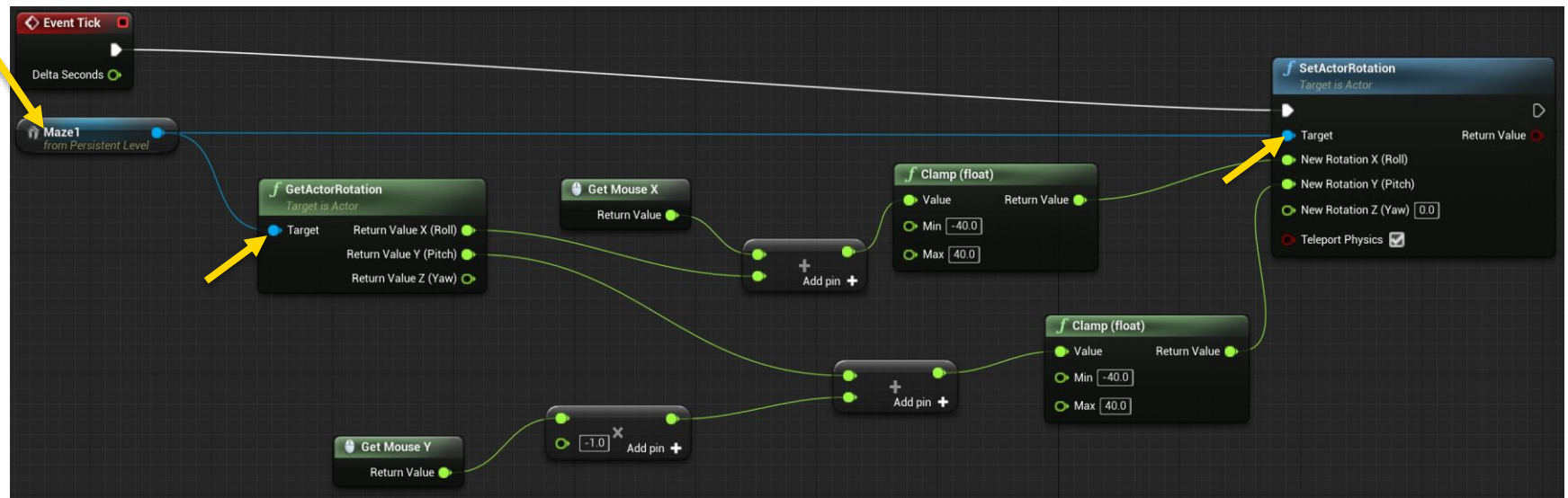
Make the maze movable with mouse input

- Click create a reference to maze 1 (the static mesh name)



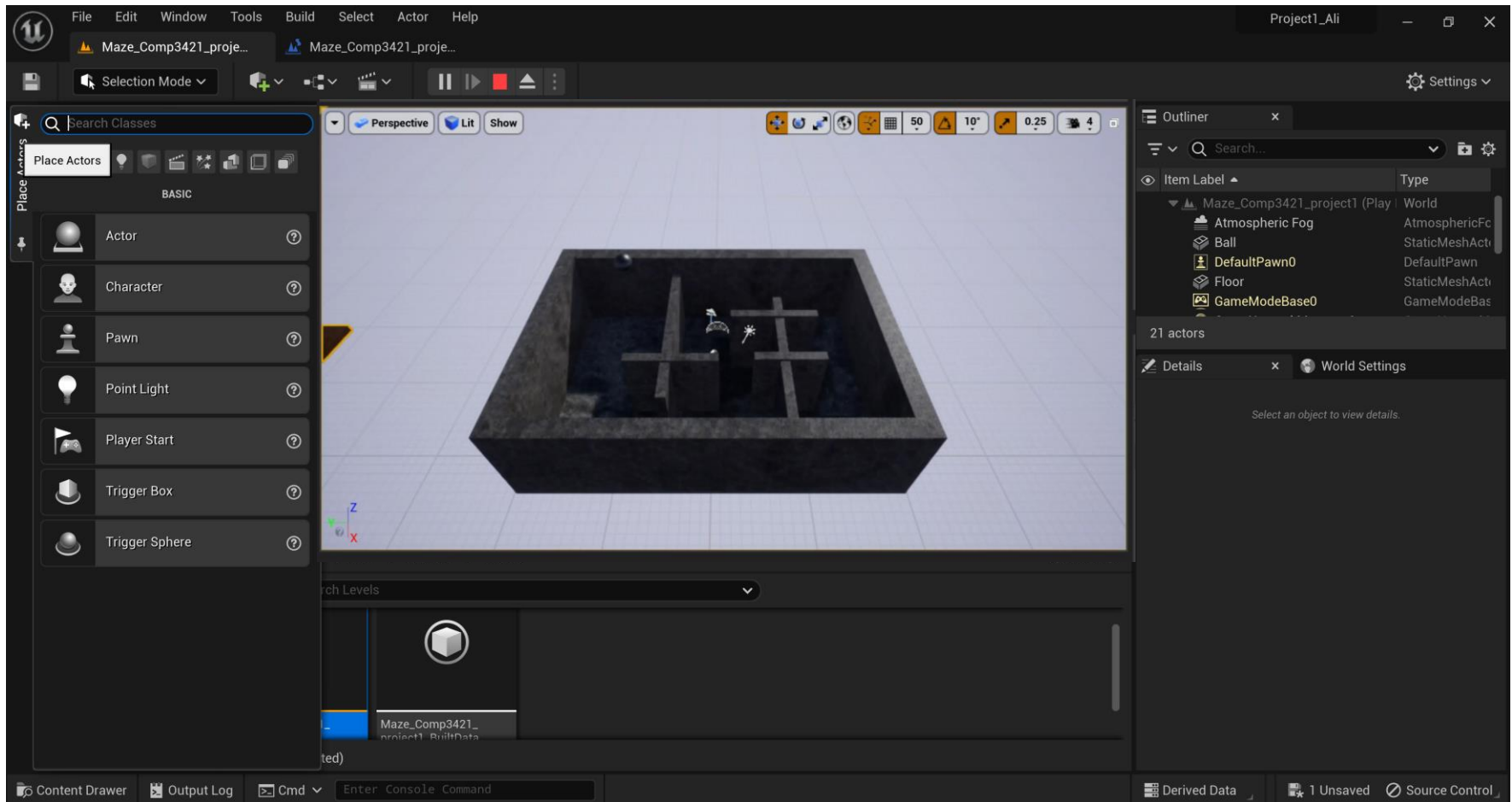
Make the maze movable with mouse input

- Replace the floor references with the maze reference.



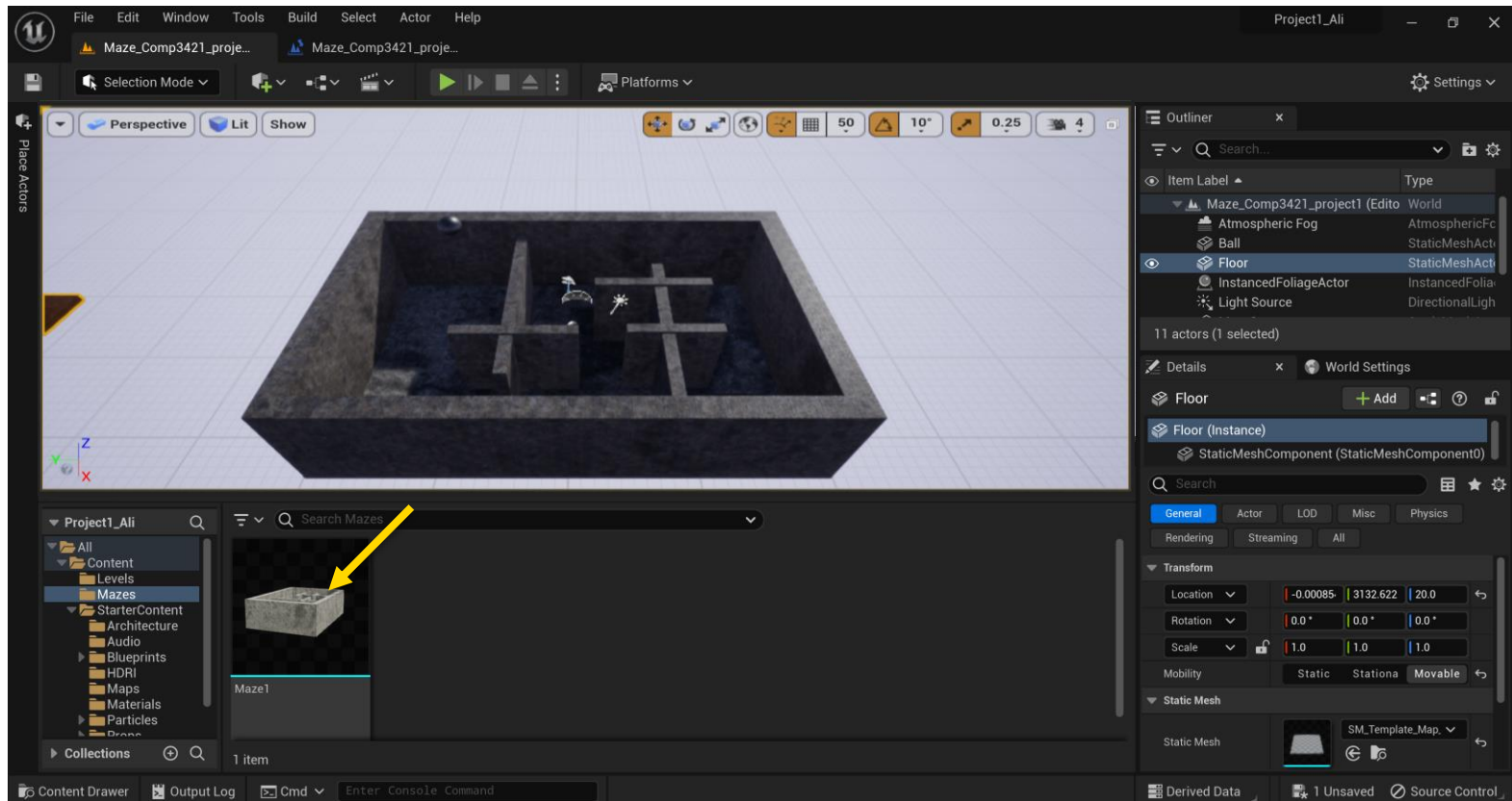
Static mesh collision

There is a problem: the ball passes through the maze floor.



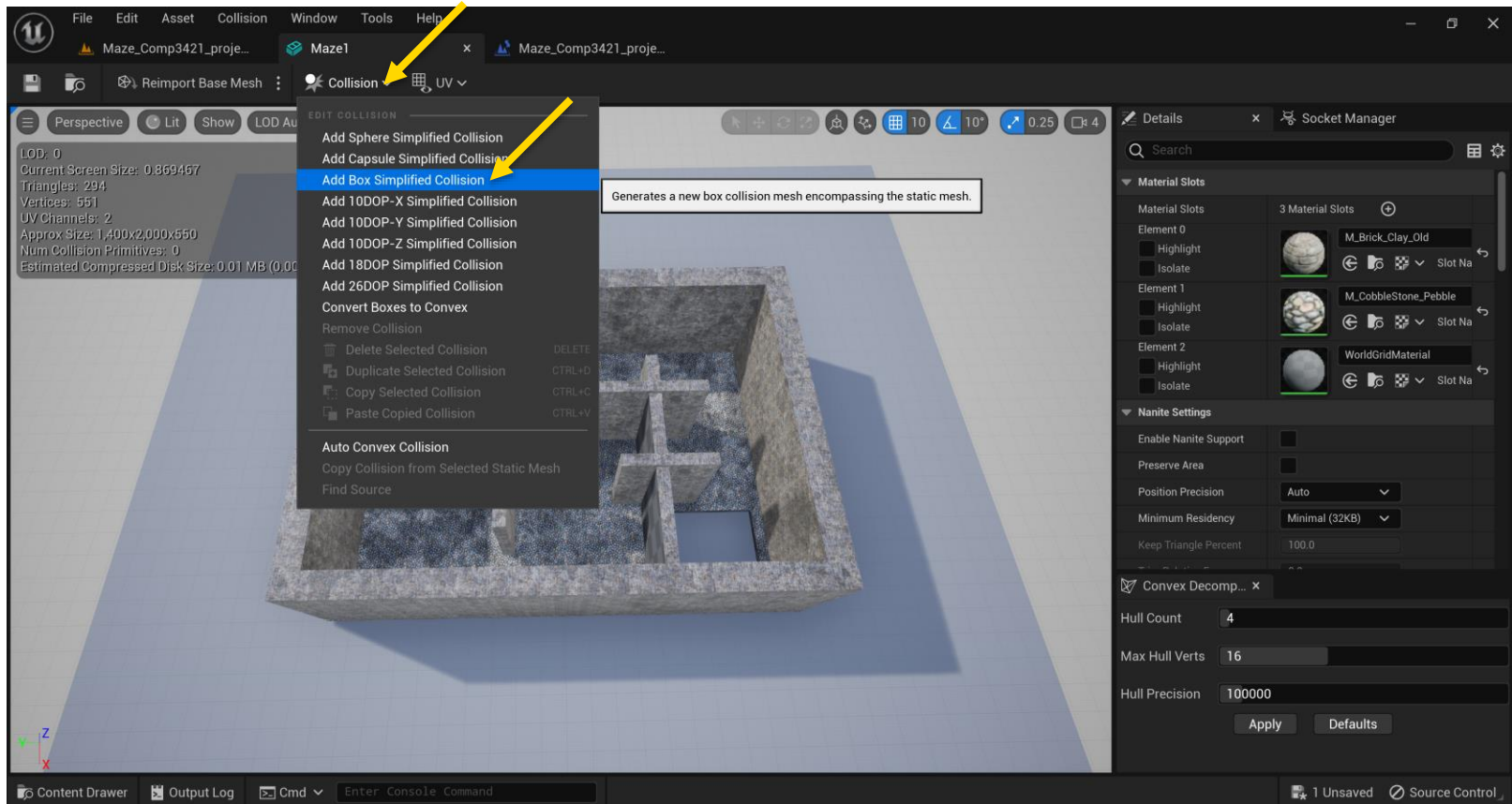
Static mesh collision

We should add collision detector to our static mesh >
Content browser > Double click the maze > Open the static
mesh editor.



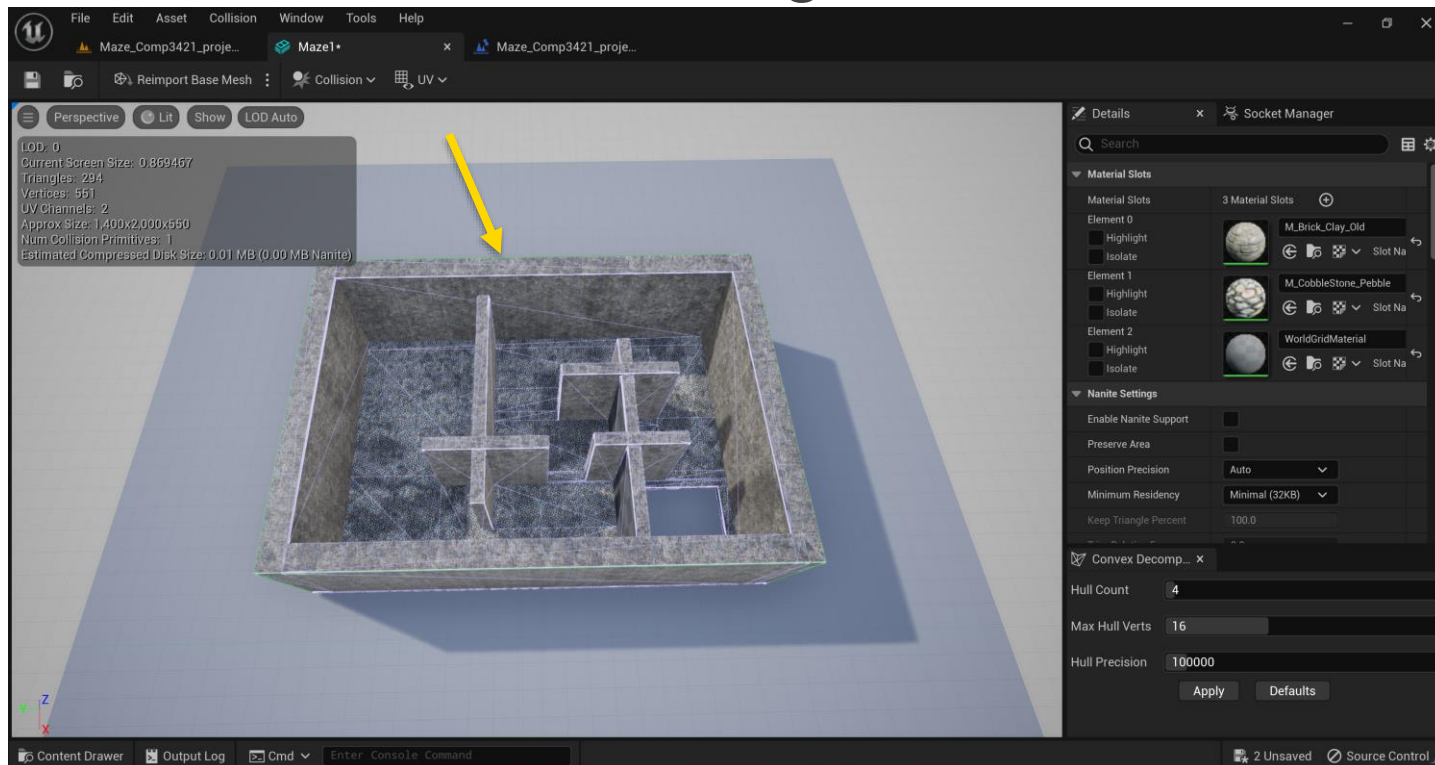
Static mesh collision

Solution 1: Collision menu > Add box simplified collision.



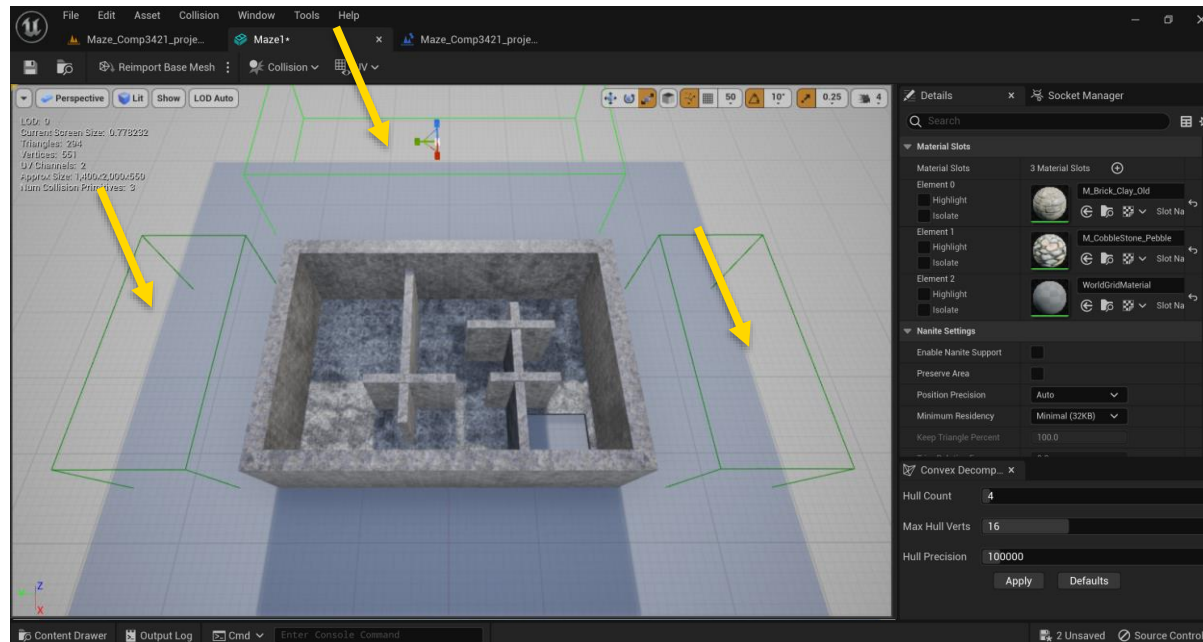
Static mesh collision

‘Add box simplified collision’ option adds a collision box (the green line) that covers the borders of the whole mesh without covering the interior walls.



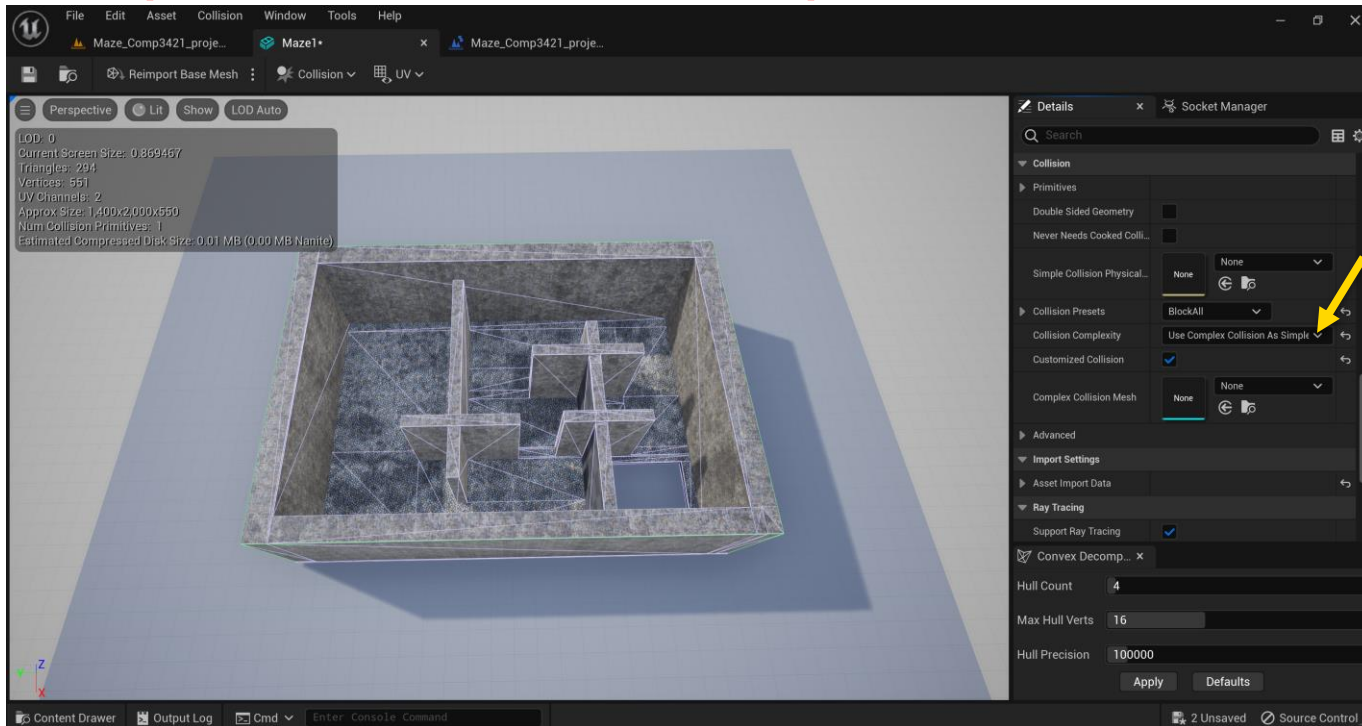
Static mesh collision

- Scale the collision primitive and make it fit the size of the walls,
- Add more box collisions and scale them to the size of all the interior walls.



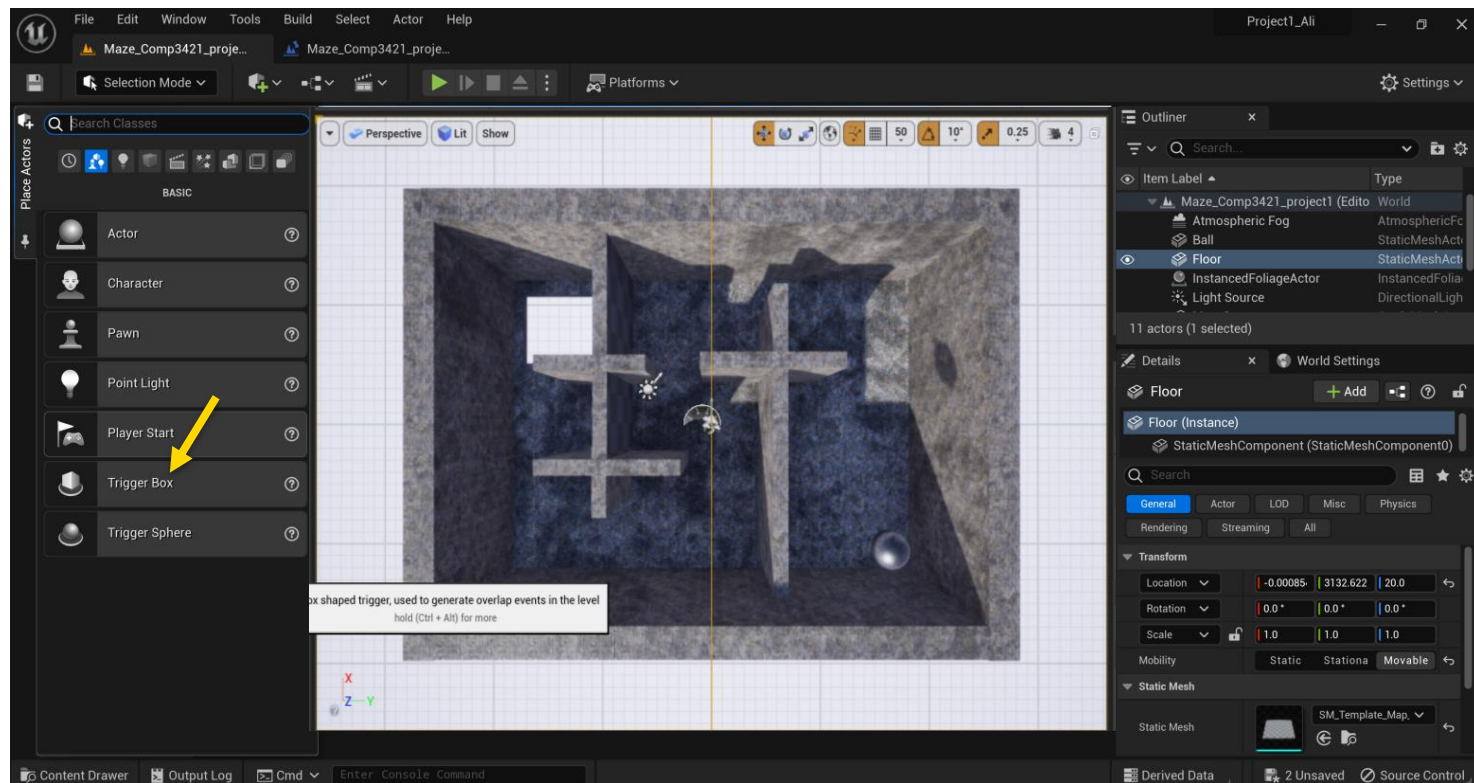
Static mesh collision

Solution 2: Details panel > Go to the collision section > From the **Collision Complexity** > Select use '**Complex collision as simple**'.



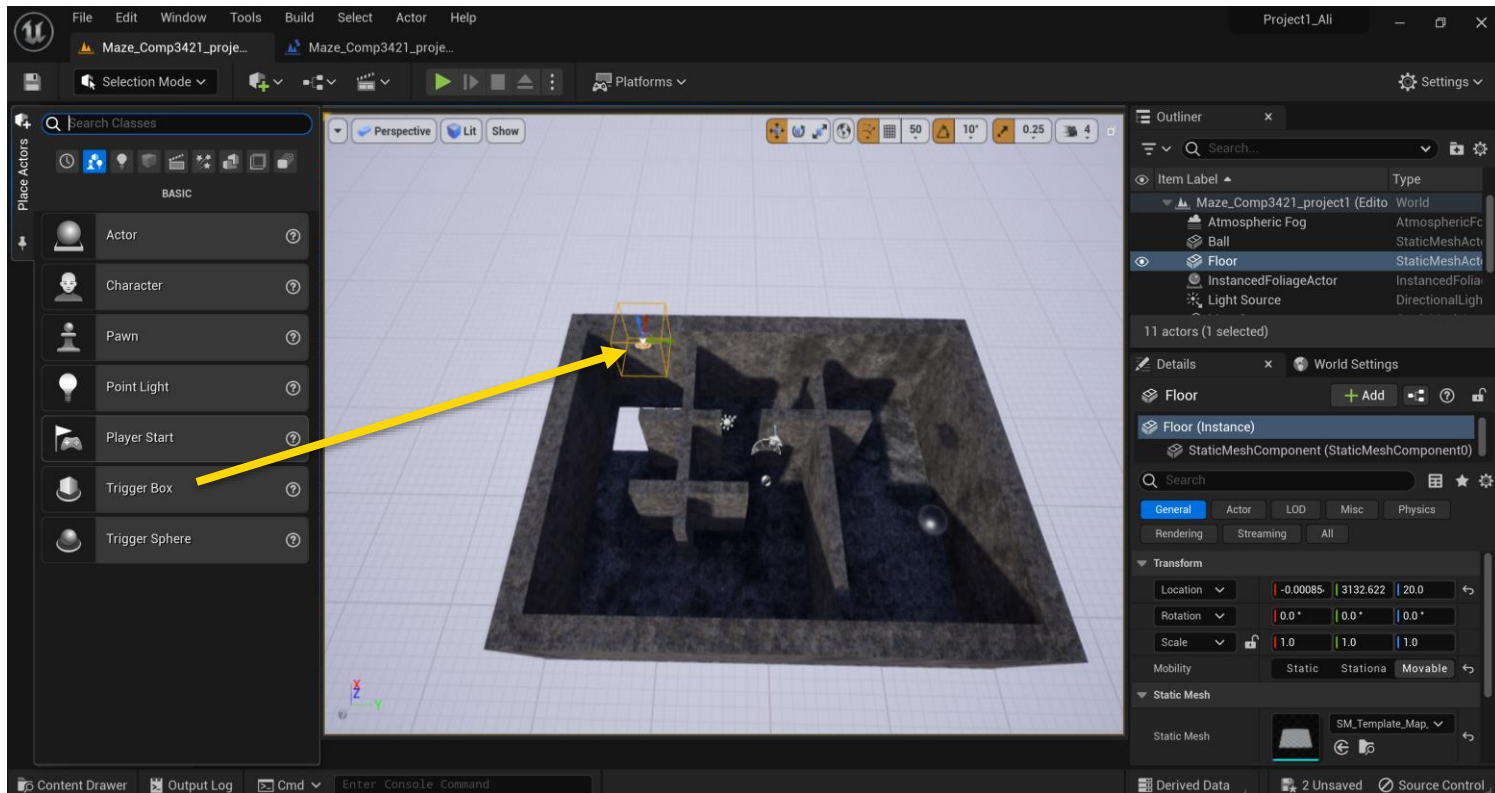
Create a win condition for our levels

- Go to the 'Place actors' panel > Add a 'Trigger Box' component.
- Trigger Box component is used to add overlap events to fire a code based on a collision.



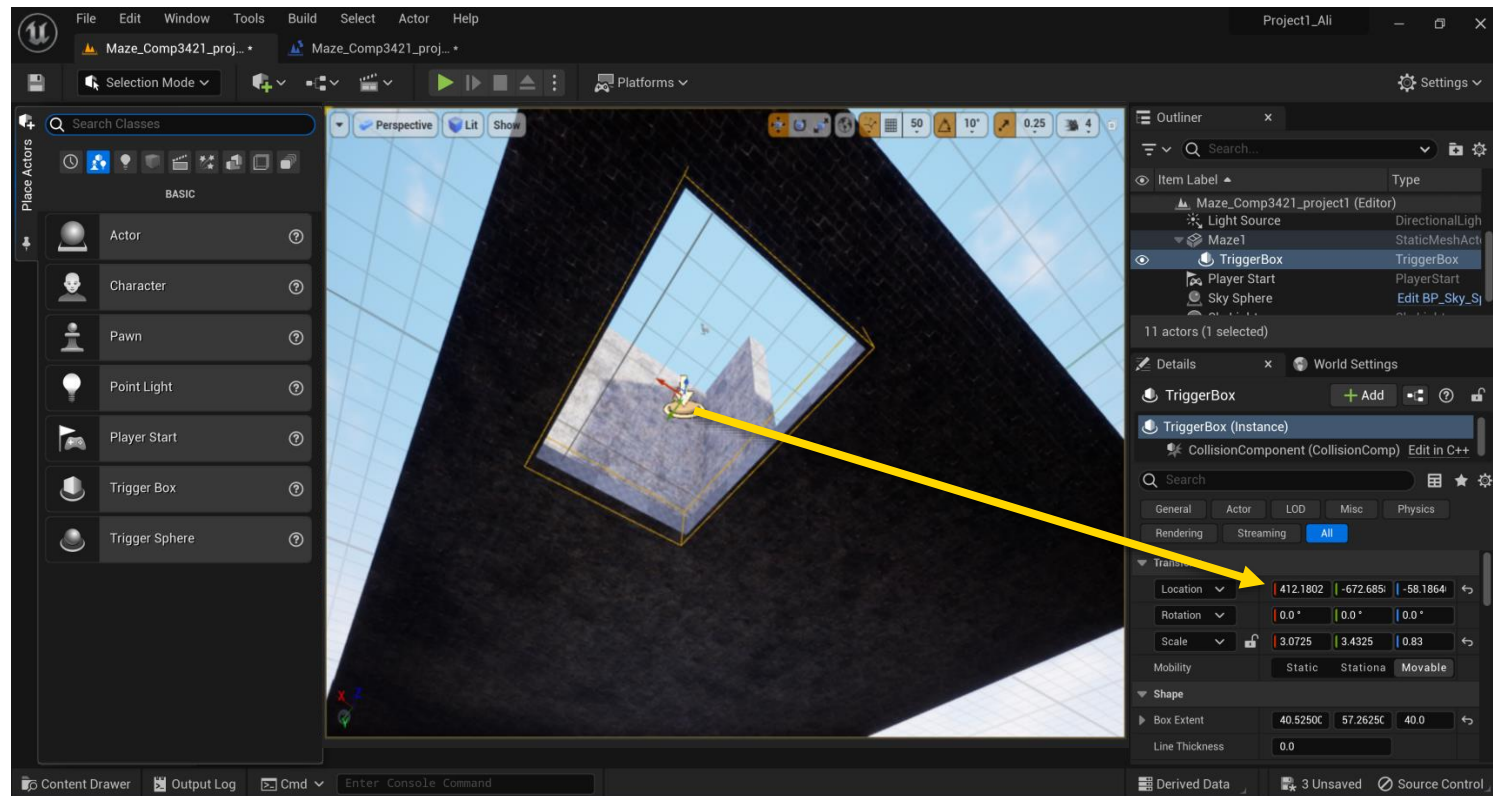
Create a win condition for our levels

- Click and drag a **Trigger Box** into your level.
- A **Trigger Box** is an invisible box.



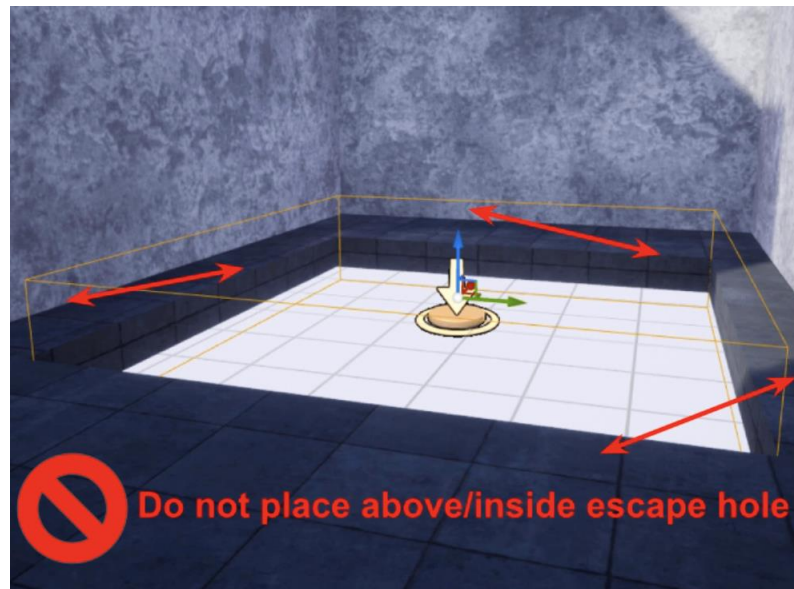
Create a win condition for our levels

- Use the Transform tools to resize it and place it inside the hole.



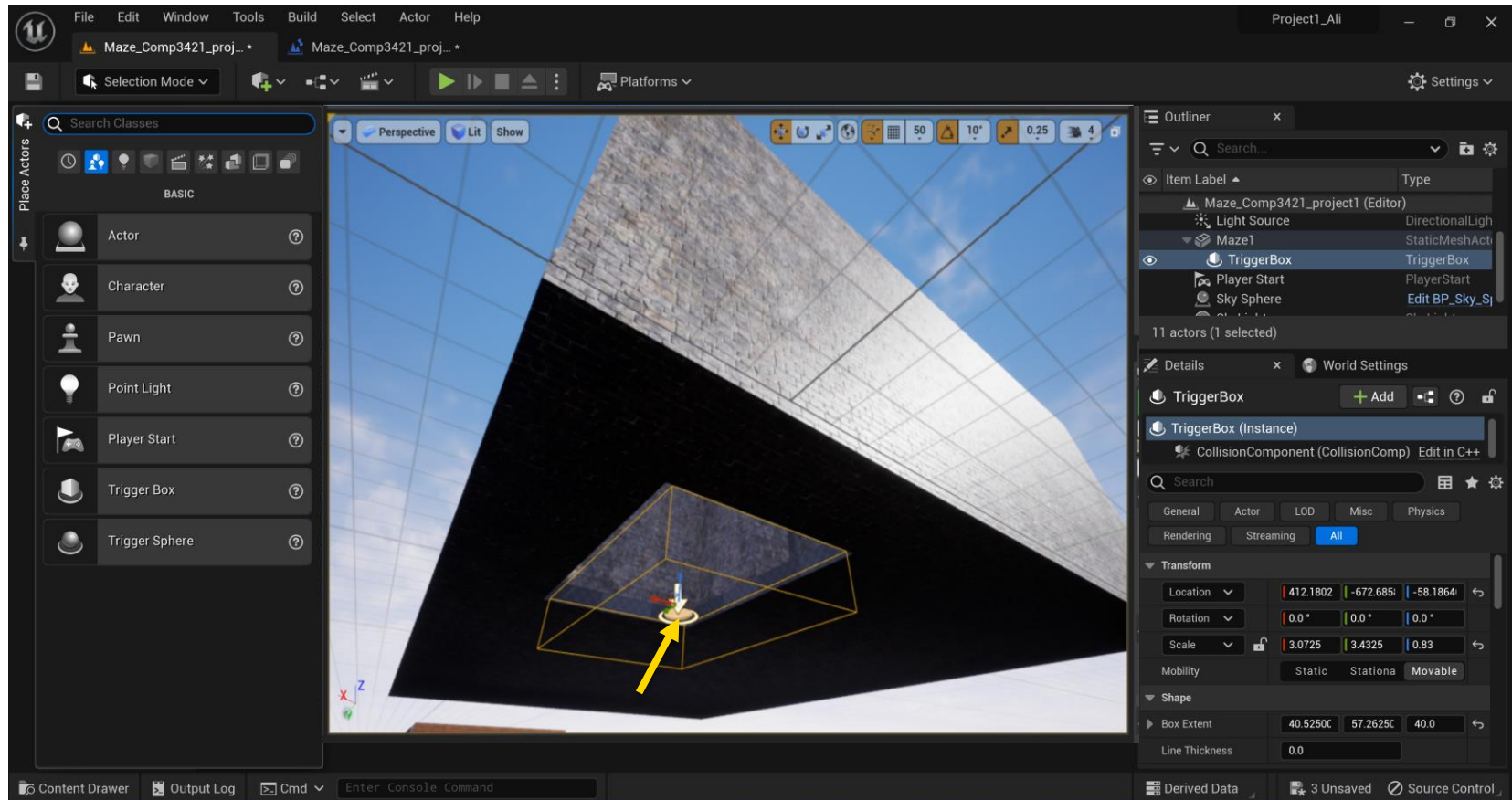
Create a win condition for our levels

- Placing the **Trigger Box** inside the escape hole is not a good solution.
- Because the ball might be moving fast and jump over the corners of the escape hole and collide with the trigger box while the ball is still inside the maze.



Create a win condition for our levels

- Place the trigger box underneath the escape hole.



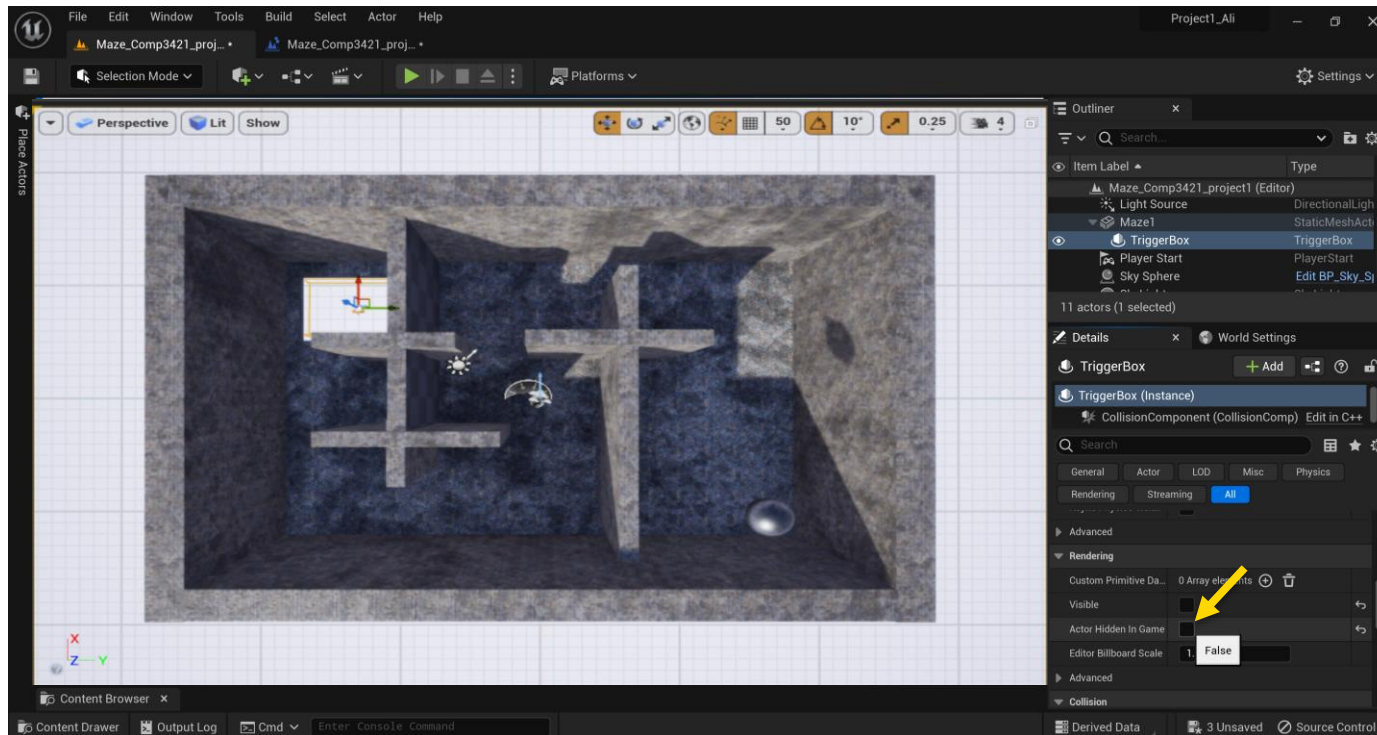
Lecture activity 3

- Add a trigger box for your maze.
- Put it underneath the maze hole.



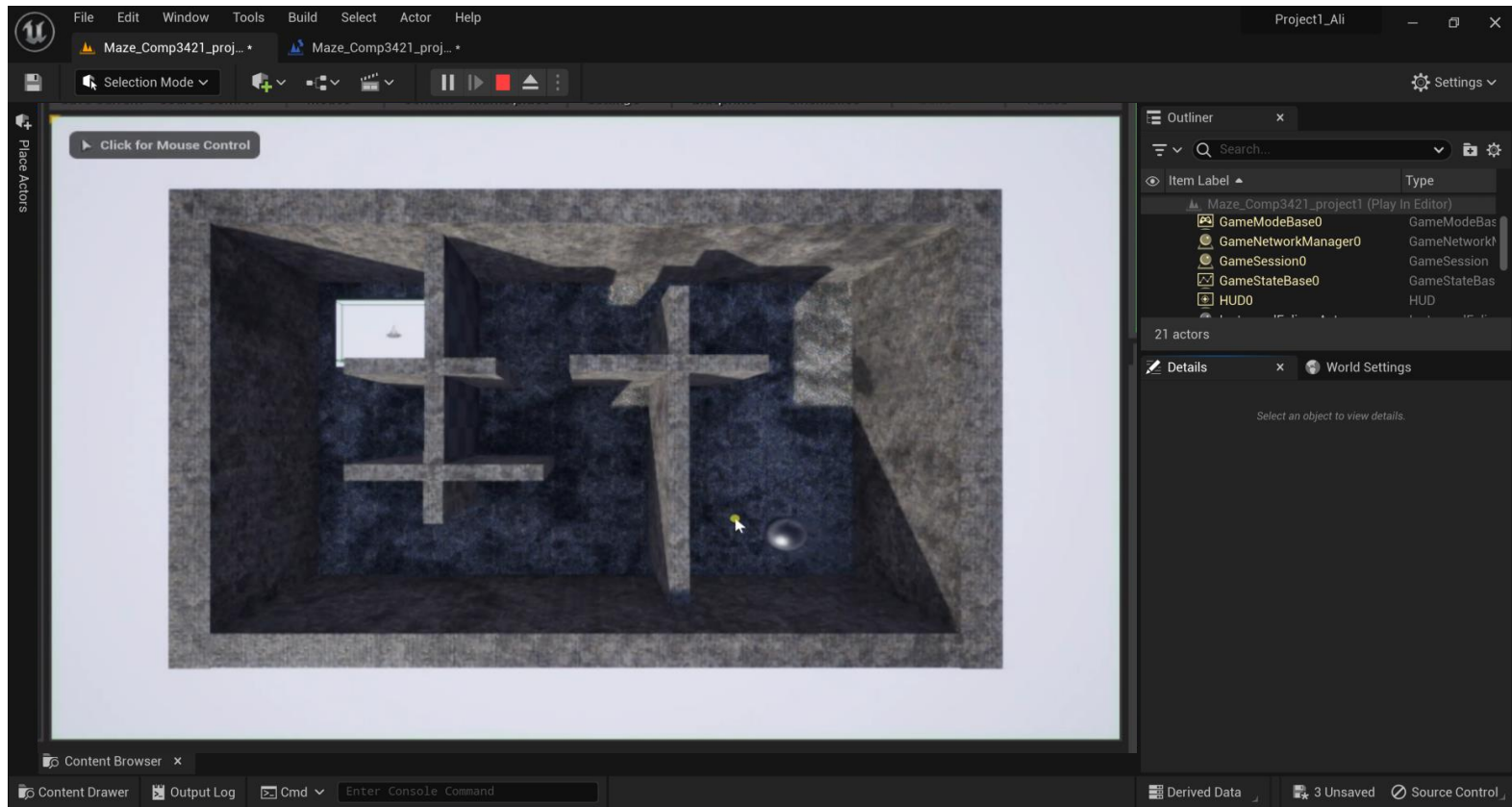
Create a win condition for our levels

- Make the Trigger box visible > Details panel > Rendering section > Uncheck the 'Actor Hidden in Game'.



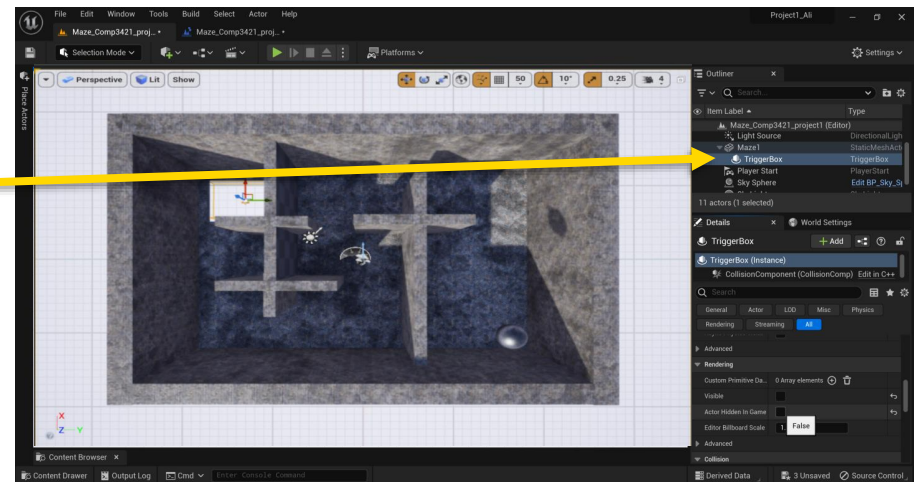
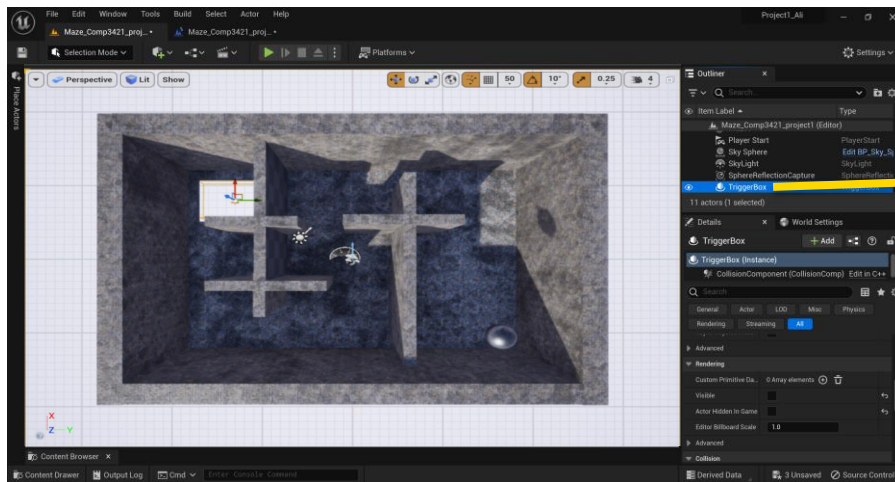
Create a win condition for our levels

By default, the trigger box doesn't move with the maze.

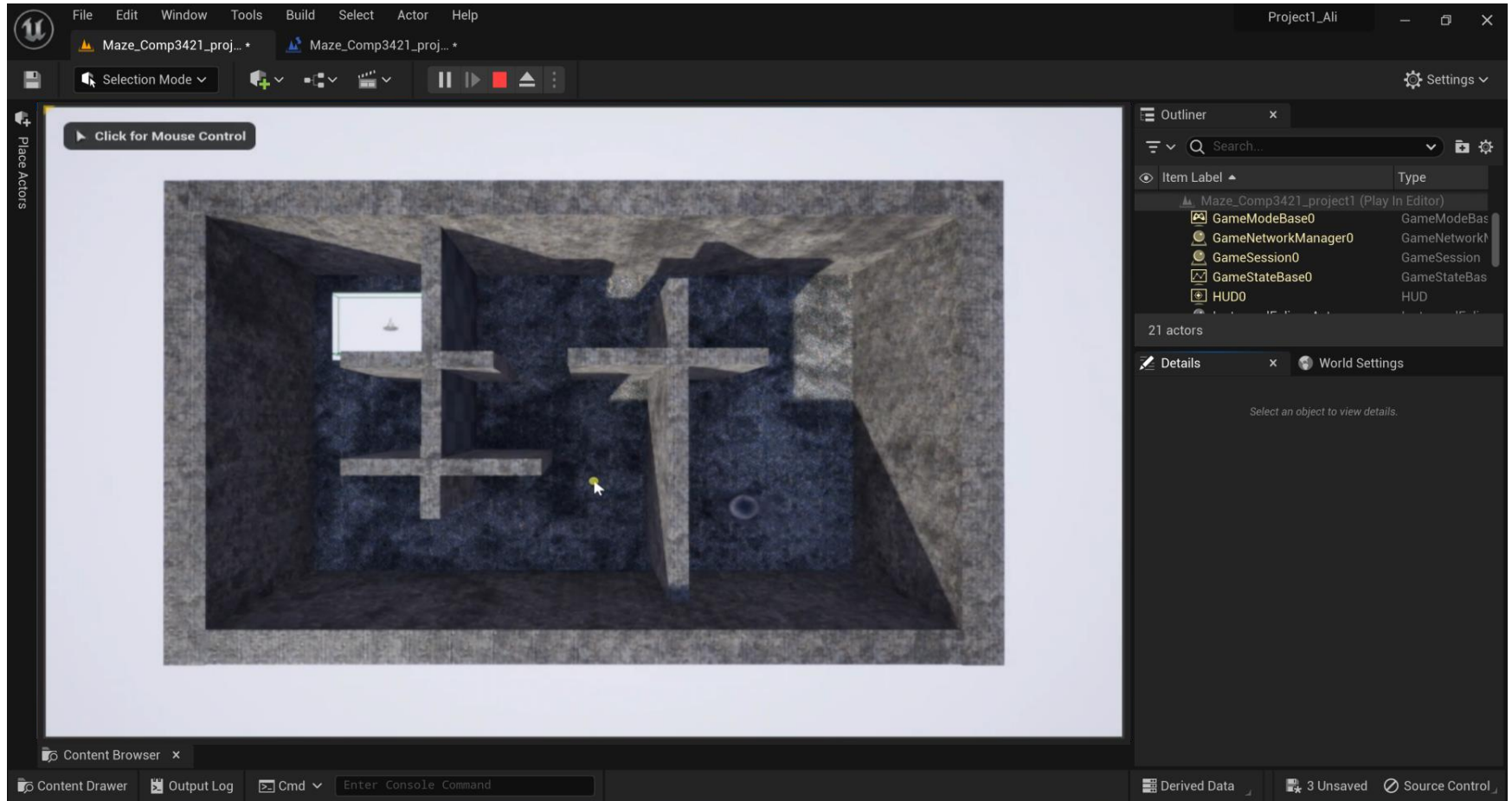


Create a win condition for our levels

- Attach the trigger box to the maze component.
- Go to the Outliner > Drag and drop the **Trigger Box** onto the maze static mesh.

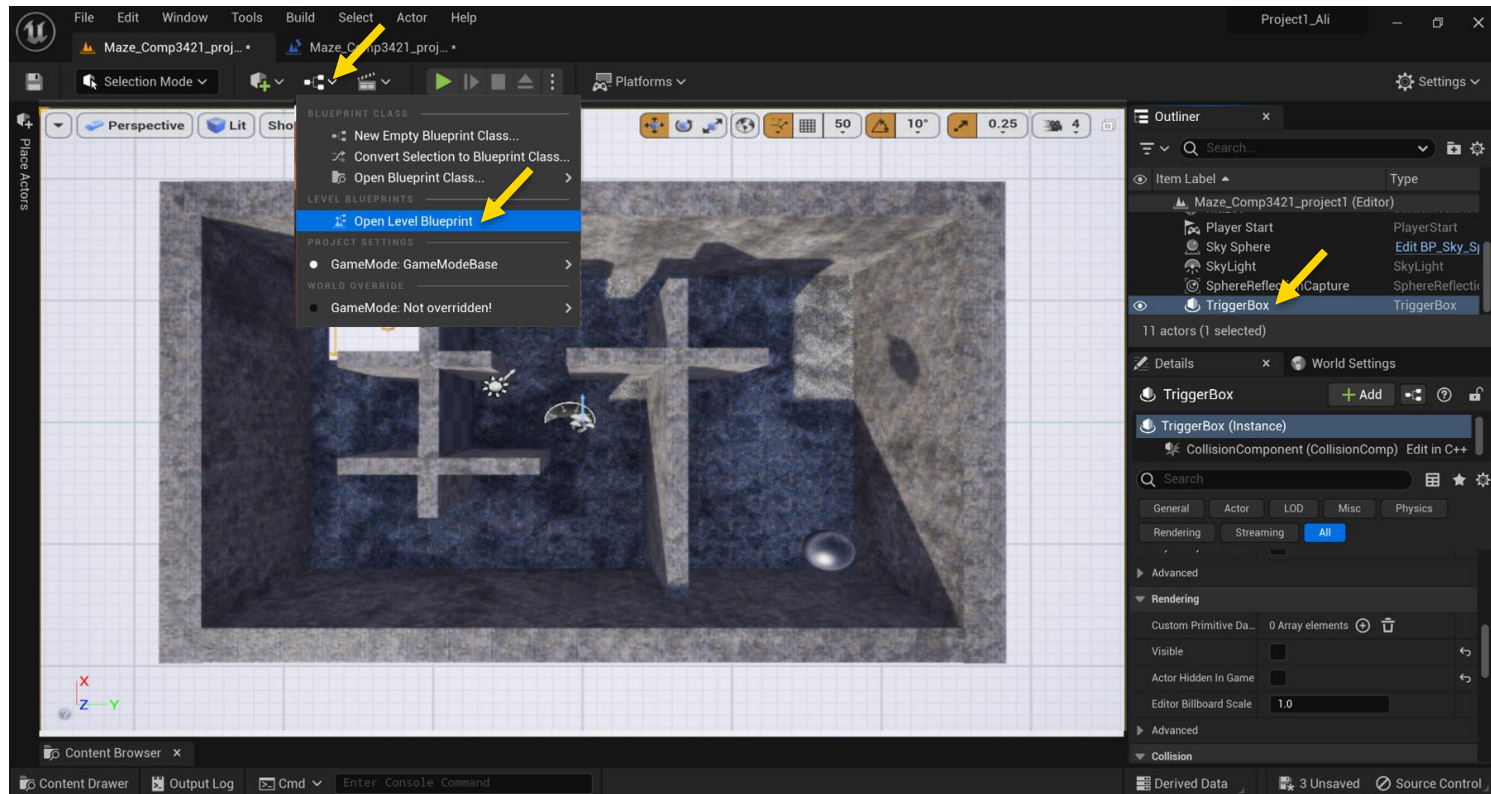


Create a win condition for our levels



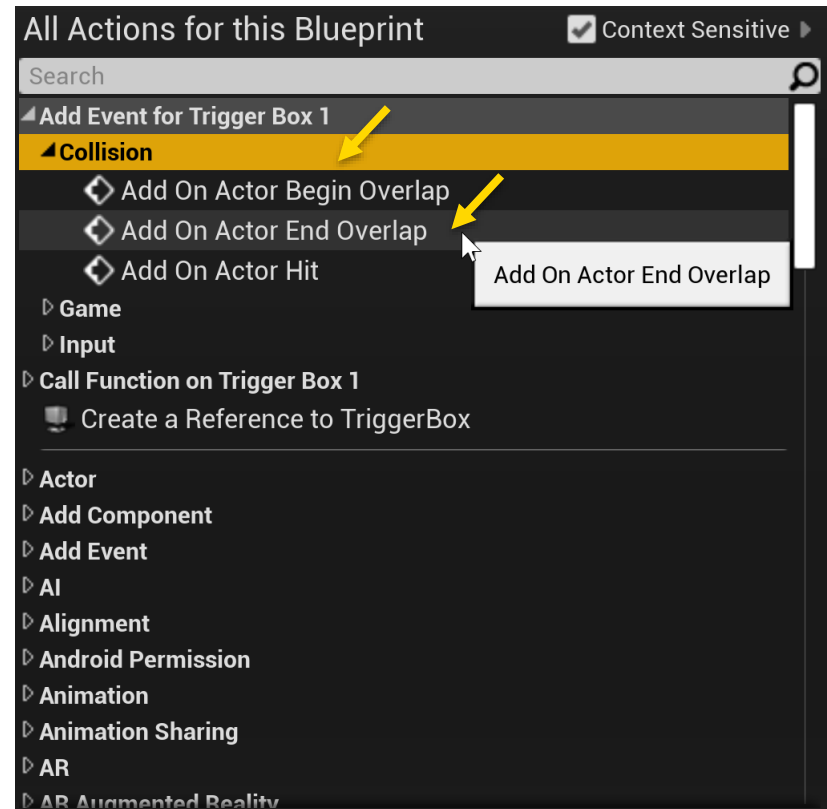
Create a win condition for our levels

- Select the **Trigger box** from the Outliner panel.
- Open the level blueprints.



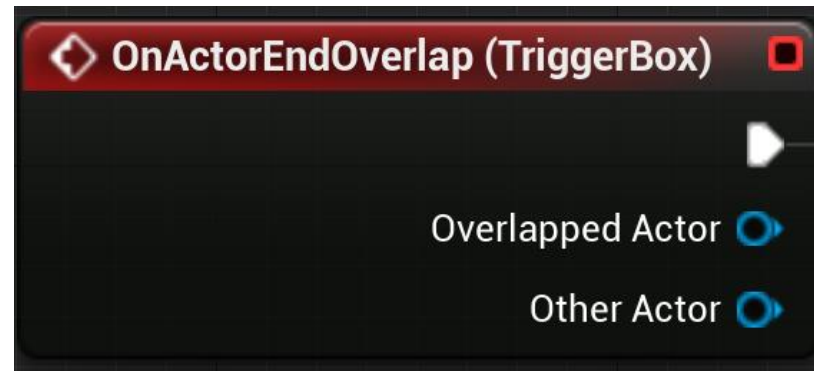
Create a win condition for our levels

We need an event for when our ball escapes the maze > Right click on the blueprint editor > Collision category > Use 'Add On Actor End Overlap' event.



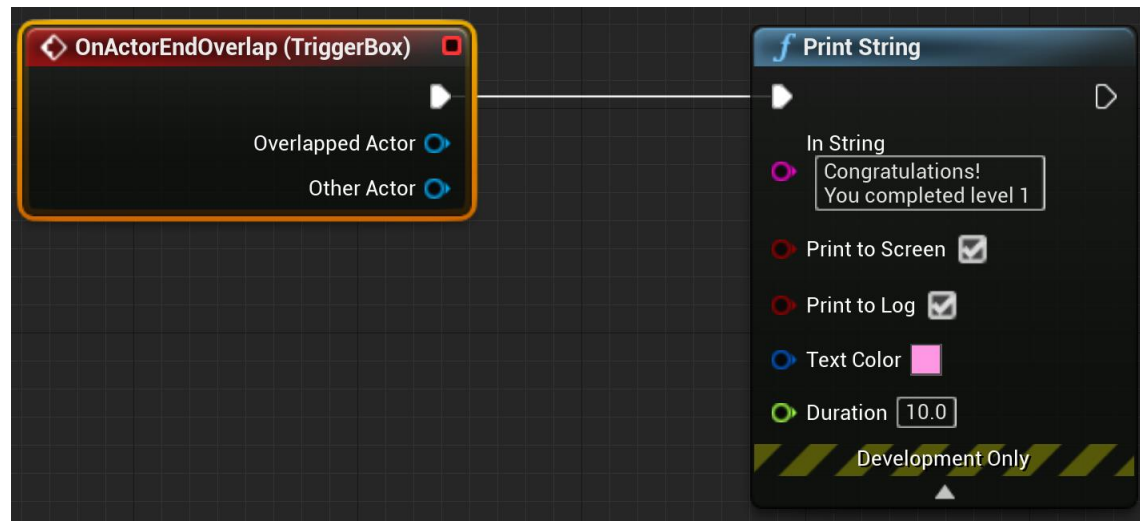
Create a win condition for our levels

- Execution pin fires a command after an overlap.
- ‘**Overlapped Actor**’ pin returns a reference to the object that overlapped (it is the trigger box itself).
- ‘**Other Actor**’ pin returns the actor that overlapped with the trigger box.

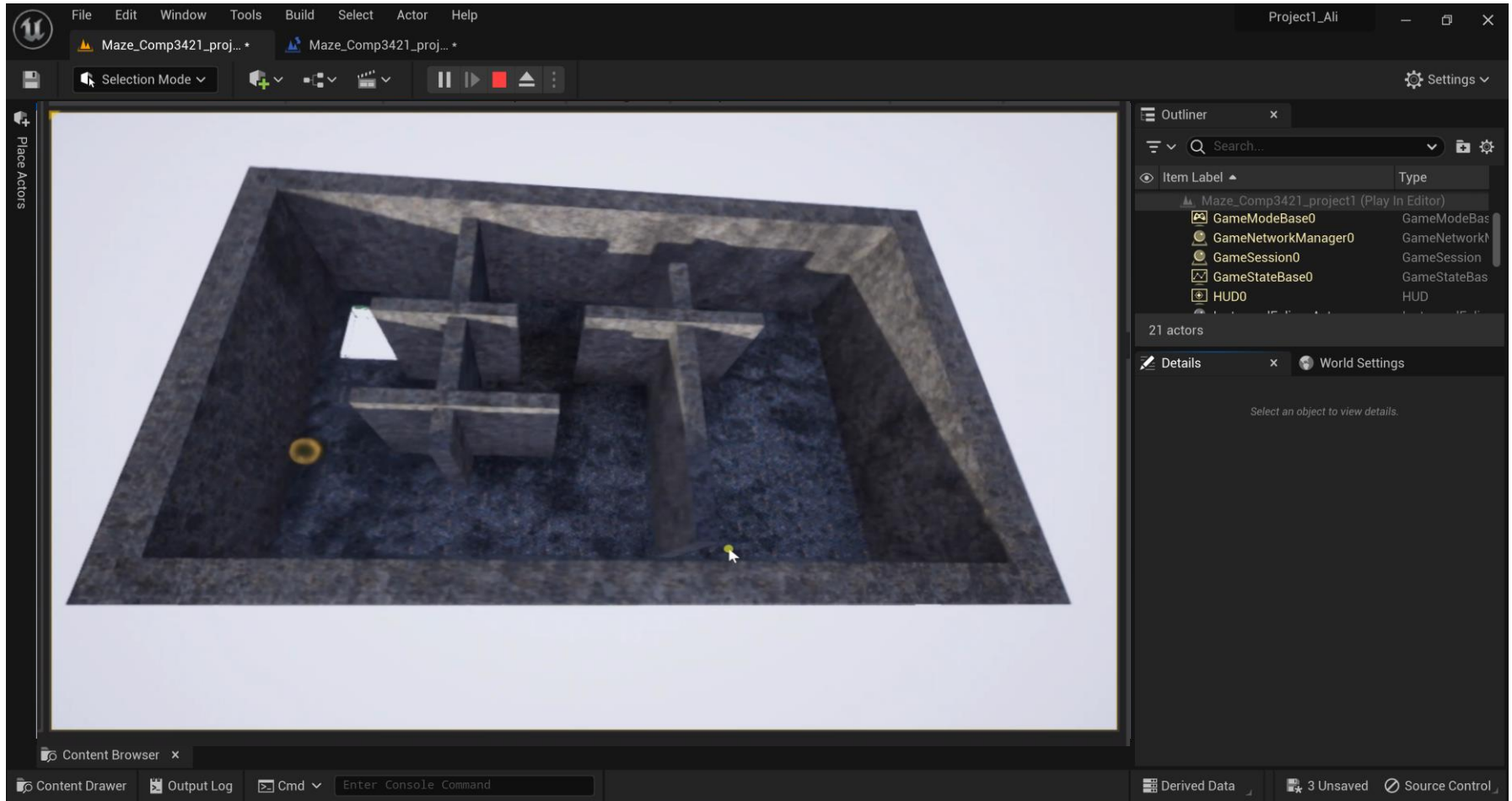


Create a win condition for our levels

- Use the execution pin to show a message to users when the ball escapes the maze.

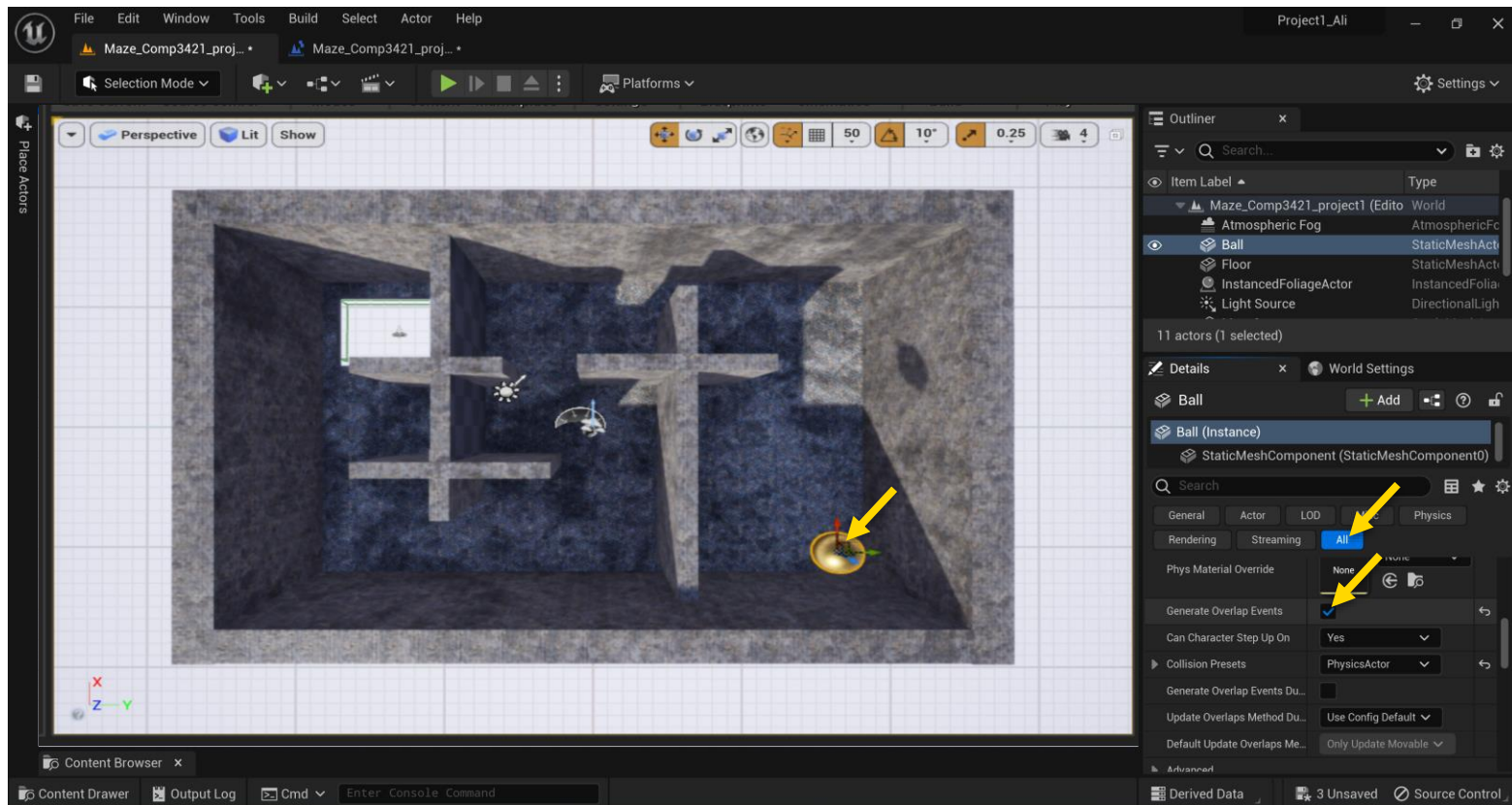


Create a win condition for our levels

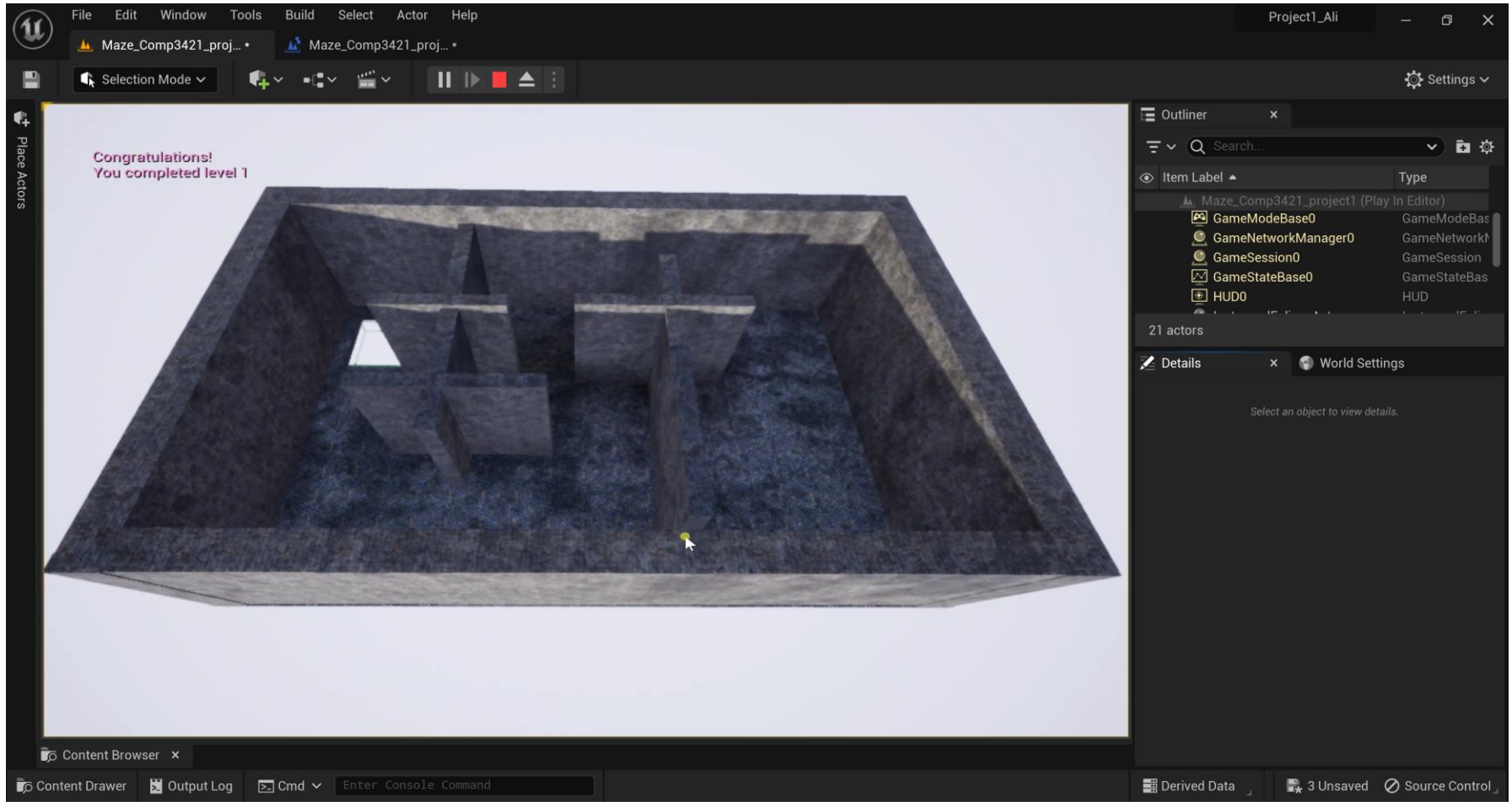


Create a win condition for our levels

- Select the ball > Details panel > Collision section > Activate the **Generate Overlap Events**.
- Without enabling **Generate Overlap Events**, the overlap event won't work.

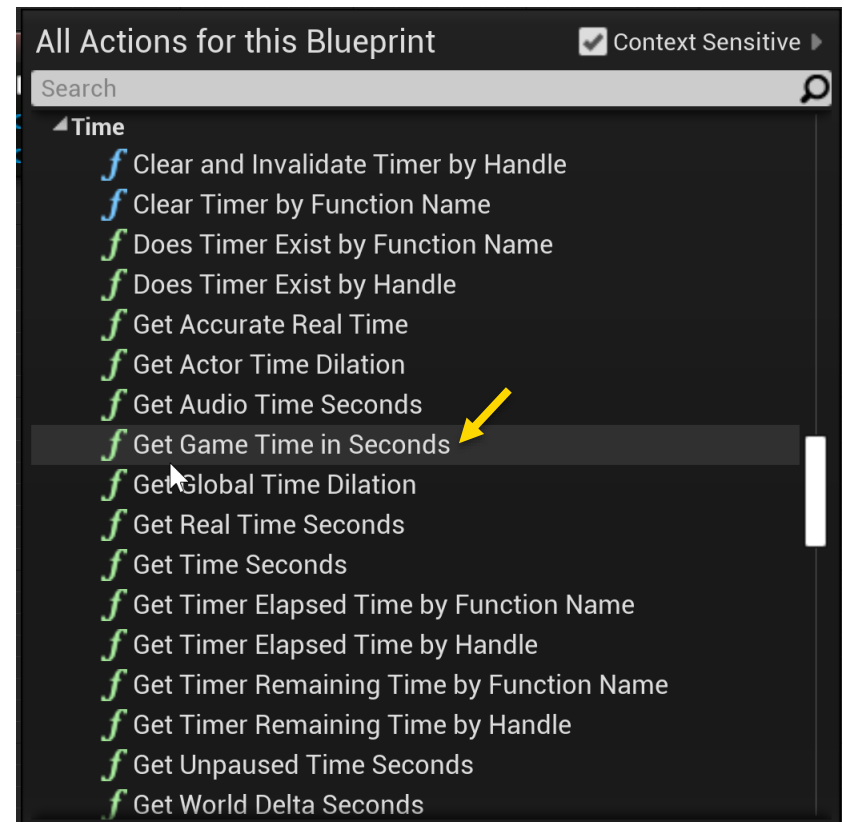


Create a win condition for our levels



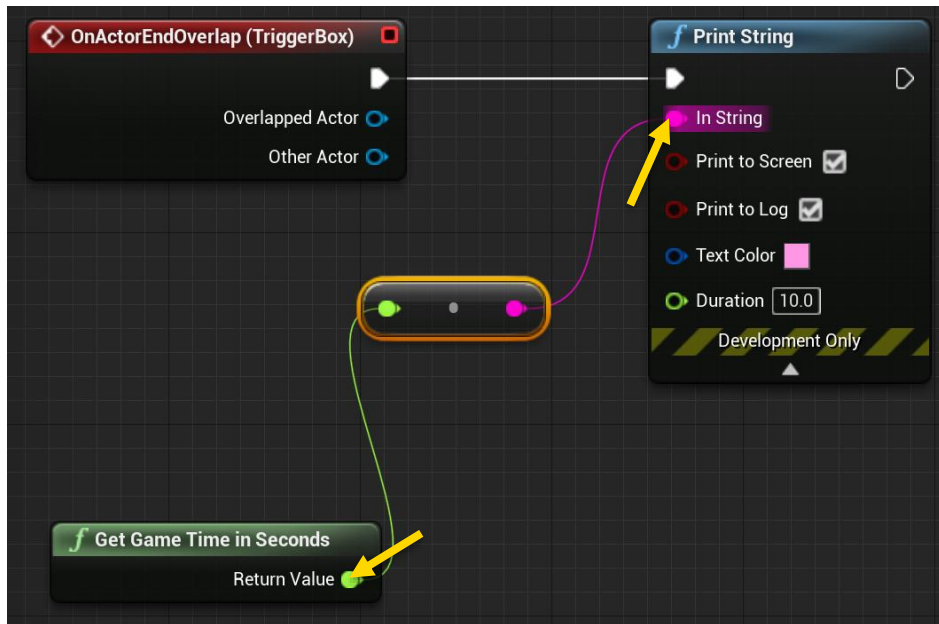
Showing the time duration, it takes for the player to make the ball escape

Right click on the blueprints editor > Utility section > Time category > select 'Get game time in seconds' function.

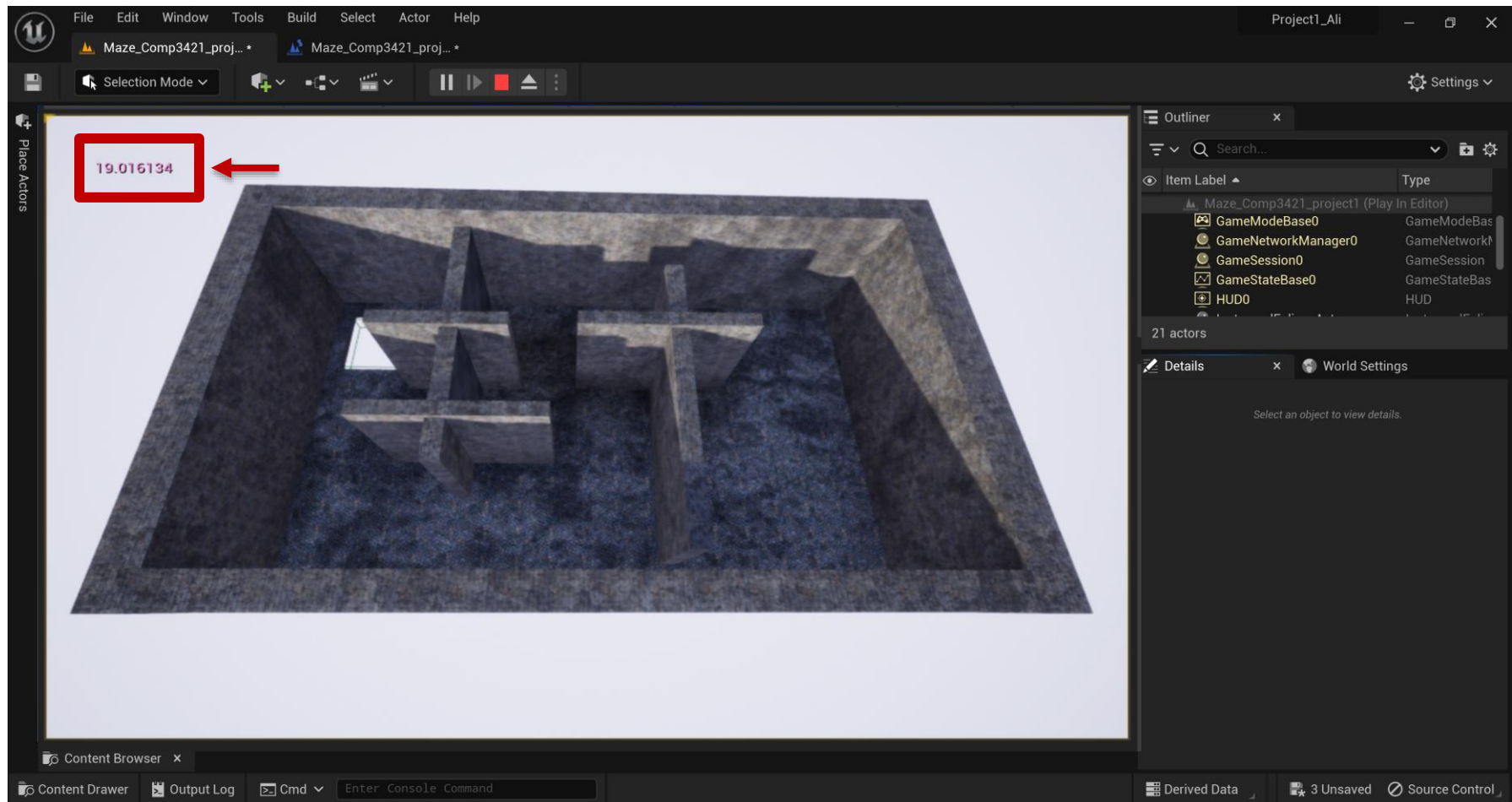


Showing the time duration, it takes for the player to make the ball escape

- **Get game time in seconds** function returns a float value.
- **Get game time in seconds** function tells us how much time has passed since we've started the game.

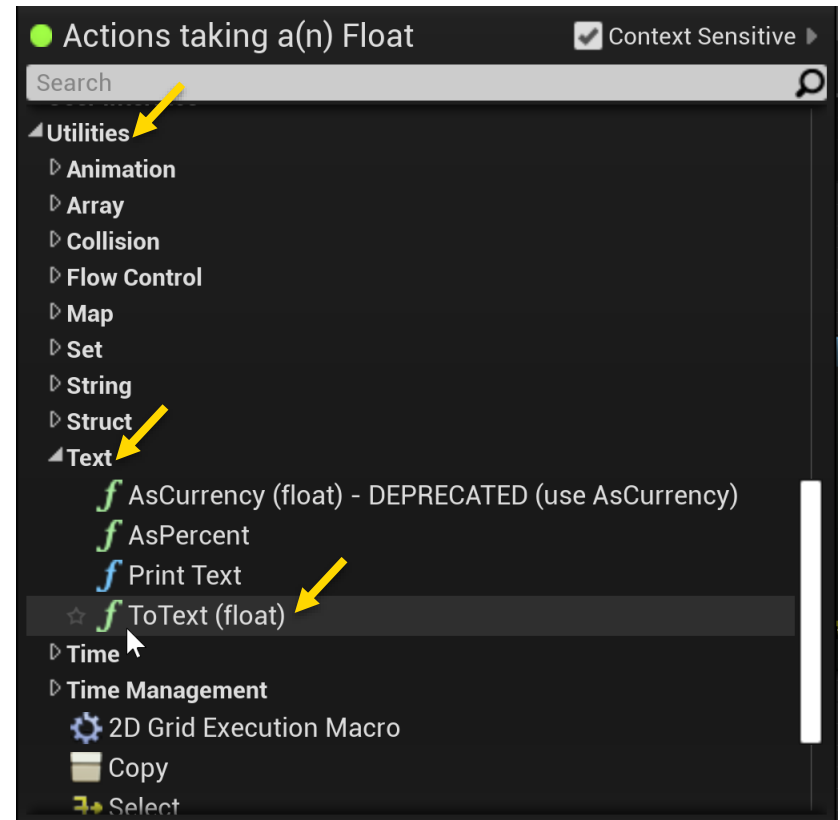


Showing the time duration, it takes for the player to make the ball escape



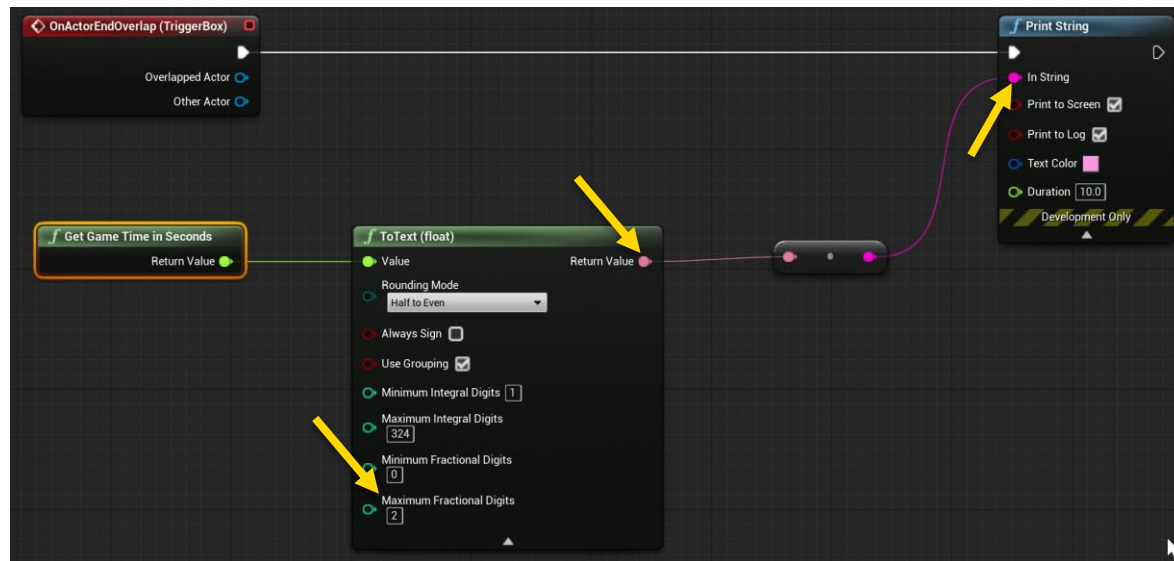
Showing the time duration, it takes for the player to make the ball escape

- Decreasing the number of decimal places of the time duration
- Utilities > Text category > **To Text (float)**



Showing the time duration, it takes for the player to make the ball escape

- Connect the return value of the 'To text (float)' function to the **String** pin of the **Print String** node.
- Expand the 'To text (float)' node settings and set the number of decimal places using 'Maximum fractional digits'.

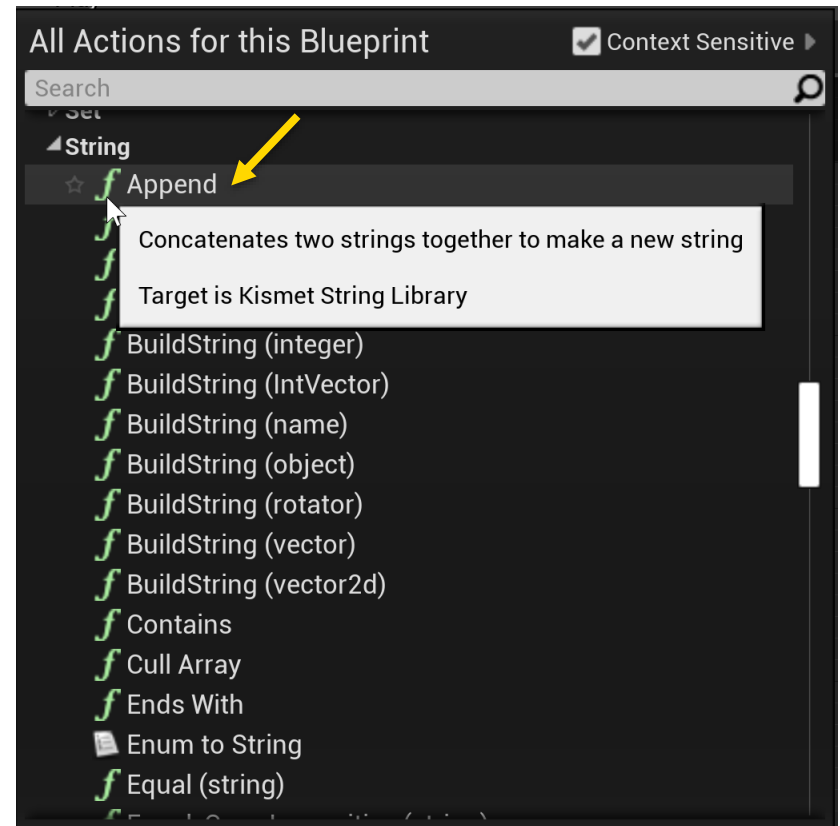


Showing the time duration, it takes for the player to make the ball escape + a text message

- Winning message = “Well done, you escaped the maze in 22.53 seconds”
- Solution: Use **Append String** function → text + time + text

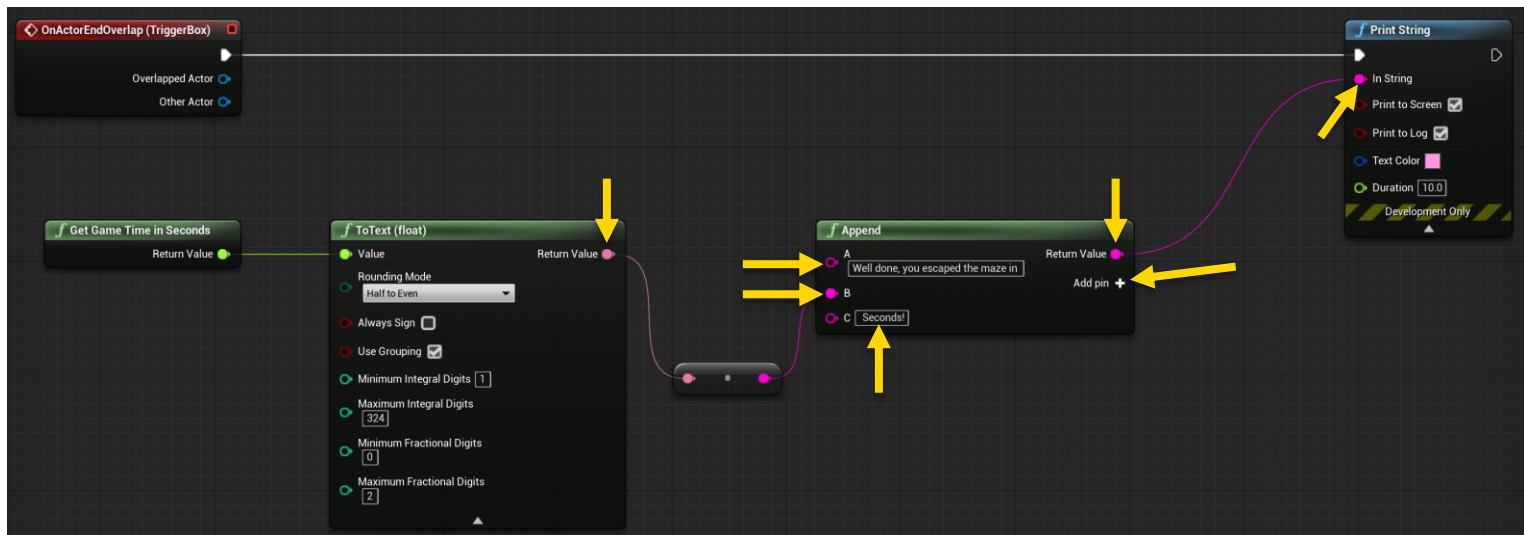
Showing the time duration, it takes for the player to make the ball escape + a text message

- Utilities category >
String section >
Append

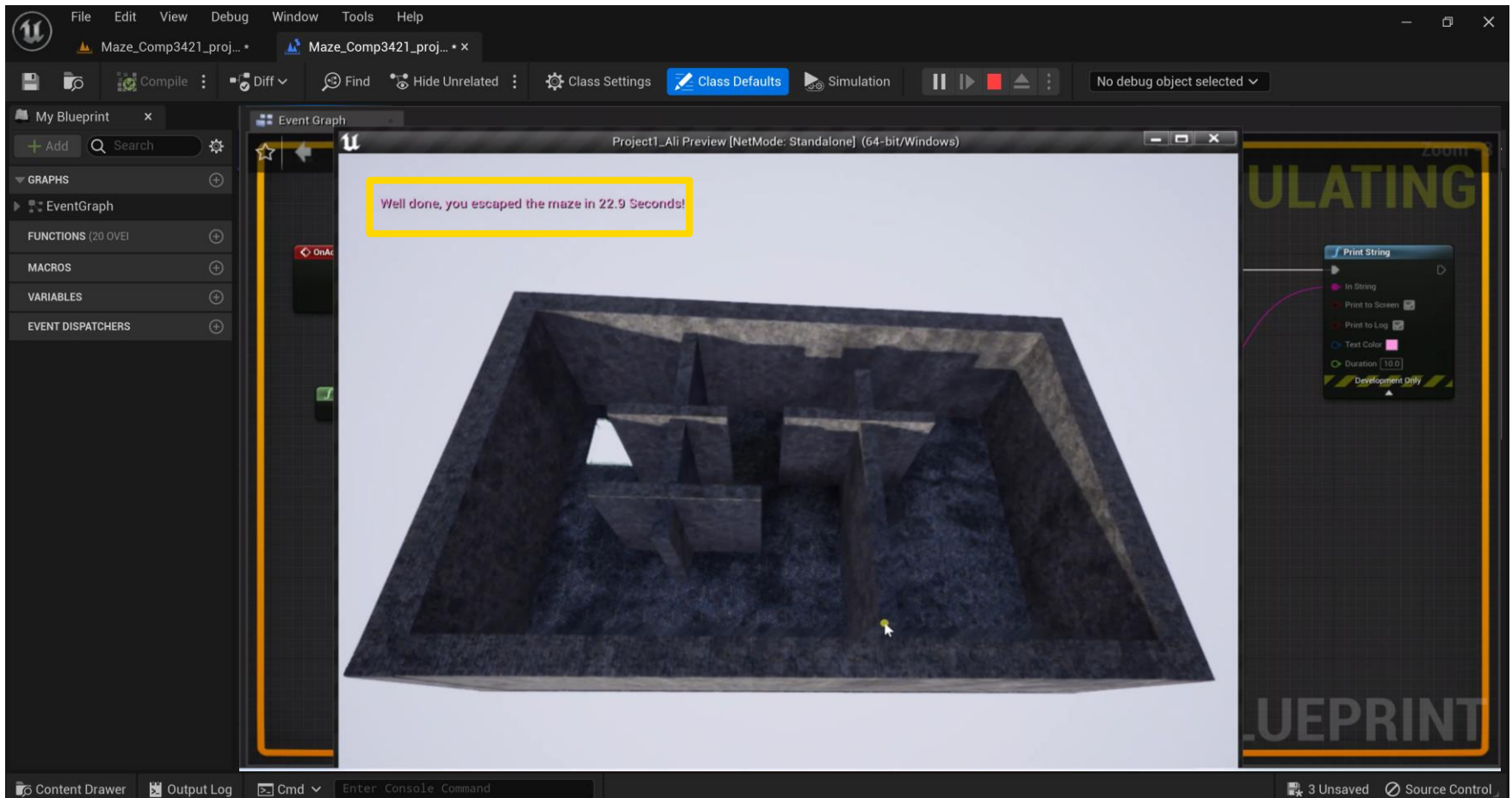


Showing the time duration, it takes for the player to make the ball escape + a text message

- In the A pin of the append function, type 'Well done, you escaped the maze in' > Connect the B pin to the **Return Value** pin of '**To Text Floats**' function > Add the third string by clicking the '**Add pin**' button > Type 'seconds' in C pin.
- Connect the '**Return value**' pin of the **Append** function to the '**In string**' pin of the '**Print String**' function.



Showing the time duration, it takes for the player to make the ball escape + a text message



Lecture activity 4

- Create a winning message for your maze.
- The message should show the time duration of completing the level.
- Decrease the number of decimal places of the time duration to two.
- Customise the text colour of the message.

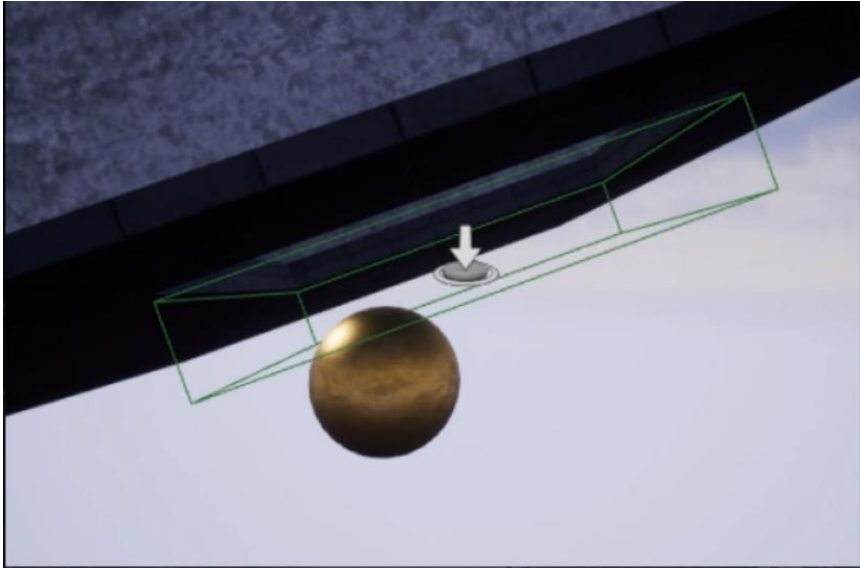


Fixing the bug of the winning message

- Potential bug: getting more than one win message.

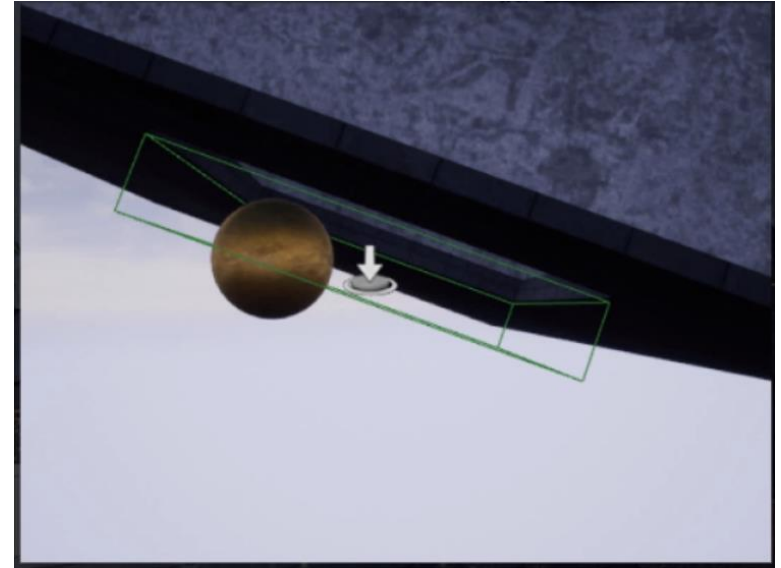


Fixing the bug of the winning message



Win message #1

Ball has ended overlap with trigger box.

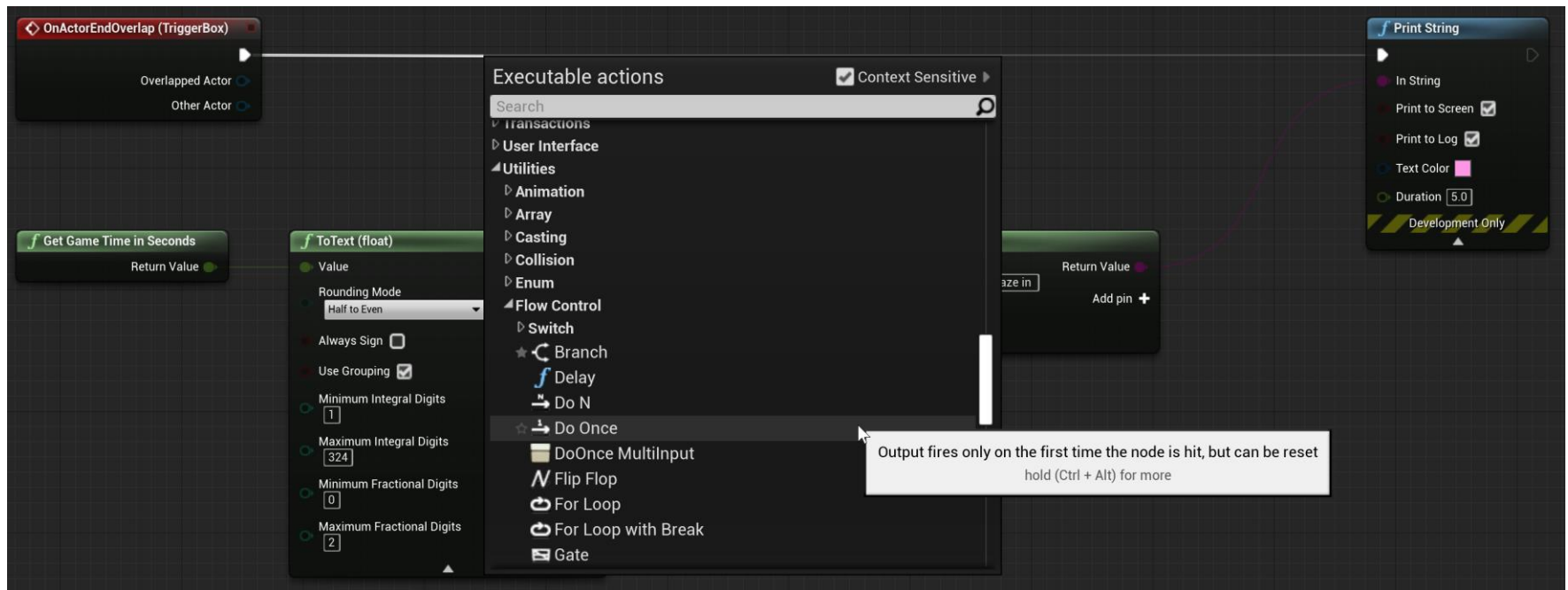


Win message #2

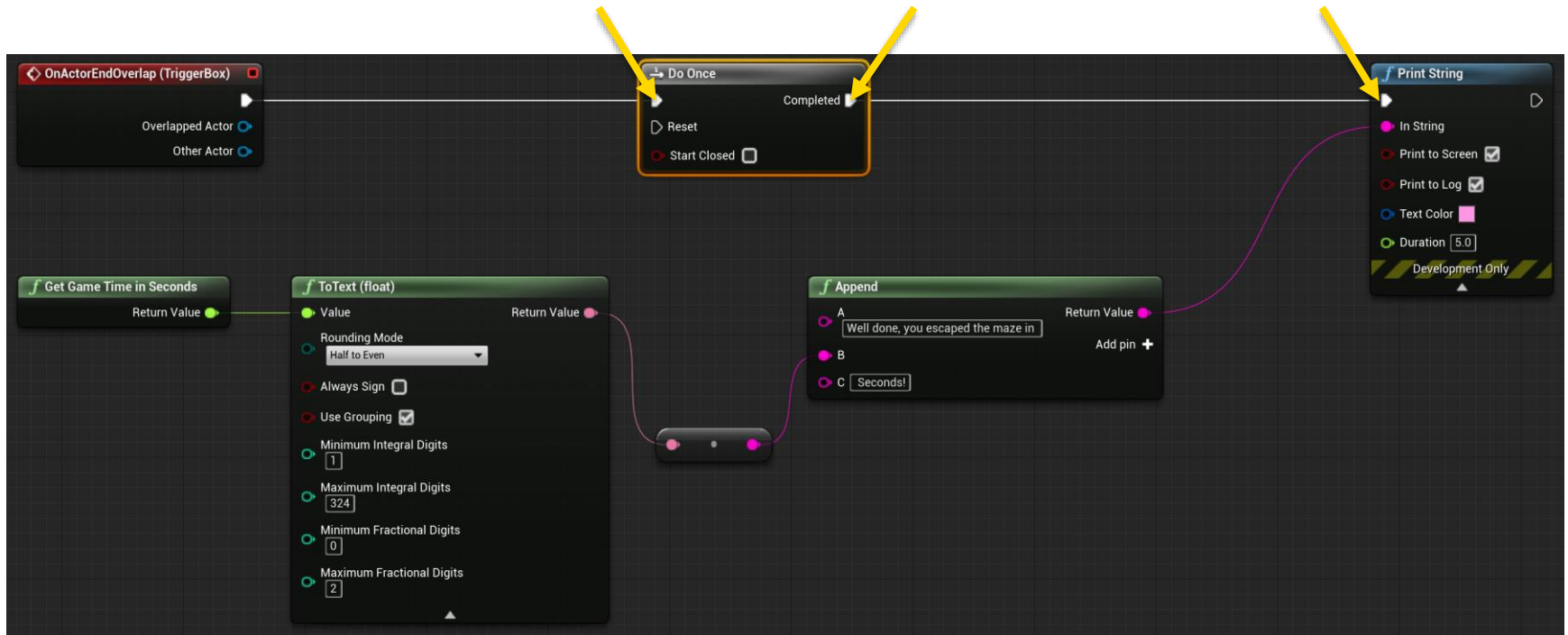
Maze moved and ball is inside trigger box again, and will fall, causing second end overlap

Fixing the bug of the winning message

- Use a blueprints command called '**Do Once**'.
- Go to utilities category > Flow control category > Click '**Do Once**'.

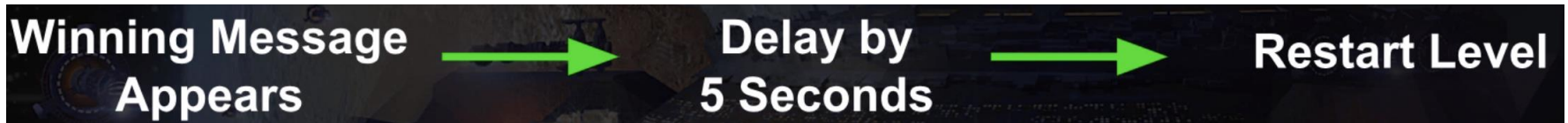


Fixing the bug of the winning message



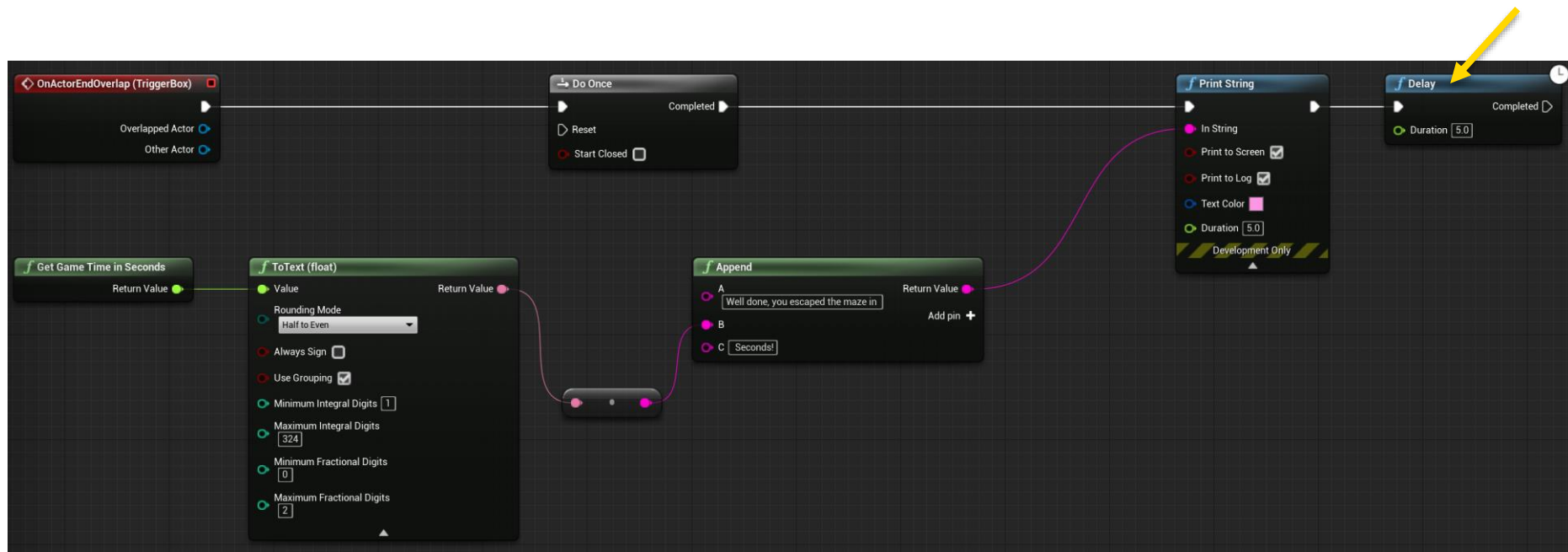
Restarting the level

Enable users to restart the level when they complete it.



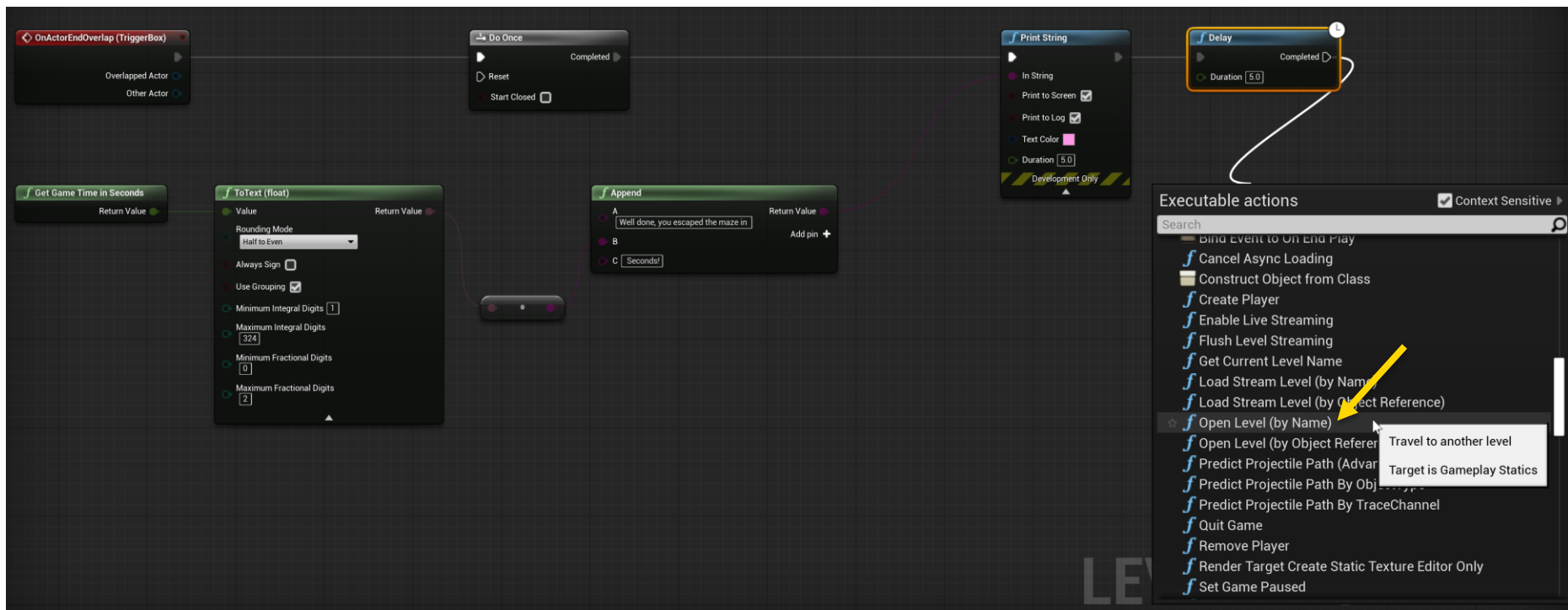
Restarting the level

- Add a Delay after the 'Print String' function to delay restarting the game by five seconds.



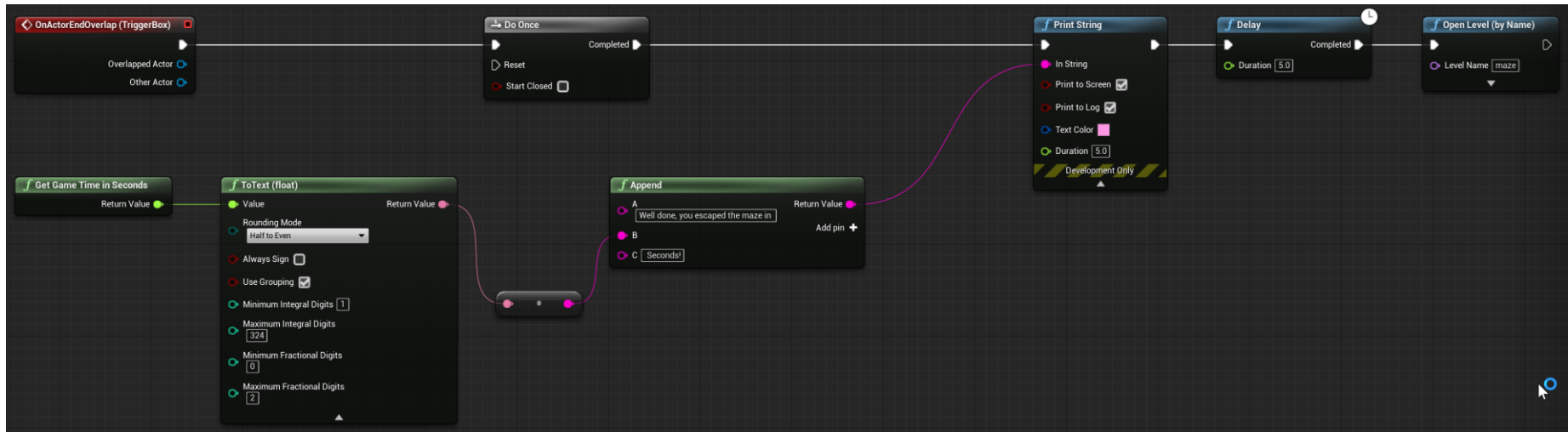
Restarting the level

- Drag out the output pin of the Delay function > Go to the Game category > Click 'Open level by name' function.

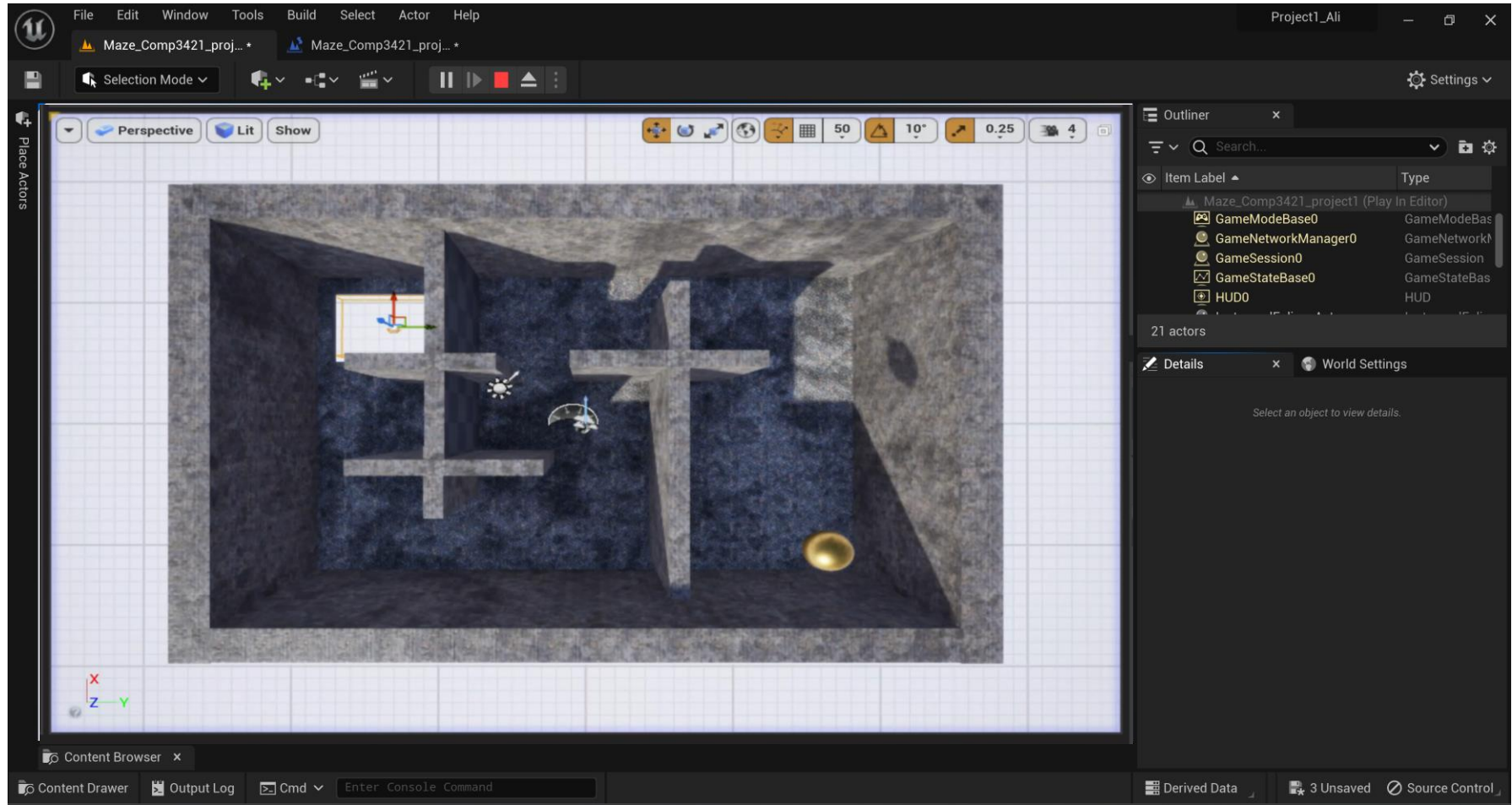


Restarting the level

- Type your level name in the open level function.
- You can find your level name in the Outliner panel and the asset repository.

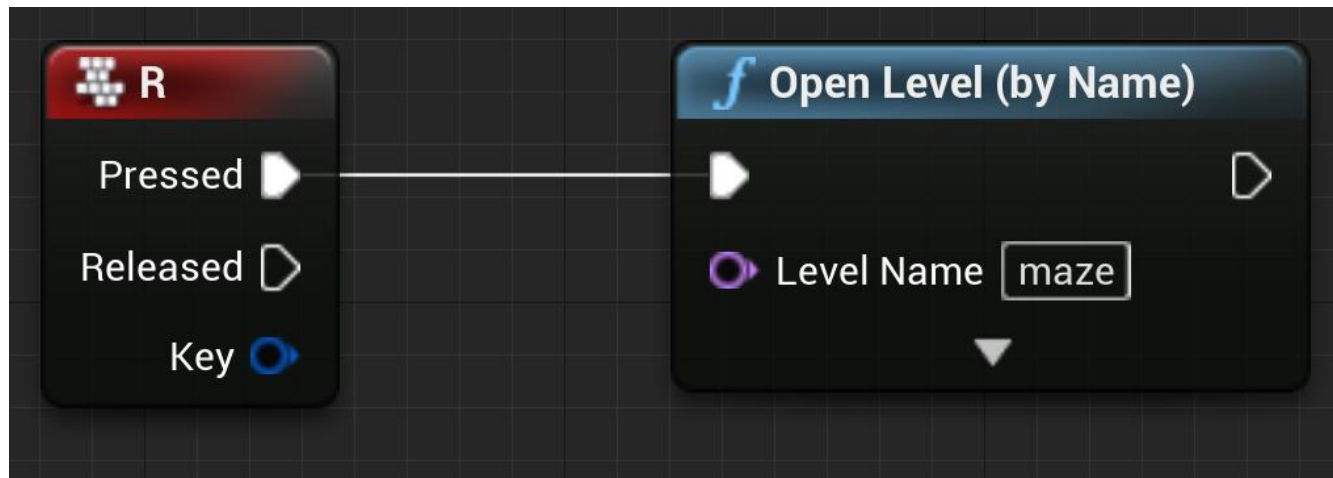


Restarting the level



Restarting the level

- Restart the level by pressing the **R key** on the keyboard.



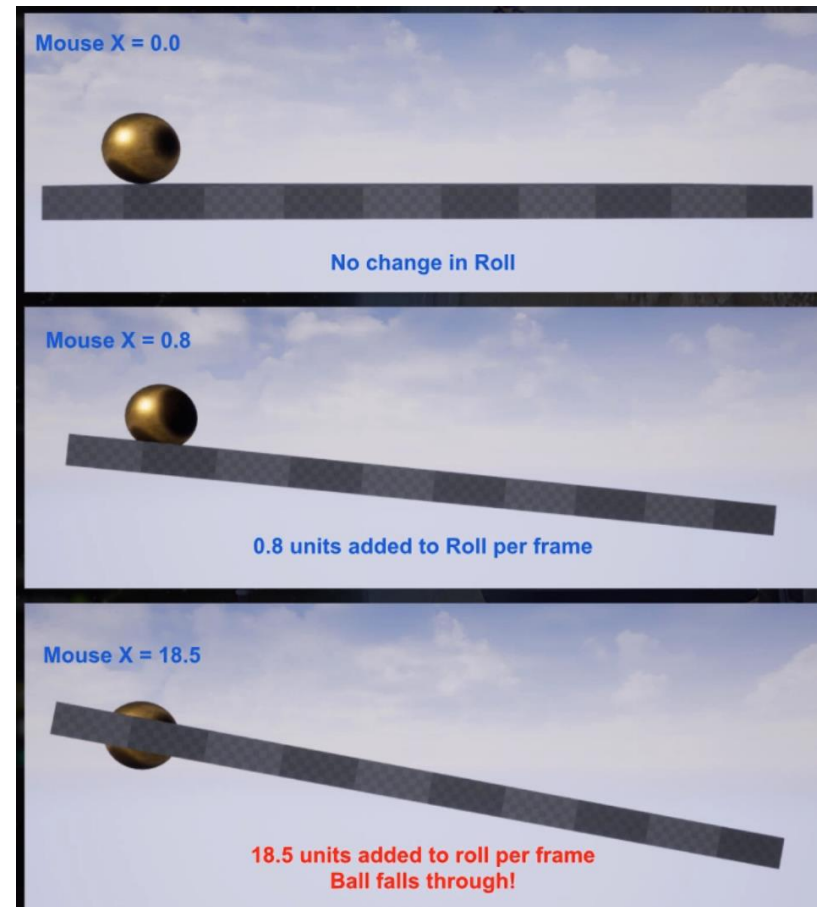
Fixing the bug of the ball falling through the maze

When we play the game and rotate the maze really quickly, the collision of the ball gives up and it falls through the maze.



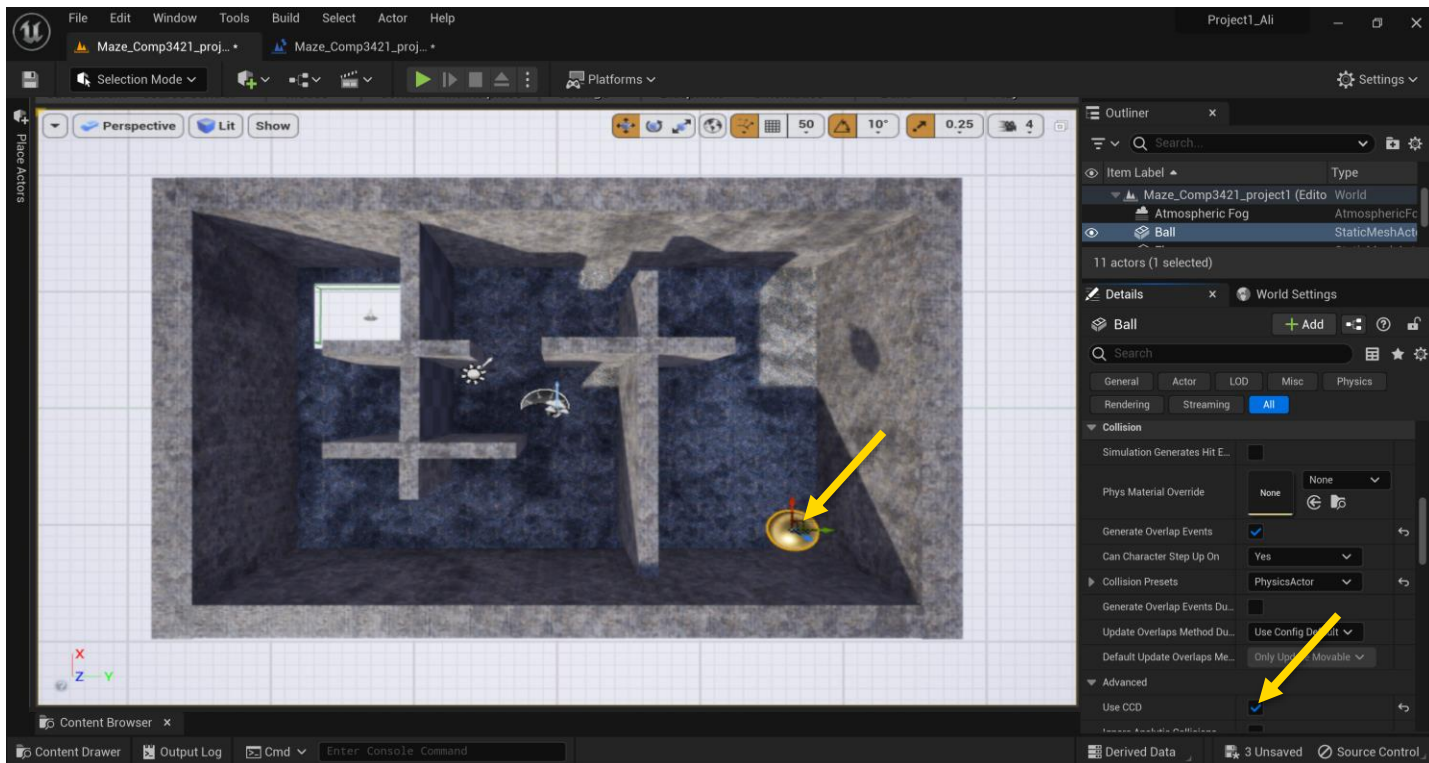
Fixing the bug of the ball falling through the maze

- Mark the ball to use continuous collision detection (CCD) helps but not a complete solution.
- The best solution is clamping mouse input values (Mouse speed). This avoid input values that are too large to be added/subtracted to maze rotation.



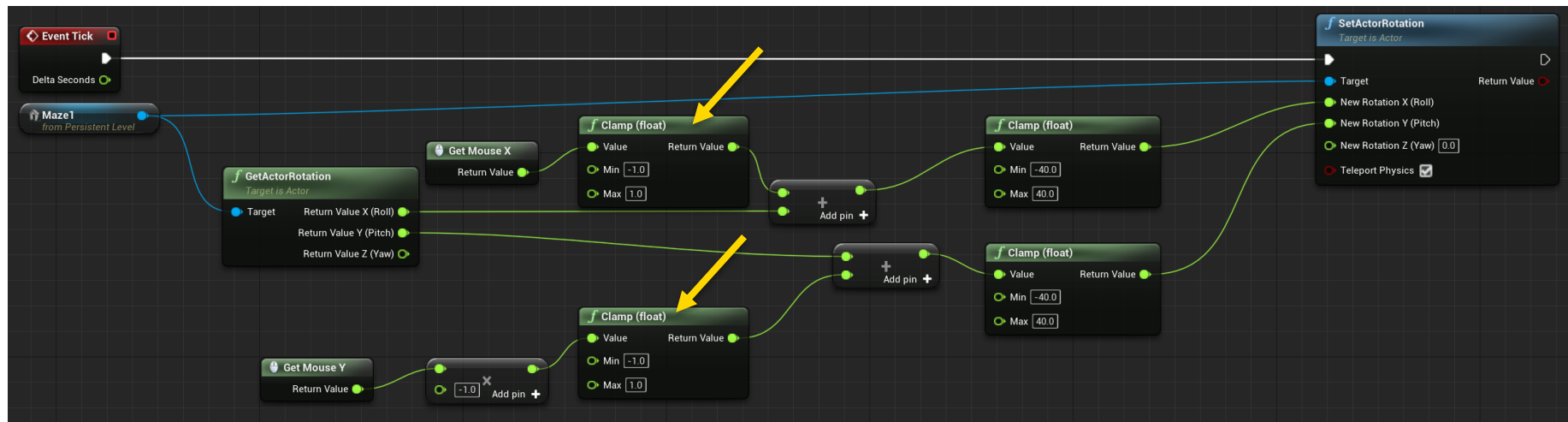
Fixing the bug of the ball falling through the maze

Details panel > Collision section > Enable 'Use CCD', which is continuous collision detection.

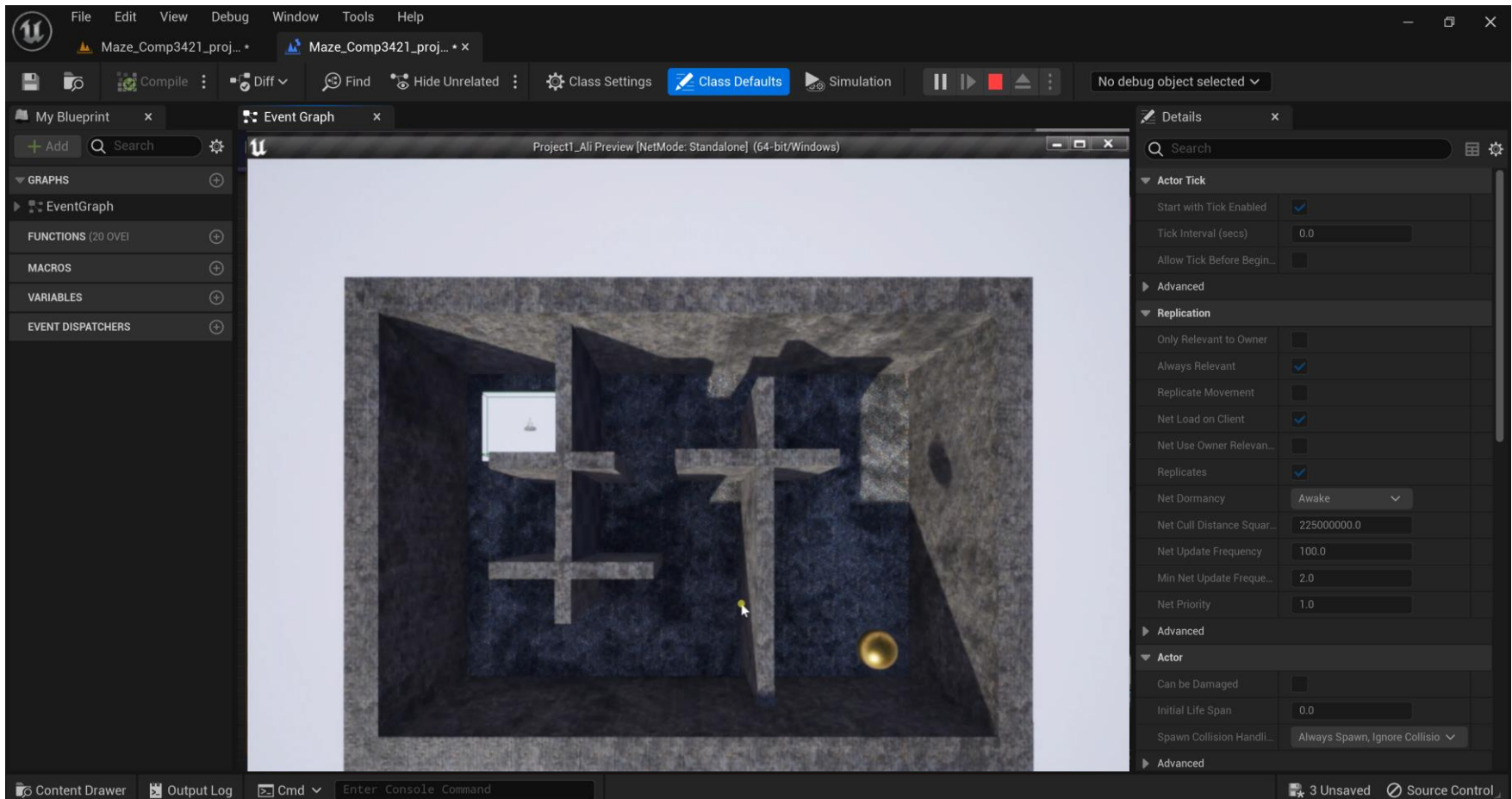


Fixing the bug of the ball falling through the maze

- Add a clamp function for 'Get mouse x' and for 'Get mouse Y'.
- Clamp the mouse movement speed between -1 and +1.

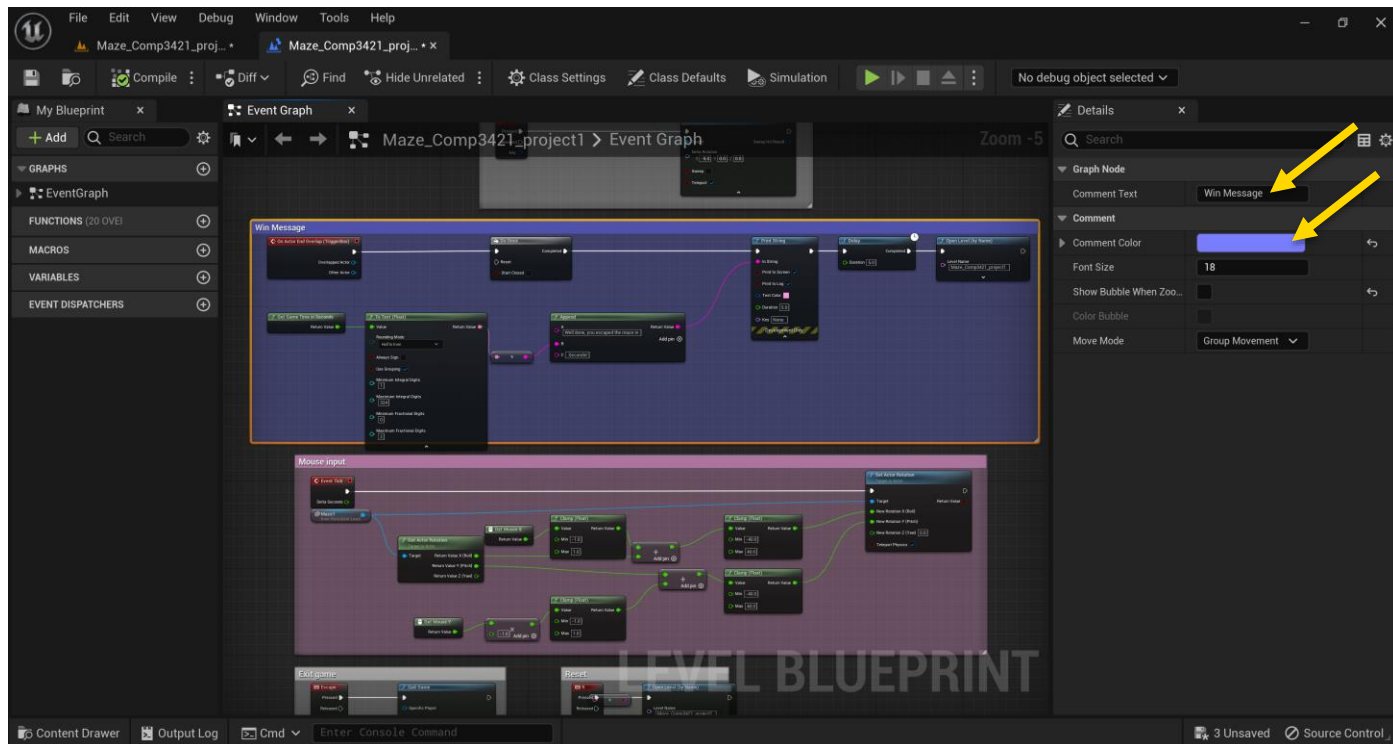


Fixing the bug of the ball falling through the maze



Organising the blueprints code

In order to group the related nodes > Select all of them > Press the **C** key to group > Add a name for the group.



Lecture activity 5

- Organize your code into comment sections.
- Change the color of the comment sections.



